

Modeling Regularization in Language Acquisition as Noise-Tolerant Grammar Selection

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Abstract

Language acquisition involves drawing systematic generalizations from messy data. On one hypothesis, this is facilitated by a domain-general bias for children to “regularize” their input, sharpening the statistical distributions in their input towards more systematic extremes. We introduce a general computational framework for modeling a different explanation: on this view, children expect that their data are a noisy realization of a restrictive underlying grammatical system. We implement a learner that evaluates a choice among composite context-free grammars, in which a restricted set of “core” rules, comprising the particular grammatical processes that the learner is currently trying to acquire, operate alongside a less restricted set of “noise” rules, representing other independent processes that have yet to be learned, and conspire to introduce distortions into the data. Our *Noisy Grammar Learner* partitions its data into portions that serve as evidence for one of the possible core grammars in its hypothesis space, and portions generated by these noise processes. It does so without knowing in advance how much noise occurs or what its properties are. We compare our learner to a common implementation of the general regularization bias approach, and show that both can account for children’s behavior in a representative artificial language learning experiment. However, we find that our approach performs better on two naturalistic case studies in early syntax acquisition: learning the rules governing canonical word-order and case-marking, given natural language data with “noise” from non-canonical sentence types. We show that our learner succeeds because its architecture allows a natural way to express linguistically-motivated expectations about the character of those rules. This suggests that, in certain domains, successful learning from messy data may be enabled by a hypothesis space comprising restrictive grammatical options.

Keywords: language acquisition, syntax, grammar, regularization, computational modelling, Bayesian reasoning

1 Introduction

Language acquisition involves drawing systematic generalizations from data that appear on the surface to be ambiguous and messy. For instance, infants acquire the system of sound categories in their language from noisy, overlapping distributions of sound tokens in the speech that they hear (Bion, Miyazawa, Kikuchi, & Mazuka, 2013; Cristia, 2018; Hitczenko & Feldman, 2022; Maye, Werker, & Gerken, 2002; Swingley, 2019). They learn the phonetic and phonological regularities that mark word boundaries in their language, despite a high degree of variability in how words are pronounced (Beech & Swingley, 2023; Cristia, Dupoux, Ratner, & Soderstrom, 2019; Jusczyk & Aslin, 1995; Mattys, Jusczyk, Luce, & Morgan, 1999; Mattys & Jusczyk, 2001). What kind of mechanisms allow for learning to abstract away from such messiness in the learner’s representation of the data?

In this paper, we examine a specific case of messiness that arises as learning proceeds incrementally over the course of development: the case of *opacity*, where evidence for one grammatical property can be hidden by other interacting grammatical properties that have not yet been acquired. One example that illustrates this issue is a problem in the domain of syntax learning, of identifying the rules governing basic word order. The canonical order of subjects, verbs, and objects in a language varies cross-linguistically and therefore must be acquired, but other properties of the grammar conspire to hide this canonical word order in certain sentences. For instance, children learning English and French must identify a grammatical rule system that canonically produces subject-verb-object (SVO) orders, whereas children learning Japanese must identify a rule system that canonically produces subject-object-verb (SOV). This task could be abstractly characterized as selecting among rules for basic clauses that place the subject noun-phrase before or after the verb-phrase, and rules that place the object noun-phrase before or after the verb. Empirical evidence suggests that children learn their language’s basic word order very early in development, in infancy (Hirsh-Pasek & Golinkoff, 1996; Gertner, Fisher, & Eisengart, 2006; Lidz, White, & Baier, 2017; Perkins & Lidz, 2020; Franck, Millotte, Posada, & Rizzi, 2013; Zhu, Franck, Rizzi, &

Gavarró, 2022; Gavarró, Leela, Rizzi, & Franck, 2015). What is intriguing is that they do so at ages before they can identify the sentences where interacting processes have caused subjects and objects to appear in non-canonical positions (Dautriche et al., 2014; Perkins & Lidz, 2020, 2021; Gagliardi, Mease, & Lidz, 2016). These kinds of processes include those that produce *wh*-questions and relative clauses, as in (1). In these sentences, a fronted phrase acts as the verb’s direct object in a non-canonical position, rather than post-verbally; they therefore depart from the basic SVO word order of English reflected in (2).

- (1) a. Which dog is she chasing?
- b. That’s the dog we like.
- (2) a. She’s chasing a dog.
- b. We like that dog.

If infants cannot yet distinguish between sentences exhibiting basic word order and those that depart from this basic word order, because they do not yet know the form that *wh*-questions and relative clauses take in the language, then the representations that they will be able to construct of these sentences do not coherently point towards the correct canonical word order. The sentences in (2), with one NP preverbally and one NP postverbally, could be taken as evidence for rules that canonically produce verb-medial orders; but the sentences in (1), with multiple preverbal NPs, could equally be taken as evidence for rules that produce SOV canonically. The existence of *wh*-questions and relative clauses therefore distorts the body of evidence that an infant might use for the basic positions of clause arguments, in effect rendering infants’ data messy and misleading at the stage of development when word order learning takes place. Infants nonetheless manage to abstract away from this messiness to draw accurate generalizations about basic word order.

This example illustrates how opacity can arise from a grammatical system with multiple interacting components, when a learner is attempting to acquire these components incrementally. This type of learning problem has received substantial previous attention in

the syntax learning literature (e.g., Fodor, 1998; Frank & Kapur, 1996; Gibson & Wexler, 1994; Howitt, Dey, & Sakas, 2021; Hyams, 1986; Lightfoot, 1991; Manzini & Wexler, 1987; Niyogi & Berwick, 1996; Sakas & Fodor, 2012, 2001). But the solutions proposed in this literature have tended to be problem-specific, designed to handle particular opaque interaction puzzles individually. For instance, much of the work in this tradition has treated word order learning as a problem that requires its own set of learning heuristics, with other heuristics specified for different phenomena to be acquired (Gibson & Wexler, 1994; Howitt et al., 2021; Lightfoot, 1991; Sakas & Fodor, 2012, 2001). Here, we show that learners’ solutions to the problem of word order acquisition, along with many other similar problems, can be characterized under a general computational framework for learning in the face of opacity.

We propose that in certain circumstances, learners face a choice among discrete hypotheses for the systems that generated their data, each of which is restrictive or deterministic in a way that is incompatible with the full messiness of the available data. Learners assume that their data result from an opaque interaction between (i) one of the restrictive hypotheses that they are currently considering, and (ii) various other processes that might introduce “noise” into the data. For a child learning the syntax of basic clauses, the data reflect a combination of signal for the restrictive rules governing the target language’s basic word order, as in (2), and noise introduced by non-canonical sentence types, as in (1). This same idea generalizes beyond word order acquisition to many other learning problems. For a child learning the morphosyntax of a case-marking system, the data reflect a combination of signal for restrictive rules governing which affixes mark subjects vs. objects, and noise introduced by non-canonical argument orders, argument-drop, and case-marker omission. Successful learning emerges when children are able to identify signal for a restrictive hypothesis within their opaque data, and adopt this hypothesis as the best explanation despite its surface mismatch with the data.

We introduce a general computational framework for performing this inference. A learner of the sort we describe below expects that its data are generated by a complex system:

a restrictive core component that the learner is attempting to acquire, operating alongside a “noise” component whose properties are currently unknown. Using rational probabilistic inference, the learner aims to separate evidence for a restrictive core grammar from the distorting effects of non-canonical noise processes. Importantly, it does so without knowing ahead of time how much noise is present in its data, or what the properties of that noise are. This mechanism therefore provides a way for learners to select among a space of restrictive grammatical hypotheses from messy data, without needing *a priori* knowledge of the kinds of other phenomena, grammatical or extragrammatical, that are responsible for that opacity.

In the case studies that we discuss below, we will show that the success of this learning mechanism depends on domain-specific expectations about the types of grammatical mechanisms that may have generated the learner’s input. Our proposal therefore falls under a broader class of approaches which view language learning as an attempt to use the distributions in the observed data as evidence for the parameters of this generative system, under the assumption that these parameters may interact in opaque ways (Chomsky, 1965, 1975; Fodor, 1998; Lidz & Gagliardi, 2015; Lightfoot, 1991; Valian, 1990; C. Yang, 2002). We show that general probabilistic reasoning can be combined with these approaches to resolve the tension between a restrictive hypothesis space and the need for noise-tolerant learning.

This proposal stands in contrast to certain alternative approaches to probabilistic learning, which argue that using general probabilistic mechanisms for noise-tolerant learning can substantially reduce or potentially eliminate the need for restrictive domain-specific grammatical expectations in a learner’s hypothesis space (e.g., Elman et al., 1996; Lewis & Elman, 2001; Perfors, Tenenbaum, & Regier, 2011; Perfors, Tenenbaum, & Wonnacott, 2010; Real & Griffiths, 2009; Real & Christiansen, 2005; Sagae, 2021; K. Smith & Wonnacott, 2010; Y. Yang & Piantadosi, 2022). In what follows, we compare our *restrictive hypotheses* approach to a prominent alternative proposal in this literature, which posits that a domain-general bias to “regularize” messy data can explain many instances where learners draw generalizations that are not transparently reflected in their input. On this view,

learners do not evaluate among discrete, restrictive hypotheses, but are instead equipped to consider a flexible range of hypotheses, including those that closely match the messy distributions in their input. Divergences between the generalizations that they end up drawing and the surface distributions that they observe come from a general preference for consistency. Some motivation for this hypothesis comes from children’s behavior in artificial language learning experiments. For instance, when exposed to an artificial language in which determiners occur inconsistently with nouns, children show tendencies to produce particular determiners all of the time or not at all (Austin, Schuler, Furlong, & Newport, 2022; Hudson Kam & Newport, 2005, 2009). On this account, children may be equipped to consider that the language allows determiners with any probability, but have a domain-general bias to prefer probabilities closer to zero or one. An influential literature proposes that this bias underlies children’s behavior both in learning probabilistic regularities in non-linguistic domains, and in the context of acquiring language from non-native speakers, where it could be a driver of language change (Austin et al., 2022; Culbertson & Kirby, 2016; Ferdinand, Kirby, & Smith, 2019; Hudson Kam & Newport, 2009, 2005; Perfors, 2012; Real & Griffiths, 2009; Singleton & Newport, 2004; K. Smith & Wonnacott, 2010).

There are two important assumptions implicit in this alternative approach, which we will call the *general regularization bias* account. The first is that learners’ regularization behavior across various linguistic and non-linguistic domains can be attributed at their core to the same set of non-linguistic cognitive factors operative in early development, such as constraints on information processing, cognitive control, or working memory (Austin et al., 2022; Culbertson & Kirby, 2016; Ferdinand et al., 2019; Keogh, Kirby, & Culbertson, 2024; Hudson Kam & Newport, 2005, 2009; Newport, 1990). The second assumption is that across all of these various learning domains, learners have a hypothesis space that can accommodate the full distribution of the data, in all of its messiness. Noise-tolerance only emerges through a bias operating within this fully-flexible hypothesis space, pushing learners away from the flexible middle and towards more skewed or systematic extremes. Importantly,

because this behavior is taken to be the product of a shared set of cognitive factors operating across different learning domains, no specific extreme within the learner’s hypothesis space is preferred *a priori* for domain-specific reasons. What might differ by domain or learning context is simply the degree to which this overall skewing or numerical sharpening occurs (Ferdinand et al., 2019; Culbertson & Kirby, 2016; Real & Griffiths, 2009).

In previous work, the general regularization bias idea has been given an explicit mathematical formulation (Culbertson, Smolensky, & Wilson, 2013; Perfors, 2012; Real & Griffiths, 2009; K. Smith et al., 2017), which can be incorporated into the learning of complex, layered grammatical systems (M. Johnson, Griffiths, & Goldwater, 2007). One might imagine that such an approach would allow children to infer grammatical structure from messy data: children would be equipped to consider that their data might come from any combination of grammatical rules applying with any probability, but regularization leads them to posit rules that apply with a high degree of consistency, rather than merely recapitulating the messiness in their data. For this mechanism to broadly account for the generalizations that children draw over the course of natural language acquisition, it would need to be able to overcome messiness of all sorts: not only the kind of inconsistent variability that might be encountered in the context of an artificial language learning task, but also the kind of messiness arising from opaque grammatical interactions. The case studies that we examine here show that this cannot be the full solution.

Our contribution in this paper is to provide a general computational implementation for the alternative, restrictive hypotheses proposal. We present a computational architecture for learning in the face of opacity, by explicitly modeling the interaction between a set of candidate restrictive hypotheses and a component that adds noisy distortions to the data. In what follows, we compare our learning architecture to the common implementation of the general regularization bias approach: an implementation of a problem-agnostic, domain-general, widely deployed bias to prefer extremes in general, that does not favor particular extremes for domain-specific reasons. We show that (i) both approaches can

account for children’s behavior in a representative artificial language experiment (Austin et al., 2022), but (ii) our approach fares better in modeling two case studies of opaque learning problems in natural language acquisition: acquiring word order and case-marking from data that appear messy from the perspective of young language learners. We argue that a key to our model’s success in these learning problems, and the reason that it performs better than the domain-general regularization bias approach, is that it can naturally encode substantive, domain-specific expectations about the nature of grammatical rules— for instance, the expectation that canonical clauses require subjects. This provides support for views in which successful learning from noisy data depends on a hypothesis space comprising restrictive grammatical options.

2 Drawing systematic generalizations from messy data

A large body of empirical work has examined cases where learners across various linguistic and non-linguistic settings show tendencies to “regularize” their data, drawing generalizations that are not transparently reflected in the messy distributions that they observe. In this section, we review these empirical findings and describe the prominent computational approach that models this regularization behavior as deriving from a domain-general bias towards consistency. We illustrate how this mechanism can be applied to a representative artificial language learning experiment from the regularization literature. Against this background, we then introduce the foundations of our alternative computational proposal. We show how it can be applied to model the same artificial language learning results, thereby offering an alternative mechanism for understanding learners’ regularization behavior. In the following sections, we then demonstrate that although the two approaches can both account for this simple case study, they produce different results when applied to more complex learning problems in syntax acquisition.

2.1 The phenomenon of regularization

Some of the earliest proposals for regularization in language acquisition come from studies of learning from non-native speakers, where children’s data may contain substantial messiness introduced by the incomplete grammatical knowledge of their parents (J. S. Johnson, Shenkman, Newport, & Medin, 1996; Singleton & Newport, 2004; Newport, 1999; Wolfram, 1985). An example is reported in Singleton and Newport (2004), who studied a Deaf child acquiring American Sign Language solely from parents who were late learners of the language. His parents’ morphological productions were inconsistently accurate, and yet his own were strikingly consistent, approaching the accuracy of children learning from native signers. Singleton and Newport posit that this illustrates a general trend for young learners to draw generalizations that are more systematic or categorical than their noisy data would seem to support. In the contexts of learning from non-native speakers, this phenomenon could be responsible for creolization and language change (Hudson Kam & Newport, 2005, 2009; Newport, 1999; see also Bickerton, 1981, 1984; Senghas & Coppola, 2001).

This kind of phenomenon has been further investigated through an extensive and influential body of experimental work (e.g., Austin et al., 2022; Culbertson et al., 2013; Ferdinand et al., 2019; Hudson Kam & Newport, 2005, 2009; Perfors, 2012; Real & Griffiths, 2009; K. Smith & Wonnacott, 2010). For instance, a classic series of artificial-language learning experiments exposed children and adults to input in which novel determiners occurred with novel nouns inconsistently (Austin et al., 2022; Hudson Kam & Newport, 2005, 2009). When asked to produce sentences in this language, most adults tended to mirror these messy distributions, producing determiners at roughly the rates at which they had heard them in the training data. By contrast, most young children tended to behave as if they had learned more systematic or skewed rules, producing one determiner at a much higher rate.

The contrast in tendencies between adults and children in these experiments resembles a developmental trend in an older literature on non-linguistic “probability learning.” In a representative task, Gardner (1957) exposed adults to two flashing lights: in one condition,

Light A flashed on a random 70% of trials, and Light B flashed on a random 30% of trials. When asked to predict which light would flash next, the participants closely matched these proportions: in 70% of trials they predicted Light A, and in 30% of trials they predicted Light B. This “probability-matching” behavior has been found in a wide variety of tasks with adults, older children, and some non-human animals (Behrend & Bitterman, 1961; Bullock & Bitterman, 1962; Estes, 1964, 1976; Gardner, 1957; Myers, 1976; Stevenson & Weir, 1959). Increasing the complexity of the task can sometimes lead adults to stop matching the distributions in their data as closely and start regularizing, producing more skewed responses (e.g., Gardner, 1957; M. W. Weir, 1964). But although adults can behave variably in these tasks, an important finding that emerges is that children tend to regularize at young ages (Bever, 1982; Craig & Myers, 1963; Derks & Paclisanu, 1967; Stevenson & Weir, 1959; M. W. Weir, 1964).

To explain young learners’ behavior in these tasks, the prior literature has primarily explored explanations within the class of accounts that we are calling the general regularization bias approach (Austin et al., 2022; Culbertson & Kirby, 2016; Ferdinand et al., 2019; Hudson Kam & Newport, 2005, 2009; Perfors, 2012; Real & Griffiths, 2009; K. Smith & Wonnacott, 2010). In the artificial language learning experiments described above, children may have considered the veridical distribution of determiner occurrences in their training data, but did not settle on this distribution due to a domain-general prior belief that probabilities close to zero or one are generally more likely to occur than intermediate values. Alternatively, children may have attempted but failed to veridically encode that distribution due to constraints on developing cognitive systems that are shared across learning domains, such as information processing, cognitive control, or working memory (Austin et al., 2022; Hudson Kam & Newport, 2005, 2009; Newport, 1990, 1999). It is possible that these cognitive constraints may conspire to skew learning towards more categorical outcomes. (For more on these issues, see Culbertson & Kirby, 2016; Ferdinand et al., 2019; Keogh et al., 2024; and Perfors, 2012.)

Here we explore an alternative interpretation of this phenomenon, where children’s regularization behavior derives from domain-specific expectations about the particular types of regularities that they are likely to encounter in a given learning context. In the domain of language, children may expect that the grammatical system generating their data contains particular sorts of restrictive rules (Bickerton, 1981, 1984; Chomsky, 1965, 1975; Lidz & Gagliardi, 2015; Lightfoot, 1991; Pinker, 1984; Senghas & Coppola, 2001). For instance, they may expect that determiner distributions are systematic. In non-linguistic domains, children may bring other domain-specific hypotheses about rule systems responsible for their data (e.g., Schulz & Sommerville, 2006). Regularization emerges when children can abstract away from apparent inconsistencies in their data— “noise” coming from areas of the domain that haven’t yet been acquired— in such a way as to identify signal for one of the restrictive hypotheses that they are considering.

It is important to note that the issue of how learners abstract away from various sorts of “noise” in their data is not identical to the issue of how they acquire both deterministic rules and nondeterministic rules. The grammatical systems that children must eventually acquire contain many types of variable processes. For instance, while languages like English and French have strict basic SVO word order, other languages allow more flexibility: children acquiring a language like Spanish must learn that subjects can appear both pre- and post-verbally in basic clauses (e.g., Torrego, 1989). There is a substantial literature on how children acquire phonetic, phonological, and morphosyntactic variability, conditioned by grammatical, contextual, or sociolinguistic factors (Labov, 1989; Miller, 2013; Miller & Schmitt, 2012; Roberts, 1997; Roberts & Labov, 1995; Shin & Miller, 2022; J. Smith, Durham, & Richards, 2013; J. Smith & Durham, 2019; Song, Shattuck-Hufnagel, & Demuth, 2015). Any theoretical approach to the phenomenon of regularization must account for the fact that not all variability in a learner’s input is regularized away; some of this variability is meaningful and must be acquired.

To account for this issue, some of the prior regularization literature has proposed that

children’s regularization biases become more attenuated over the course of development. On this proposal, children’s regularization tendencies may apply most strongly at initial stages of development, leading them to favor one variant— perhaps the most frequent— when multiple are available. Over time, this tendency to use one variant to the exclusion of others becomes weaker, and children incrementally learn the full range of variable processes that apply within their language and the factors that govern when those processes apply (e.g., Austin et al., 2022; see also Shin & Miller, 2022). Our proposal offers an alternative account, where learners even at early stages of development may distinguish between variability to be acquired and variability that should be regularized away. Although the hypotheses that we consider in the following case studies are restrictive, they still allow for a type of genuine variability: even in a strict word-order language like English, children must learn that some verbs do not deterministically require a direct object, but can freely alternate between transitive and intransitive frames (e.g., *She’s chewing* vs. *She’s chewing her food*). We consider cases where learners evaluate among different strict word orders, while still allowing for verb phrases to vary in whether they have a direct object. These simple hypotheses can be further extended to include cases where learners additionally consider more variable word order rules, including Spanish-like options that produce both SVO and VOS (Maitra & Perkins, 2023). Thus, while we use the term “restrictive hypotheses” to label the core grammatical processes that the learner is attempting to acquire, our proposal is compatible with the possibility that there is variability in the output of these processes. Importantly, our proposal allows for a learner to encode a distinction between variability that lives within one of the restrictive hypotheses under consideration, and variability in the form of distortions to the patterns expected from any of those hypotheses, the kind of variability that we are referring to as “noise.”

In what follows, we will describe the existing computational approach for implementing a general regularization bias (e.g., Culbertson et al., 2013; Real & Griffiths, 2009; Perfors, 2012; K. Smith et al., 2017), and illustrate its application to a representative artificial

language learning experiment in Austin et al. (2022). Using the same case study, we will then introduce the alternative computational approach that we propose in this paper.

2.2 Learning with a numerical regularization bias

We will consider children’s behavior in the “inconsistent language” condition from Austin et al.’s (2022) Experiment 1. Participants were trained on an artificial language with VSO word order and novel vocabulary items. The noun-phrases (NPs) in subject and object position each comprised a noun followed by a determiner. There were two determiners in the language, ‘ka’ and ‘bo’;¹ one was designated as the primary determiner, and appeared in 67% of determiner positions, and the other determiner appeared in the other 33% of determiner positions. Apart from this proportional requirement, the choice between ‘ka’ and ‘bo’ in any given determiner position was unpredictable. Sentences in this artificial language were used to label scenes with puppets interacting in short video clips. In the training sessions, participants were taught the names of these puppets and were asked to repeat the sentences that they heard. The scene to be described dictated whether the sentence was transitive or intransitive, and dictated the choices of nouns and verbs, but left open the choice of determiner. A single session of “sentence exposure” consisted of 24 sentences, each labeling a scene, of which 18 were transitive (two NPs each) and 6 were intransitive (one NP each). Each one of these sessions therefore exposed participants to 42 determiner-noun pairs in total. Of these, 28 had the primary determiner; for concreteness, we’ll take ‘ka’ to be the primary determiner, so participants saw 28 occurrences of ‘ka’ and 14 of ‘bo’.

After three training days, participants were then asked to produce their own sentences to describe scenes with new combinations of the puppets and actions. A single session of “production test” consisted of 12 scenes to be labeled, of which 8 were transitive (two NPs each) and 4 were intransitive (one NP each). Three of these production test sessions were conducted over the course of the experiment (on Days 3–5). Given sufficient training, a

¹ These determiners were actually ‘ka’ and ‘po’, but we’re calling the second one ‘bo’ to avoid notational confusions with $p(\cdot)$.

participant could learn the word order of this language and the labels for the puppets and actions, and therefore learn to appropriately label scenes like the ones that they had been trained on. The interesting question is how they chose between ‘ka’ and ‘bo’ in determiner positions in these NPs. In line with the experimental results surveyed above, adult participants matched the proportions of ‘ka’ and ‘bo’ in their training data: they produced ‘ka’ in 67% of determiner positions, and ‘bo’ in 33% of determiner positions. The key finding for our purposes is that five- and six-year-olds strengthened the observed dominance of ‘ka.’ In the aggregate, they produced ‘ka’ about 86% of the time and ‘bo’ 14% of the time. Moreover, 6 of the 15 children at this age showed categorical behavior, using ‘ka’ all of the time and ‘bo’ not at all.

Prior approaches have formally modelled scenarios like this one as the task of estimating an unknown parameter that governs the probability of some variant appearing in the learner’s input: for instance, ‘ka’ vs. ‘bo’ (Real & Griffiths, 2009; Perfors, 2012). Suppose that learners assume that a given determiner position will have ‘ka’ with some single unknown probability θ , and ‘bo’ with probability $1 - \theta$. For any particular probability θ , the *likelihood* of observing k instances of ‘ka’ and b instances of ‘bo’ is the binomial probability $\binom{k+b}{k} \theta^k (1 - \theta)^b$. This is the same as the probability of tossing a coin with a weight θ of coming up heads, and observing k total heads out of $k + b$ total tosses. Given a particular estimate of θ , the likelihood of the learner’s training data before the first production test on Day 3 (84 instances of ‘ka’ and 42 instances of ‘bo’)² is

$$(3) \quad P\left(\begin{smallmatrix} 84 \text{ ‘ka’} \\ 42 \text{ ‘bo’} \end{smallmatrix} \mid \theta\right) = \binom{126}{84} \theta^{84} (1 - \theta)^{42}$$

The value of θ that maximizes this likelihood is 0.67, which is the proportion of ‘ka’ in the training data. In order to account for children’s higher rate of ‘ka’ production, we need a learning model that does something other than simply maximizing likelihood.

² We model learning prior to the first production test, as Austin et al. (2022) find no differences in children’s behavior from the first through the third day of production testing. However, this approach could be generalized to model learning across the remaining days of the experiment.

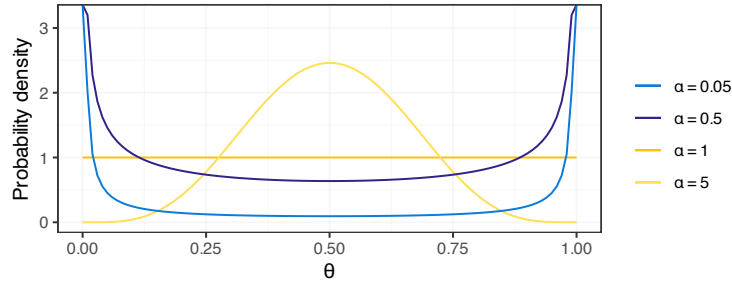


Figure 1. Illustration of symmetric Beta prior distributions, with different shape parameters

Rational Bayesian reasoning can be applied to this problem by combining the likelihood of the data with a learner’s *prior* beliefs about θ , before having seen any data. Bayes’ Rule tells us that the *posterior* probability of a particular hypothesis about θ given the observed data depends on the likelihood of the data under θ and the learner’s prior degree of belief in θ , expressed in the prior probability distribution $p(\theta)$. For data consisting of k observations of ‘ka’ and b observations of ‘bo’, the posterior probability distribution over θ is

$$(4) \quad p(\theta \mid k, b) = \frac{P(k, b \mid \theta) p(\theta)}{\int_0^1 P(k, b \mid \theta) p(\theta) d\theta} = \frac{\binom{k+b}{k} \theta^k (1-\theta)^b p(\theta)}{\int_0^1 \binom{k+b}{k} \theta^k (1-\theta)^b p(\theta) d\theta}$$

To model children’s behavior as deriving from a domain-general regularization bias, approaches like those in Reali and Griffiths (2009) and Perfors (2012) posit that young learners’ prior beliefs about θ are skewed. In particular, we might suppose that a learner’s prior distribution $p(\theta)$ takes the form of a symmetric Beta(α, α) distribution, whose α shape parameter governs the type and degree of skew (Figure 1). When $\alpha = 1$, the distribution is uniform, so a learner with this prior will have no preference *a priori* for any particular value of θ . When $\alpha > 1$, the distribution is unimodal, so a learner with such a prior will have some degree of preference for θ values close to 0.5. When $0 < \alpha < 1$, the distribution is U-shaped; a learner with this form of prior is biased towards regularization, preferring θ values close to zero or one. The strength of this bias is controlled by the numerical value of α . As α becomes smaller, the learner’s prior preference for these extremes becomes stronger. But importantly, the prior is symmetric, so the learner has no prior belief about which of these

two extremes is more likely.

A child with a U-shaped Beta prior will combine a preference for the endpoints of the distribution with the preference to fit the 67% observed proportion of ‘ka’ in the Austin et al. training data. For example, after observing 6 instances of ‘ka’ and 3 instances of ‘bo,’ such a learner will infer that θ is greater than 0.67: if $\alpha = 0.05$, the value of θ with that maximizes the posterior probability density under (4) is 0.71.³ However, in this model, the learner’s prior bias will play less and less of a role as the amount of data increases, eventually resulting in posterior estimates that are very close to the maximum-likelihood estimate. Given the large amount of training data that children observe in Austin et al.’s (2022) experiment, accounting for their regularization behavior with this approach requires a further assumption, which is that young learners’ general cognitive limitations significantly limit the amount of data that they learn from (Keogh et al., 2024; Perfors, 2012; see also Newport, 1990, 1999). Only in combination with a mechanism that limits the size of the learner’s data will this form of prior numerical regularization bias exert its influence.

For the purposes of illustration, we use Perfors’ 2012 simple model of memory limitations, whereby a certain proportion of a learner’s data is forgotten. For a learner with a “forgetting rate” of r , each data point that the learner observes in the input is dropped with probability r . (For more sophisticated ways to model memory limitations, see the further simulations in Perfors 2012, which find qualitatively similar results with several more complex models.) Note that this forgetting process does not asymmetrically target any particular variant in the learner’s data; it merely reduces the size of the learner’s intake (the remembered data) while leaving the proportion of ‘ka’ vs. ‘bo’ approximately unchanged. In Figure 2, we plot the maximum *a posteriori* estimates of θ in Austin et al.’s (2022)

³ More concretely, if a learner’s data consist of k observations of ‘ka’ and b observations of ‘bo,’ with both $k, b > 0$, and the learner’s prior takes the form of a symmetric Beta with parameter α , the value of θ that maximizes the posterior probability density under (4) is $\hat{\theta}_{\text{MAP}} = \frac{k+\alpha-1}{k+b+2\alpha-2}$. Because the posterior incorporates the learner’s prior beliefs about θ , the value of θ that maximizes this posterior does not correspond to the proportion of ‘ka’ in the learner’s data, but rather corresponds to what would be the proportion of ‘ka’ in a collection of $(k + \alpha - 1)$ ‘ka’ observations and $(b + \alpha - 1)$ ‘bo’ observations.

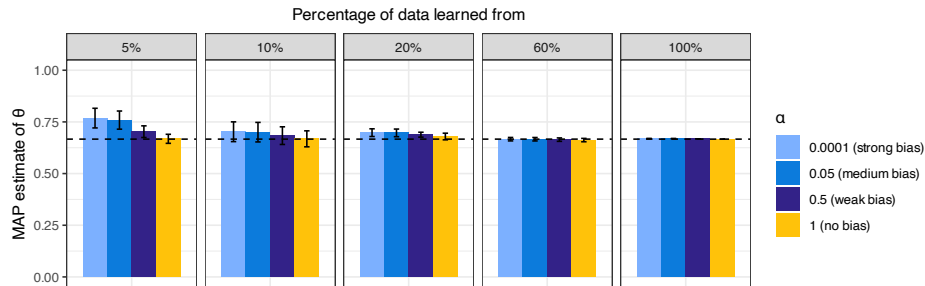


Figure 2. Maximum *a posteriori* estimates of θ across varying regularization biases, and varying amounts of data randomly sampled from 84 observations of ‘ka’ and 42 observations of ‘bo’. The dashed line indicates the proportion of ‘ka’ in the training data.

experiment for models with varying degrees of prior regularization bias, and varying rates at which data is forgotten. Each bar represents an average across 10 randomly-generated intake datasets at a particular forgetting rate, from actual input data consisting of 84 ‘ka’ and 42 ‘bo’. When the learner’s regularization bias is weak or a large amount of data is remembered, the posterior estimates of θ are around 0.67, matching the actual proportion of ‘ka’ in the data (represented by the dashed line). But with an increasing bias towards regularization and less data remembered, the learner’s posterior estimates of θ are more extreme.

This illustration shows us that a learner with a skewed Beta prior, in combination with cognitive limitations that reduce the amount of data to learn from, would infer that the true posterior probability of ‘ka’ in the language is greater than the rate of ‘ka’ in the training data. There are several ways that one might apply this result to model the patterns of children’s responses in Austin et al. (2022). One might suppose that each child identifies the maximum *a posteriori* estimate of θ given her remembered training data, and uses that estimate to generate her own productions of ‘ka’ and ‘bo’ in the testing sessions. Or, a child might sample from the entire inferred posterior distribution over θ , choosing a value of θ on each trial in proportion to its posterior probability under (4). (For the details of these two approaches, see Real and Griffiths 2009.) In either case, a child whose inferred posterior distribution is centered on a value sufficiently higher than 0.67 will show some degree of regularization at test, producing ‘ka’ at a higher rate than attested in the training data.

In summary, a prominent approach to modelling children’s regularization behavior applies Bayesian inference with a symmetrical Beta prior distribution, whose degree of skew is given by the numerical α parameter. When the amount of data that children learn from is sufficiently small, possibly as a result of limited memory or other cognitive resources, this skewed prior enforces a preference for extreme points in a gradient space. This type of prior has been applied to many cases of regularization in experimental contexts (Culbertson et al., 2013; Perfors, 2012; Real & Griffiths, 2009; K. Smith et al., 2017). There are two primary assumptions inherent in this approach: (i) learners operate with a flexible hypothesis space that can accommodate any degree of variability, but (ii) they are biased *a priori* to expect that particular outcomes will occur in given contexts at rates close to zero or one. Varying the parameter of this prior can alter the strength of a learner’s regularization bias, which may be needed to account for different degrees of regularization in different individuals or domains (e.g., Culbertson & Kirby, 2016; Ferdinand et al., 2019). But importantly for this account, the shape of the prior need not vary across domains: each of these cases of numerical regularization assumes a symmetrical skew in the learner’s prior distribution, with no preference for one extreme over another. A learner comes to regularize simply by balancing this symmetrical prior with a desire to fit observed skews in the data.

2.3 Learning as selection among noisy hypotheses

We will now introduce the distinct approach to explaining learning from messy data that is the main focus of this paper, which we offer as an alternative to the numerical regularization approach described above. We will start with a simple illustration involving coin flips that will introduce the important intuitions, and then show how the Austin et al. (2022) experiment can be seen as an instance of the very same coin-flipping scenario. In Section 3, we will scale up the idea in order to apply it to the learning of grammars: finite systems that generate unbounded collections of sentences. The grammar-based system that we introduce there serves as the basis for the detailed case studies in Sections 4 and 5.

2.3.1 Illustration with coin flips. Suppose we have two bags of coins: one, which we'll call Bag H, contains coins where both sides are heads, and the other, Bag T, contains coins where both sides are tails. We will consider a scenario where we are trying to decide which bag's coins were responsible for generating some observed sequence of coin flips. The catch is that each bag also contains some unknown proportion of “noise coins”, which we'll call Ψ coins, which all have some single unknown probability ψ of coming up heads (and $1 - \psi$ of tails). To decide which bag provides a better explanation of the observed collection of heads and tails, we will calculate the likelihood of the data under each bag, which will involve guesses about how many of the observed coin flips were flips of a Ψ coin — in other words, how many of the observed coin flips were “signal” bearing on the decision between Bag H and Bag T, and how many were noise. The difference in how this partitioning into signal and noise plays out across the two bags will be the basis of two bags' differing likelihoods, which will in turn lead the learner to choose one bag over the other. In the linguistic case studies that follow, the analogues of Bag H and Bag T will be the restrictive core grammatical hypotheses that a learner is entertaining. Similar to how the “core” coins in these bags always produce heads or tails, the learner might consider the hypothesis that the “core” determiner in an artificial language is ‘ka’ vs. ‘bo,’ or that the core basic word order of a natural language is SVO or SOV. The analogues of the unpredictable “noise” coins will be other grammatical processes that may introduce distortions into the learner's data, such as non-canonical constructions that produce divergences from a core word order. Choosing between Bag H and Bag T will be analogous to choosing which core hypothesis provides the best explanation for the observed data.

We'll start by focusing attention on Bag H, where the core coins — we'll call these Φ coins — always produce heads. Suppose that ten times, a coin is drawn from this bag (with replacement) and flipped, producing eight heads and two tails. How many of these flips should we guess came from the Φ core coins, and how many from the Ψ noise coins? There is a wide range of options, including the possibility that all ten flips came from the Ψ coins.

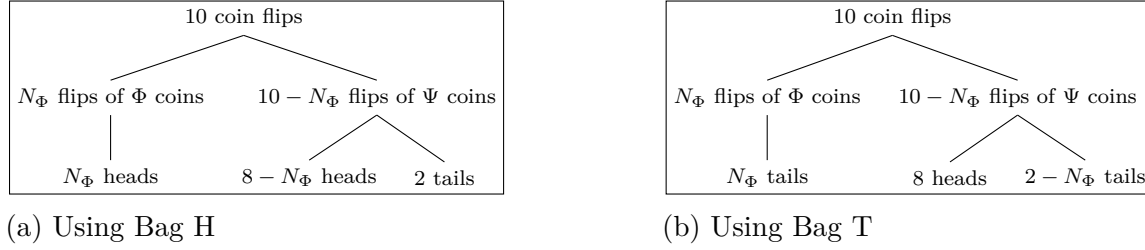


Figure 3. Partitioning 8 heads and 2 tails into signal and noise

But given the observed skew towards heads, there is a clear intuition that the double-headed Φ coins were probably responsible for a significant portion of the observations. Why is this?

Figure 3a illustrates the possible ways of breaking down the ten observed coin flips, where N^Φ is the number of flips of Φ coins. Under the hypothesis that all ten flips came from Ψ coins ($N^\Phi = 0$), eight of those flips would need to come up heads and two to come up tails in order to generate the observed data. Contrast this with the more intuitively plausible hypothesis that there were six Φ -flips and four Ψ -flips ($N^\Phi = 6$). Under this hypothesis, the six Φ -flips need to come up heads, which is guaranteed to happen; so, generating the observed data just amounts to having the four Ψ -flips produce two heads and two tails. This is less “costly” than the first hypothesis’s requirement that ten Ψ -flips produce eight heads and two tails. By positing six Φ -flips, six of the heads that we need to generate “come for free”; with only Ψ -flips, however, we get no such head start.

More precisely, given a particular probability ψ that the Ψ -coins produce heads, then the likelihood of the data under the hypothesis that relies on only four Ψ -flips ($N^\Phi = 6$) is $\binom{4}{2}\psi^2(1 - \psi)^2$. Under the hypothesis that leaves all the work to ten Ψ -flips ($N^\Phi = 0$), this likelihood is $\binom{10}{8}\psi^8(1 - \psi)^2$. Because we don’t know the value of ψ , we calculate the likelihood under each hypothesis by marginalizing over all possible values of that parameter. These details are given in the Appendix, but the key result can be understood intuitively in the following way. If we assume that nothing is known about how frequently a Ψ -flip will come up heads, then this means that we have no prior belief that any particular value of ψ is more likely than any other value; each value of ψ has equal prior probability. With this

assumption, our likelihood calculations have particularly simple solutions. If there are four Ψ -flips, then each of the possible numbers of heads that these flips might produce (ranging from 0 to 4) is equally likely, and has probability $\frac{1}{5}$. So, under the $N^\Phi = 6$ hypothesis, the likelihood of observing eight heads out of ten flips, just like any other number out of ten flips, is $\frac{1}{5}$. On the hypothesis that there are ten Ψ -flips, then the likelihood of the data is smaller: there are eleven possible numbers of heads that could have been observed, so the likelihood of seeing this particular number of heads under the $N^\Phi = 0$ hypothesis is only $\frac{1}{11}$.

This is the point that will be central to our approach: when we make no commitments about the probability that Ψ -coins will produce heads, then *all* that matters about a particular hypothesis is how many Ψ -flips it must appeal to. A learner of the sort that we model here considers how to partition the data into portions coming from core vs. noise processes, without any prior assumptions about whether noise processes are more likely to produce certain types of data over others— here, whether noise is more likely to produce heads vs. tails; in what follows, whether it is more likely to produce ‘ka’ vs. ‘bo,’ or subject-initial vs. subject-final clauses. In each case, the hypotheses that will be favored are those that invoke the noise processes as few times as possible. In this case, we’ve seen that four noise flips is better than ten, but two is even better: the very best hypothesis is that there were eight Φ -flips and two Ψ -flips, for a likelihood of $\frac{1}{3}$.⁴ More generally, if we hypothesize n^Φ flips of Φ coins, then we must invoke $10 - n^\Phi$ flips of Ψ coins, and so the likelihood is $\frac{1}{(10 - n^\Phi) + 1} = \frac{1}{11 - n^\Phi}$. But $N^\Phi = 8$ is as far as we can go. With more than eight Φ -flips, the likelihood of the data under Bag H would be zero, because there are only eight heads in the data. The observed number of heads puts a cap on the degree to which Φ -flips can be used to achieve high likelihoods.

Now let’s consider how the same observed data might have been generated by Bag T,

⁴ If this is surprising, it may be because of an intuition that we expect the noise flips to yield a roughly equal number of heads and tails, contrary to our assumption of a uniform prior on ψ . But if we have no reason to suspect that the probability of a head is around $\frac{1}{2}$, then we have no reason to suspect that the number of heads from ten flips will tend to be around the middle of the $\{0, 1, 2, \dots, 11\}$ range.

in which Φ coins always produce tails rather than heads. As with Bag H, the hypothesis that there were zero Φ -flips is the most costly, and alternatives that make use of larger numbers of Φ -flips provide higher likelihoods. But since Φ -flips in Bag T produce certain *tails*, the best we can do is to suppose that both of the observed tails came from the two-tailed coins, and rely on eight Ψ -flips to do the rest of the work (likelihood of $\frac{1}{9}$). So, there is no way for the two-tailed Φ coins in Bag T to contribute to particularly good explanations of the observed high proportion of heads in the data. Figure 4 shows the full range of ways that Bag H and Bag T can explain the observed data. The explanations made available by Bag H range from very good ones that invoke more flips of Φ -coins, to more costly ones that invoke fewer flips. Since Bag T makes available a subset at the costly end of that range, it is intuitive that Bag H is a better explanation overall — even though the presence of head-tail Ψ coins ensures that any observed combination of heads and tails is compatible with both bags.

To make this even more precise, we can ask about the likelihood of the data under each bag by considering all possible hypotheses about N^Φ . This means marginalizing (summing) over the full range of possibilities, from zero Φ -flips to ten Φ -flips, but Figure 4 shows that only some subsets of this range will contribute non-zero values to the sum.

$$\begin{aligned}
 (5) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag H}\right) &= \sum_{n^\Phi=0}^{10} \left[P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N^\Phi = n^\Phi, \text{Bag H}\right) \times P(N^\Phi = n^\Phi \mid \text{Bag H}) \right] \\
 &= \sum_{n^\Phi=0}^8 \left[\frac{1}{11 - n^\Phi} \times \frac{1}{11} \right] = 0.138
 \end{aligned}$$

$$\begin{aligned}
 (6) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag T}\right) &= \sum_{n^\Phi=0}^{10} \left[\frac{1}{11 - n^\Phi} \times P(N^\Phi = n^\Phi \mid \text{Bag T}) \right] \\
 &= \sum_{n^\Phi=0}^2 \left[\frac{1}{11 - n^\Phi} \times \frac{1}{11} \right] = 0.027
 \end{aligned}$$

The product of $\frac{1}{11}$ and $\frac{1}{11-n^\Phi}$ inside these sums corresponds to combining (i) a choice of how to split 10 coin flips into Φ -flips and Ψ -flips, shown at the root of the tree diagrams in Figure 3, and (ii) a choice of how to split $10 - n^\Phi$ flips of Ψ coins into heads and tails, shown on the right-hand side of the tree diagrams. We've seen why each possible outcome of this latter choice has probability $\frac{1}{11-n^\Phi}$, given some n^Φ . The same logic applies when considering

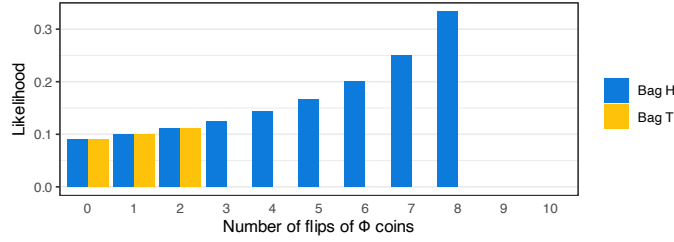


Figure 4. Likelihood of 8 heads and 2 tails

the probability that any particular number n^Φ of the 10 coin flips were Φ -flips. We have assumed that we know nothing about the ratio of Φ coins to Ψ coins in these bags. This means that the probability of each value of n^Φ , $P(N^\Phi = n^\Phi \mid \dots)$, is the same across the eleven possible options from 0 to 10, and equals $\frac{1}{11}$.

Considering Figure 4 again, what we can now see is that each bag’s likelihood, in (5) and (6), is proportional to the sum of its corresponding bars. This likelihood is about five times larger under Bag H than under Bag T. Bayes’ Rule tells us that the posterior probability of a particular bag is proportional to the likelihood of the data under that bag times the prior probability of the bag; this is analogous to the reasoning that we introduced in (4) earlier, but here we have only two discrete hypotheses rather than a continuous range of hypothesized values for the parameter θ . Assuming that we have no reason to prefer Bag H or Bag T *a priori*, then the higher likelihood of Bag H will lead to a correspondingly higher posterior probability. We’ll conclude that Bag H is about five times more likely to have generated the data than Bag T.

In summary, in the learners that we describe below, the choice between competing restrictive hypotheses is essentially a choice between two-headed coins and two-tailed coins, where either choice is embedded in a system (“bag”) where head-tail coins also produce some noise: divergences from what would be generated by the core mechanisms (the Φ coins) alone. When comparing such composite systems, our learner will prefer the one whose core mechanisms predict the skew in the data. This will provide the least costly solution, even though the shared noise possibilities (the Ψ coins) ensure that all the competing systems can

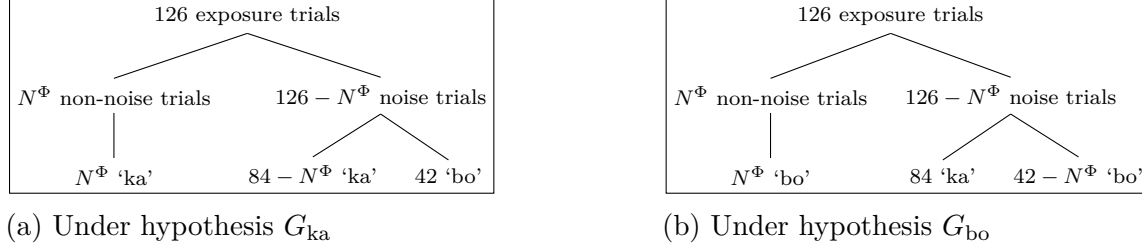


Figure 5. Partitioning 84 ‘ka’ and 42 ‘bo’ into signal and noise

account for the data as a whole. And the proposed learner will do this without knowing *a priori* how much of the data is noise (how much of the data came from Ψ coins) or what the contribution of noise looks like (the probability ψ of noise contributing a head).

2.3.2 Applying this to the Austin et al results. Suppose the participant is construing the task in Austin et al.’s (2022) experiment as one of choosing between two highly restrictive grammars, one of which uses ‘ka’ in all determiner positions, and the other of which similarly uses only ‘bo’. Obviously neither of these grammars is directly compatible with the observed data in the exposure sessions. But we will consider each to be embedded in a noisy system, in the same way that the always-heads coins were embedded in Bag H and the always-tails coins were embedded in Bag T. One system, which we’ll call G_{ka} , contains the always-‘ka’ grammar along with some noise-generating machinery that flips a coin to decide between ‘ka’ and ‘bo’, while the other, G_{bo} , contains the always-‘bo’ grammar along with that same noise-generating machinery. The observed data with 84 occurrences of ‘ka’ and 42 of ‘bo’ is therefore compatible with both G_{ka} and G_{bo} , but G_{ka} will provide a better explanation of the data, via exactly the same logic that we saw with the coins.

The ways in which the 126 total determiner observations could break down under each of these two hypotheses are illustrated in Figure 5, where again N^Φ is the number of observations for which the core mechanisms were responsible. Then following (5) and (6), the likelihood of the data under one of these hypotheses is a sum over possible values of this variable, where each contribution to the sum represents a particular choice among 127 ways to split the total exposure trials and among $(127 - n^\Phi)$ ways to split the noise trials. The

two hypotheses differ only in the range of contributing values in the summation.

$$(7) \quad P\left(\begin{smallmatrix} 84 & \text{'ka'} \\ 42 & \text{'bo'} \end{smallmatrix} \mid G_{\text{ka}}\right) = \sum_{n^{\Phi}=0}^{84} \left[\frac{1}{127 - n^{\Phi}} \times \frac{1}{127} \right] = 8.65 \times 10^{-3}$$

$$(8) \quad P\left(\begin{smallmatrix} 84 & \text{'ka'} \\ 42 & \text{'bo'} \end{smallmatrix} \mid G_{\text{bo}}\right) = \sum_{n^{\Phi}=0}^{42} \left[\frac{1}{127 - n^{\Phi}} \times \frac{1}{127} \right] = 3.24 \times 10^{-3}$$

Since the data imposes a cap of 42 on the number of observations that can be attributed to the core mechanisms of G_{bo} , this hypothesis's likelihood is significantly lower than that G_{ka} hypothesis, where up to 84 of the observations can be attributed to the core mechanisms.

Using the likelihoods in (7) and (8), and assuming equal prior probabilities $P(G_{\text{ka}}) = P(G_{\text{bo}}) = 0.5$, then Bayes' Rule tells us that the posterior probabilities of each hypothesis are $P(G_{\text{ka}} \mid \begin{smallmatrix} 84 & \text{'ka'} \\ 42 & \text{'bo'} \end{smallmatrix}) = 0.728$ and $P(G_{\text{bo}} \mid \begin{smallmatrix} 84 & \text{'ka'} \\ 42 & \text{'bo'} \end{smallmatrix}) = 0.272$. This is shown in the rightmost panel of Figure 6. To set up a direct comparison with the simulation in Section 2.2, we also plot the inferred posterior distributions for the previously-created intake datasets of different sizes, from actual input consisting of 84 'ka' and 42 'bo'. For the majority of these datasets, we see that the learner arrives at posterior distribution where G_{ka} is more than 2 times as likely as G_{bo} , on the basis of data where 'ka' was only 2 times as frequent as 'bo'. With 100% of the data, the learner infers that G_{ka} is 2.7 times as likely.

Thus, like we found in the numerical regularization approach, the learner arrives at a grammatical hypothesis that strengthens the dominance of 'ka' over 'bo' observed in the training data. However, here we see a difference in how learning interacts with the size of data. In the numerical regularization approach, a sharper skew in favor of 'ka' is a property of learning with very small amounts of data. We saw this most prominently in the leftmost bars in Figure 2, where we modeled a context where a learner remembers very few of the training trials. This skew diminishes when more data are available to learn from, as the likelihood overcomes the learner's prior regularization bias. In the current approach, a sharper skew in favor of G_{ka} only emerges as more data are observed, as the learner gains confidence in how to split the data into noise vs. non-noise. In the case studies that follow, we will test the learner's inference across datasets of varying sizes, but will remain agnostic

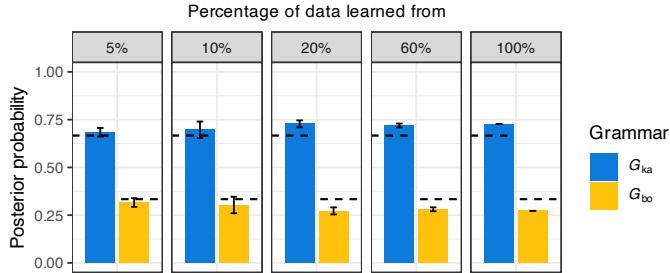


Figure 6. Posterior distribution over G_{ka} and G_{bo} across varying amounts of data randomly sampled from 84 observations of ‘ka’ and 42 observations of ‘bo’. The dashed lines indicate the proportions of ‘ka’ and ‘bo’ in the training data.

about the precise amount of data being learned from; our method will be to examine the abstract skews in the posterior distributions that the learner infers.

There are a number of ways that we might take this approach to be the core of an explanation of the observed phenomenon of regularization in an artificial language experiment. We might imagine that each child arrives at a posterior distribution over grammars on the basis of the training data, and chooses to adopt either G_{ka} or G_{bo} in accord with this distribution; if 100% of the data is learned from, then 72.8% of children would produce only ‘ka’ at testing, and 27.2% would produce only ‘bo’. Alternatively, to more closely model the reported results of the Austin et al. (2022) experiment, one can imagine that children choose (sample) either G_{ka} or G_{bo} on each production trial, with these same probabilities; this will again lead to production of ‘ka’ at a higher rate than observed in the training data. What we would like to emphasize is the following broader point: this approach, where the learner chooses among restrictive core mechanisms embedded in a system that also produces some noise, provides a candidate explanation for the phenomenon of regularization, and an alternative to the better-known numerical approach outlined in Section 2.2.

Perkins, Feldman, and Lidz (2022) applied this approach to model a naturalistic phenomenon in language acquisition that resembles regularization: how learners identify the core transitivity properties of verbs in their language— which verbs require objects— despite “noise” from non-canonical clause types. This type of noise might arise when a young child

encounters an obligatorily-transitive verb in a sentence with a displaced object (e.g., *What did you bring?*) but is unable to parse it as such. By hypothesizing that unknown noise processes cause the data to be a distorted reflection of verbs’ core argument-taking properties, their model was able to successfully identify that certain verbs deterministically require or deterministically disallow objects—for roughly the same reason that Bag H above provides a good explanation for data that do not consist entirely of heads, and that G_{ka} provides a good explanation for data that do not consist entirely of ‘ka’.

In the naturalistic case studies that we consider below, we will be asking how learners identify a grammar of basic syntax that generates subjects and objects according to some canonical order (SVO, SOV, etc.), and/or with some canonical case-marking, yielding surface strings of verbs and noun phrases. Just like in Perkins et al. (2022), unknown grammatical processes, such as argument movement, operate alongside this basic syntax, with the result that the observed strings of verbs and noun phrases are a distorted reflection of the core grammatical rules that generated them. Learning will therefore require abstracting away from the noise in the data in order to identify the core rules of the target restrictive grammar.

3 Noisy Grammar Learners

In this section, we show how the idea of restrictive hypotheses operating alongside noise mechanisms can be incorporated into the learning of a system of grammatical rules. To illustrate, we’ll first recast our account of the Austin et al. (2022) results in terms of a choice between two extremely simple grammars. Then we will show how to generalize the idea to more complex grammars made up of collections of interacting rules. We will see that the task of the learner is to consider three questions: (1) What do the data from core grammatical rules look like? (2) What do the distortions introduced by noise look like? (3) How is responsibility for the observed data distributed among the core rules and the distortions? The learner considers these three questions simultaneously in seeking the best explanation of the observed data.

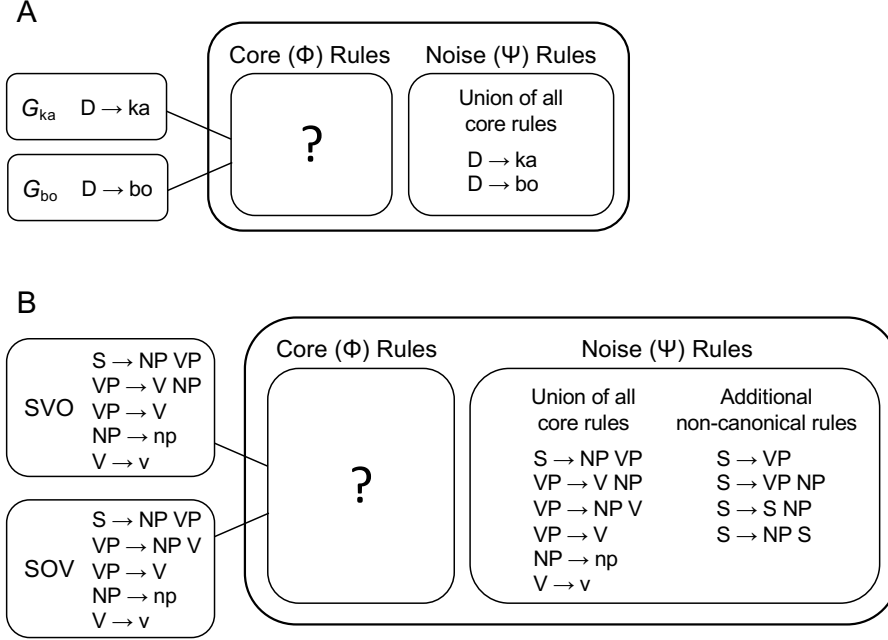


Figure 7. The hypothesis spaces of two example Noisy Grammar learners

Probability	Rule	Probability	Rule
$1 - \epsilon_D$	$D \rightarrow D_\Phi$	$1 - \epsilon_D$	$D \rightarrow D_\Phi$
ϵ_D	$D \rightarrow D_\Psi$	ϵ_D	$D \rightarrow D_\Psi$
$\phi_{D \rightarrow ka}$	$D_\Phi \rightarrow ka$	$\phi_{D \rightarrow bo}$	$D_\Phi \rightarrow bo$
$\psi_{D \rightarrow ka}$	$D_\Psi \rightarrow ka$	$\psi_{D \rightarrow ka}$	$D_\Psi \rightarrow ka$
$\psi_{D \rightarrow bo}$	$D_\Psi \rightarrow bo$	$\psi_{D \rightarrow bo}$	$D_\Psi \rightarrow bo$

(a) For G_{ka} (b) For G_{bo}

Figure 8. Compiled-out PCFGs for the two grammars in Fig. 7a

3.1 From surface forms to grammar rules

To begin, we can represent the relationship between the G_{ka} and G_{bo} hypotheses from the previous section as in Figure 7a, where the two possible realizations of a determiner, ‘ka’ and ‘bo’, are expressed as two possible rules for rewriting the nonterminal symbol D ; more precisely, these are rules of that sort that appear in a Context-Free Grammar (CFG).⁵ Both of these rules are available as “ Ψ rules” under either hypothesis; where the two hypotheses

⁵ A Context-Free Grammar specifies a set of rules for rewriting nonterminal symbols (in uppercase) as sequences of other nonterminal or terminal symbols (in lowercase). In the examples we consider here, the terminal symbols are words, and the nonterminal symbols are grammatical and phrasal categories.

differ is which of the two rules they choose to include in their Φ component.

To implement the idea that each realization of D is mediated by a choice between whether it should be realized noisily or not, we can think of G_{ka} and G_{bo} as each taking the form of a Probabilistic Context-Free Grammar (PCFG; Booth & Thompson, 1973; Wetherell, 1980) that “compiles out” the signal/noise distinction into two additional intermediate nonterminals, D_Φ and D_Ψ . These two PCFGs are shown in Figure 8.

Each rule in a PCFG is associated with a particular probability. The rule probabilities define a local distribution over possible rewrites for each nonterminal, so in each of these two grammars the probabilities of the two rules rewriting D_Ψ must sum to one (i.e., $\psi_{D \rightarrow ka} + \psi_{D \rightarrow bo} = 1$). Similarly, in these very simple examples that each only include one possible rewrite of D_Φ , the parameter $\phi_{D \rightarrow ka}$ in G_{ka} and the parameter $\phi_{D \rightarrow bo}$ in G_{bo} will necessarily be 1. But the generic notation in Figure 8 for these ϕ and ψ parameters will be useful when we extend to more involved examples. The parameter ϵ_D controls the choice between a noisy or non-noisy realization of D , and the ψ parameters specify the probabilities of ‘ka’ and ‘bo’ if a noisy realization is chosen. In terms of our earlier example, we can think of G_{ka} and G_{bo} as two bags of coins, where for each bag there is a parameter ϵ_D that expresses the ratio of Φ coins to Ψ coins, and there are ψ parameters that express the distribution over outcomes of the Ψ coins.

Each of the two compiled-out PCFGs in Figure 8 allows exactly three derivations. Those allowed by G_{ka} are shown in (9), along with their probabilities. Note that these rules implement precisely the branching structure in Figure 5a. An observed ‘ka’ might have been generated either via this grammar’s Φ component or its Ψ component, but an observed ‘bo’ necessarily came from the Ψ component.

(9)	Derivation	Probability
	$D \rightarrow D_\Phi \rightarrow ka$	$(1 - \epsilon_D) \times \phi_{D \rightarrow ka}$
	$D \rightarrow D_\Psi \rightarrow ka$	$\epsilon_D \times \psi_{D \rightarrow ka}$
	$D \rightarrow D_\Psi \rightarrow bo$	$\epsilon_D \times \psi_{D \rightarrow bo}$

The total probability assigned to the realization of D as ‘ka’ is the sum of the two corresponding derivations’ probabilities.

The example we worked through in the previous section can be thought of as asking which of the two PCFGs in Figure 8 provides a better explanation for a corpus of 126 observed “sentences”, each of which was just one word long; 84 were ‘ka’, and 42 were ‘bo’. The system we propose below extends this idea to allow for noise not only in choices of surface forms (e.g. $D \rightarrow \text{‘ka’}$ or $D \rightarrow \text{‘bo’}$), but also higher level choice points such as the order in which a sentence arranges its subject noun phrase and its verb phrase (e.g. $S \rightarrow NP\ VP$ or $S \rightarrow VP\ NP$). Unlike the choice to realize D as ‘ka’, a choice to realize S as $NP\ VP$ does not introduce observable surface forms, but rather other nonterminals, which themselves will be realized either noisily or non-noisily via other rewrite rules. The observed data will be unboundedly long strings, derived via sequential expanding rewrites of nonterminals, with the possibility of noise at each rewriting step. Our learner will ask which of a range of hypothesized grammars like the two in Figure 8 provides the best explanation for a corpus of such strings.

3.2 Noisy CFG Learners

Here we instantiate the notion of a Noisy Grammar Learner for the particular case where the grammatical rules involved take the form of CFG rules. The mathematical properties on which our learning model relies are compatible with many different choices of grammatical formalism.⁶ We use the CFG formalism here because this is helpful for the purposes of illustration, and is also sufficient for formalizing the learning problems that we address in both of the case studies in Sections 4 and 5.

Specifying the hypothesis space of a Noisy CFG learner involves specifying (i) a

⁶ For example, finite-state grammars (which are most likely appropriate for phonological and morphological acquisition) correspond to a sub-case of CFGs that only allows right-branching. Many different kinds of “mildly context-sensitive” grammars (Joshi, 1985; Stabler, 2004) can be characterized in ways that share the same abstract branching structure of CFG derivations, while extending their linguistic expressivity in a way that might be needed to characterize the learning of more complex syntactic structure (e.g. D. Weir, 1988; Seki, Matsumara, Fujii, & Kasami, 1991; Stabler, 2011).

collection $R_1, R_2, R_3, \dots$ of sets of context-free rules, and (ii) a further set R of context-free rules that has all the others as subsets (i.e. each $R_i \subseteq R$).⁷ The learner will see data that has been generated by a composite system that has the full set R as its Ψ component, and has one of the subsets R_i as its Φ component; the learner’s task is to choose which of the subsets is playing this role. In the introductory example above, $R_1 = \{D \rightarrow ka\}$, $R_2 = \{D \rightarrow bo\}$, and $R = \{D \rightarrow ka, D \rightarrow bo\}$; but in general there may be additional noise rules in R that are not part of any candidate Φ component. In the first of our two case studies below, the learner will have four different candidate sets of Φ rules, corresponding to the word orders SVO, SOV, VOS and OVS, and the set of Ψ rules will include the rules for all four word orders plus some additional rules allowing for possibilities that are non-canonical under all four hypotheses, such as clauses that lack a subject. The outcome of the simulations we run using this learner will be a posterior distribution over these four sets of Φ rules, i.e. over these four different ways a canonical, or non-noise, (transitive) sentence can be realized.

Asking how well a particular hypothesized Φ component accounts for the observed data amounts to asking how likely those data are given the combination of that particular set of Φ rules and the shared set of Ψ rules — marginalizing over the individual ϕ parameters associated with the Φ rules, the individual ψ parameters associated with the Ψ rules, and also the ϵ parameters that control how likely each nonterminal is to be realized (non-)noisily. Just as before, we will assume that the learner has no prior commitments to these parameters taking any particular value over another, which means that the same logic that applied in working out the likelihood of the data under a bag of coins will also apply here.

The choice of what goes into the hypothesis space of a Noisy CFG learner reflects theoretical commitments about the knowledge that a learner brings to the language acquisition task. We will use the very small hypothesis space in Figure 7b to illustrate the basics of the model here. This amounts to equipping the learner with two basic word orders

⁷ More minimally, the system is completely determined by the collection of sets $R_1, R_2, \dots$ and the set $R - \bigcup_i R_i$ of remaining rules, if any.

to choose from— SVO and SOV— corresponding to the two candidate Φ components. In either case, a sentence is made up of a noun phrase (NP) followed by a verb phrase (VP). A verb phrase may be a bare verb (V) or may in addition contain another NP. Where the two candidate Φ components differ, and therefore the crux of the question facing this learner, is in whether this optional NP occurs before or after the V. The labels SOV and SVO for these two respective options are based on the purely structural classifications of the optional VP-internal NP as an object, and of the VP-external NP as a subject. What the learner also brings to bear on deciding between these two options is knowledge that its observations will contain not only data from the to-be-identified Φ component but also data generated by the particular noise rules listed in the Ψ component. In Figure 7b, these rules generate sentences where the VP either has no sister NP or has a sister NP on its right (as opposed to its left), and sentences with additional NPs. The learner being modeled here is aware that these are the divergences from the core SVO/SOV word order that will complicate the task of using its data to make the SVO/SOV decision. The hypothesis space for our first case study below, where we consider a choice between four possible word orders, uses an expanded version of the range of possibilities in Figure 7b.

All of these assumptions correspond to hypotheses about the prior expectations about linguistic structure that the learner brings to a particular learning task. That is, these correspond to hypotheses about the learner’s *inductive biases*, at least some of which may be domain-specific in nature. The linguistics literature often uses the term “Universal Grammar” (e.g. Chomsky, 1965, 1980, 1986) to refer to the innate, domain-specific components of the knowledge and formal capacities that a learner’s prior expectations derive from.⁸ However, this framework is agnostic about the sources of the hypotheses in the learner’s hypothesis space— in particular, whether these derive from what might be called Universal Grammar, from linguistic knowledge gained at an earlier stage of a learner’s development, or from another area of cognition. What the framework provides is a way to formally express these

hypotheses, whatever their source.

3.3 Likelihoods of trees in a Noisy CFG Learner

In the simulations that we report in Sections 4 and 5, the observed data take the form of a corpus of strings. The Φ and Ψ rules control the generation of tree structures that underlie those observed strings, and the probability of a string is a sum over the probabilities of its possible tree structures. To illustrate how different choices of Φ rules lead to better or worse explanations of the data, we will consider here only the task of calculating a likelihood for a collection of *trees* for a given hypothesized Φ component. The tree-based calculations here serve as the basis for calculating string likelihoods (see the Appendix for more detail).

Consider the task of working out the likelihood of the collection of trees in Figure 9a, for example, for each of the two Φ components in Figure 7b. Via the standard independence assumptions of context-free grammars, observing the collection of trees in Figure 9a amounts to observing the counts of rewrites shown in the table in Figure 9b. Furthermore, we can treat the rewrites of S nodes independently from the rewrites of VP nodes: the likelihood of the trees is exactly the product of the probability of seven S rewrites breaking down as shown and the probability of five VP rewrites breaking down as shown. (The rewrites of the NP and V nodes have probability one, since there is no optionality, and therefore can be ignored.)

We will see shortly that these likelihoods can be calculated directly from the summary counts shown in Figure 9b, but underlyingly each choice of Φ component actually corresponds to a compiled-out PCFG of the sort in Figure 8. These PCFGs generate trees that include additional layers of structure encoding the choice to treat each rewrite as either signal or noise. Two examples are shown in Figure 10; we call these *articulated trees*. The probability of a standard tree like the ones in Figure 9 is the sum of the probabilities of its

⁸ The term “Universal Grammar” is often associated with more intricate kinds of grammatical rules than the ones shown in Figure 7b, but the general computational framework that we describe here is compatible with a wide range of choices regarding the format of the Φ rules and Ψ rules, including many that resemble those more common in the contemporary syntax literature and would be necessary for modeling the learning of more complex grammatical phenomena. We return to this point in Section 6.

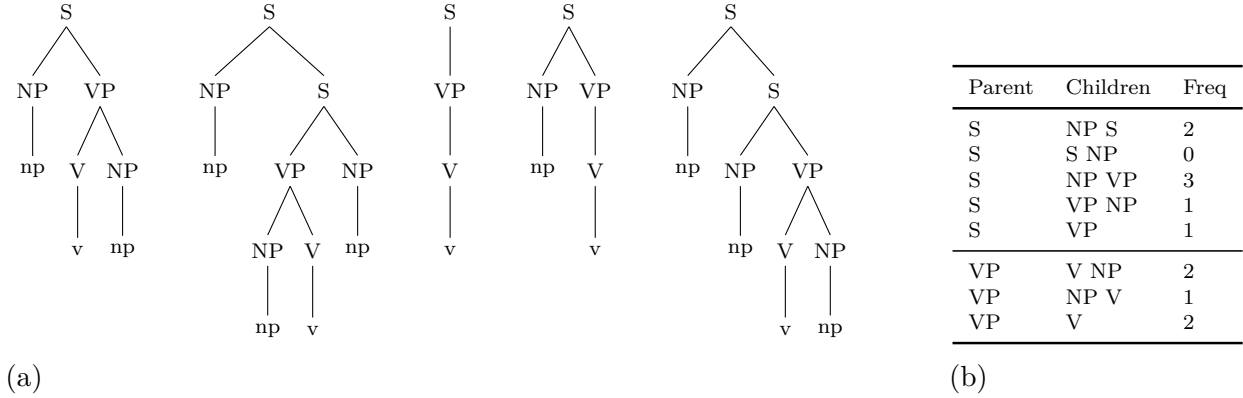


Figure 9

articulated trees, just as the probability of ‘ka’ in (9) is the sum of the probabilities of two derivations, one proceeding via D_Φ and one proceeding via D_Ψ . What is more interesting about Figure 10 is that this choice between signal and noise mechanisms can be made independently at each node, even within the same tree. In both of these trees, the $S \rightarrow NP S$ rewrite at the top is analyzed as noise (i.e. via S_Ψ) — as it must be, since this rewrite does not occur in either of the two hypotheses’ Φ components — but the S node that is introduced by that rewrite is itself realized *non-noisily* (via S_Φ). The two articulated trees differ in their treatment of the $VP \rightarrow V NP$ rewrite: the one on the left attributes this to the Φ component, and is therefore only compatible with the SVO hypothesis, whereas the one on the right attributes it to the Ψ component and is therefore compatible with either SVO or SOV.

Given the compiled-out PCFG for, say, the Φ component representing SVO word-order, and given specific values for all of the corresponding rule probabilities — one for $S \rightarrow S_\Psi$, which we could call ϵ_S , and one for $VP_\Phi \rightarrow V NP$, which we could call $\phi_{VP \rightarrow V NP}$, etc. — we could calculate a probability for each of the articulated trees in Figure 10. These are two of the many articulated trees whose probabilities we would sum to calculate the probability of the one corresponding tree in Figure 9. But what we really want is a likelihood, given the choice of the SVO Φ component, with the values of the ϵ , ϕ and ψ parameters (i.e., the PCFG’s rule probabilities) marginalized out. We assume a uniform prior over all the possible settings of those parameters, which means that the desired likelihood can be calculated by

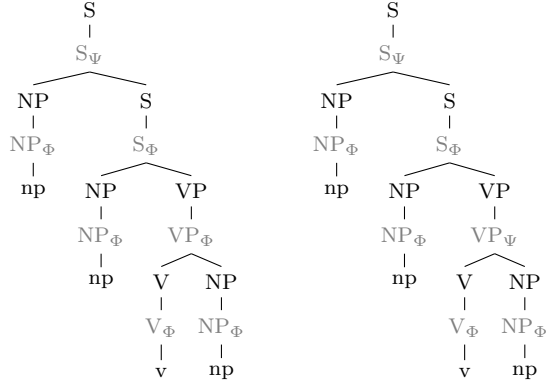


Figure 10. Two possible articulated trees for the rightmost tree shown in Fig. 9

making use of only rewrite counts like those shown alongside the trees in Figure 9, via essentially the same logic that we introduced in the simpler coins and ‘ka’/‘bo’ examples.

Turning to the VP nodes first, we see that the five rewrites of VP resulted in two occurrences of the right-hand side V NP, one of NP V and two of just V. For this to have been generated by the SVO hypothesis, the NP V rewrite would necessarily have been noise, and the other four could be either noise or non-noise. The details of how this partitioning could work are shown in Figure 11, where N_1^Φ and N_2^Φ are random variables expressing the counts of the two non-noise possibilities.

Let us consider the likelihood of this observed collection of VP rewrites arising via a specific combination of noise and non-noise, i.e. via specific values $n_1^\Phi \in \{0, 1, 2\}$ and $n_2^\Phi \in \{0, 1, 2\}$. There are six possible ways that the five VP rewrites could have been split between noise and non-noise, so the particular split that has $n_1^\Phi + n_2^\Phi$ non-noise rewrites has a probability of $\frac{1}{6}$. Similarly, the probability of the $n_1^\Phi + n_2^\Phi$ non-noise rewrites breaking into two groups as they did (on the left of Figure 11) is $\frac{1}{n_1^\Phi + n_2^\Phi + 1}$. For the breakdown of the noise rewrites (on the right of Figure 11), we must consider the ways things can be split into *three* bins, rather than two as we’ve had in all other examples to this point. Just as in the two-bin case, however, all the different ways of doing the splitting have equal probability. The number of ways to split n objects into k bins is $\binom{n+k-1}{k-1} = \frac{(n+k-1)!}{n!(k-1)!}$. For $n = 5 - (n_1^\Phi + n_2^\Phi)$

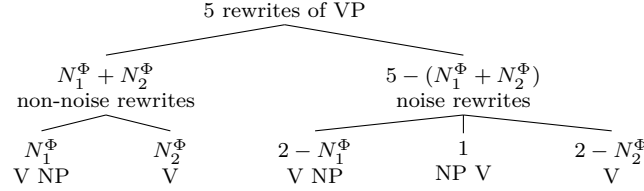


Figure 11. Partitioning the VP rewrites in Fig. 9 into signal and noise under the SVO hypothesis

and $k = 3$, this is $\frac{(7-(n_1^\Phi+n_2^\Phi))!}{(5-(n_1^\Phi+n_2^\Phi))!2!}$. The overall likelihood is therefore

$$(10) \quad P\left(\begin{array}{c} 2 \text{ VP} \rightarrow \text{V NP} \\ 1 \text{ VP} \rightarrow \text{NP V} \\ 2 \text{ VP} \rightarrow \text{V} \end{array}, \begin{array}{c} N_1^\Phi = n_1^\Phi \\ N_2^\Phi = n_2^\Phi \end{array} \middle| G_{\text{SVO}}\right) = \frac{1}{6} \times \frac{1}{n_1^\Phi + n_2^\Phi + 1} \times \frac{(5 - (n_1^\Phi + n_2^\Phi))!2!}{(7 - (n_1^\Phi + n_2^\Phi))!}$$

and from here we can marginalize over the unknown values of n_1^Φ and n_2^Φ to find a likelihood conditioned only on the choice of G_{SVO} .

$$(11) \quad P\left(\begin{array}{c} 2 \text{ VP} \rightarrow \text{V NP} \\ 1 \text{ VP} \rightarrow \text{NP V} \\ 2 \text{ VP} \rightarrow \text{V} \end{array} \middle| G_{\text{SVO}}\right) = \sum_{n_1^\Phi=0}^2 \sum_{n_2^\Phi=0}^2 \left[\frac{1}{6} \times \frac{1}{n_1^\Phi + n_2^\Phi + 1} \times \frac{(5 - (n_1^\Phi + n_2^\Phi))!2!}{(7 - (n_1^\Phi + n_2^\Phi))!} \right]$$

$$= 6.07 \times 10^{-2}$$

The three fractions inside the summation in (11) correspond to the three branch-points in Figure 11, just as the fractions inside the summations in (7) and (8) correspond to the branch-points in Figure 5. The situation in (11) differs from the earlier examples in having non-determinism inside the Φ component — unlike the G_{ka} and G_{bo} hypotheses where there is only one non-noise outcome — but what matters is that the range of non-noise options is *more restricted* than the range of noise options in the Ψ component. As long as there is this asymmetry, explanations that invoke more non-noise mechanisms will be preferred, because the non-noise options share probability mass with fewer competitors. The earlier examples illustrated the special case where the Φ component was completely deterministic, so there was *no* competition among non-noise options.

The situation for G_{SOV} is similar, except that n_1^Φ , rather than expressing how many of the two V NP rewrites to treat as non-noise, will now express “how many of” the one NP V rewrite to treat as non-noise. As in (7) and (8), this distinction has no effect on the fractions

inside the summation, but constrains the range of values to sum over for n_1^Φ : rather than the cap of 2 in (11), for the SOV hypothesis it is capped at 1.

$$(12) \quad P\left(\begin{array}{c} 2 \text{ VP} \rightarrow \text{V NP} \\ 1 \text{ VP} \rightarrow \text{NP V} \\ 2 \text{ VP} \rightarrow \text{V} \end{array} \middle| G_{\text{SOV}}\right) = \sum_{n_1^\Phi=0}^1 \sum_{n_2^\Phi=0}^2 \left[\frac{1}{6} \times \frac{1}{n_1^\Phi + n_2^\Phi + 1} \times \frac{(5 - (n_1^\Phi + n_2^\Phi))!2!}{(7 - (n_1^\Phi + n_2^\Phi))!} \right]$$

$$= 3.71 \times 10^{-2}$$

An analogous calculation can be done for the frequencies of the various rewrites of S nodes in Figure 9. The likelihood of the trees in Figure 9 under any hypothesized Φ component is the product of the probabilities of the VP rewrites and the S rewrites. Since G_{SVO} and G_{SOV} include the same S rewrite rules in their Φ components, the S rewrites contribute the same probability for each hypothesis.

$$(13) \quad P(\text{trees in Figure 9} \mid G_{\text{SVO}}) = P\left(\begin{array}{c} 2 \text{ VP} \rightarrow \text{V NP} \\ 1 \text{ VP} \rightarrow \text{NP V} \\ 2 \text{ VP} \rightarrow \text{V} \end{array} \middle| G_{\text{SVO}}\right) \times P\left(\begin{array}{c} 2 \text{ S} \rightarrow \text{NP S} \\ 0 \text{ S} \rightarrow \text{S NP} \\ 3 \text{ S} \rightarrow \text{NP VP} \\ 1 \text{ S} \rightarrow \text{VP NP} \\ 1 \text{ S} \rightarrow \text{VP} \end{array} \middle| G_{\text{SVO}}\right)$$

$$= (6.07 \times 10^{-2}) \times (1.45 \times 10^{-3})$$

$$= 8.82 \times 10^{-5}$$

$$P(\text{trees in Figure 9} \mid G_{\text{SOV}}) = (3.71 \times 10^{-2}) \times (1.45 \times 10^{-3})$$

$$= 5.39 \times 10^{-5}$$

The result is therefore that the posterior probability of G_{SVO} given the trees in Figure 9 is

$$\frac{8.82}{8.82+5.39} = 0.621, \text{ compared to } 0.379 \text{ for } G_{\text{SOV}}.$$

3.4 Full inference from strings

We have seen that a Noisy CFG learner corresponds to a collection of compiled-out PCFGs. From a collection $\vec{w}$ of observed strings, the goal is to infer the posterior distribution over this collection of grammars, $P(G \mid \vec{w})$.

We will use $\vec{\theta}^G$ to represent the vector of rule probabilities in a compiled-out PCFG G ; this represents the full collection of ϕ , ψ , and ϵ probabilities. Let $\vec{\theta}^{A_G}$ be the weights of the allowable expansions of a given nonterminal A . We can represent the prior over $\vec{\theta}^{A_G}$ as a

Dirichlet distribution with parameters $\vec{\alpha}^{A_G}$; the Dirichlet distribution is a generalization of the Beta distribution to cases with more than two possible outcomes. We assume here that all components $\alpha_i^{A_G}$ are equal to 1, which results in a uniform prior distribution: the model has no preference for or against assigning probability mass to any particular expansions of A . This differs from the way in which the numerical regularization approach imposes a skewed Beta prior, by setting α to a small value less than 1. We'll return to this comparison in the case studies that follow.

In the small example that we worked through in the section above, where we assumed that the strings came with observable tree structures (a set of trees $\vec{t}$, e.g., Figure 9), it was possible to calculate the posterior $P(G \mid \vec{t}, \vec{w})$ analytically by marginalizing over the rule weights ($\vec{\theta}^G$) and the partitionings of rewrites into signal and noise (the choice of articulated trees for each tree). But calculating the posterior $P(G \mid \vec{w})$ given only the strings would require further marginalizing over all the possible ways to choose a tree for each string in the data ($\vec{t}$). This calculation is intractable. So we instead infer the posterior $P(G, \vec{t} \mid \vec{w})$ via Gibbs sampling (Geman & Geman, 1984). After randomly initializing a set of possible trees for the observed strings, we alternate between sampling a new grammar according to $P(G \mid \vec{t}, \vec{w})$, and sampling new trees according to $P(\vec{t} \mid G, \vec{w})$. This process will converge to the joint posterior distribution over G and $\vec{t}$.

The step of sampling a grammar from $P(G \mid \vec{t}, \vec{w})$ involves exactly the calculations illustrated in Section 3.3. Note that any treeset $\vec{t}$ is compatible with all of the grammars in the learner's hypothesis space: it might be generated by core rules in certain grammars, or by some combination of noise and core rules, or by *only* noise rules, which are shared across all grammars. Thus, for every grammar G , $P(\vec{t}, \vec{w} \mid G)$ is always non-zero, and so $P(G \mid \vec{t}, \vec{w})$ is also always non-zero, allowing us to draw samples from this posterior in a feasible way.

We sample trees from the posterior $P(\vec{t} \mid G, \vec{w})$ with a Hastings proposal (Hastings, 1970), using a variant of an algorithm introduced by M. Johnson et al. (2007) and marginalizing over $\vec{\theta}^G$. See the Appendix for details.

3.5 Summary

Section 2.3 introduced the idea of a learner that makes a choice among discrete restrictive hypotheses, each of which is embedded in a system that also produces some noisy, non-canonical output. In this section we have shown how to extend this approach to systems of interacting rules that can model the fundamentals of natural language syntax, to define what we call a *Noisy Grammar Learner*. Given some data that reflects a mixture of core and noise mechanisms, the learner evaluates the following three questions, corresponding to the three branch-points in a choice for how to partition its data into noise and non-noise components: (1) What do the data from the core rules look like? (2) What do the data from the noise rules look like? (3) What is the right division into signal vs. noise? For each of the core grammars in its hypothesis space, the learner considers the possible answers to these three questions in order to determine how well that grammar can explain the observed data. In the case studies below, we show how two particular aspects of syntax acquisition can each be set up as a problem of the form in Figure 7, with a choice among certain sets of core rules against the backdrop of a certain set of noise rules.

4 Simulation 1: Learning basic word order

In the following simulations, we show that the approach of deciding among competing Noisy CFGs can be applied to model naturalistic phenomena in language acquisition that involve drawing systematic generalizations from messy data. Our first simulation models the problem of acquiring the basic positions of subjects and objects in the face of opacity introduced by non-canonical word orders. Our second simulation models the problem of identifying the affixes that mark subjects vs. objects in a language with case-marking.

To recapitulate the learning problem that we introduced in Section 1, children identify the canonical word order of their language in infancy, at ages even before they can identify the processes that produce non-canonical word orders in sentences like *wh*-questions and relative clauses (Hirsh-Pasek & Golinkoff, 1996; Lidz et al., 2017; Gagliardi et al., 2016;

Perkins & Lidz, 2020, 2021). These “non-basic” clause types are common— *wh*-questions comprise approximately 15% of the input to English-learning 1-year-olds— and therefore create substantial distortions in the body of evidence that children have available to identify the structure of basic clauses⁹ (Cameron-Faulkner, Lieven, & Tomasello, 2003; Stromswold, 1995). How do learners manage to abstract away from the messiness in their representations of their data in order to draw an accurate inference about basic word order? Some accounts assume that learners have the ability to “filter” non-basic sentences, ignoring them when drawing early syntactic inferences (Pinker, 1984). But if learners do not yet know what counts as basic, it is not obvious how they identify which sentence types count as *non*-basic, in order to filter them out (Gleitman, 1990; Perkins et al., 2022). Our model provides a way to implement the essence of this filtering idea, while avoiding potential issues of circularity.

In our first simulation, we tested our model on child-directed English, French, and Japanese. Both English and French are canonically SVO, but allow different types of argument dislocation. Japanese is canonically SOV, but also has a large amount of argument dislocation, and moreover allows both subject and object drop. We show that our model successfully identifies the target grammars for its noisy data in all three languages. Moreover, our model out-performs a learner that is capable of encoding the full messiness of its data— its hypothesis space allows all word-order rules with some probability— but is numerically biased to prefer extreme points in that hypothesis space, favoring rule weights that are close to 0 or 1. Thus, we show that our approach fares better in this learning problem than the numerical regularization approach taken in prior literature.

⁹ The notion of a “basic clause” has various characterizations in the literature. Pinker (1984), following Keenan (1976), characterizes basic clauses as “roughly, those that are simple, active, affirmative, declarative, pragmatically neutral, and minimally presuppositional.” Here, we are using the term “basic word order” to mean the word order in something similar to Chomsky (1957)’s notion of a kernel sentence, one where no obligatory transformations have applied. In languages like English, French, and Japanese, learners must come to identify that one primary order of clause arguments is possible in these sentences, although other orders may be possible through optional transformations such as *wh*-movement and scrambling. In a more modern framework, this notion of “basic word order” could be taken to mean the surface word order that most transparently reflects the dependencies that an argument must obligatorily enter into (e.g. theta role assignment and Case).

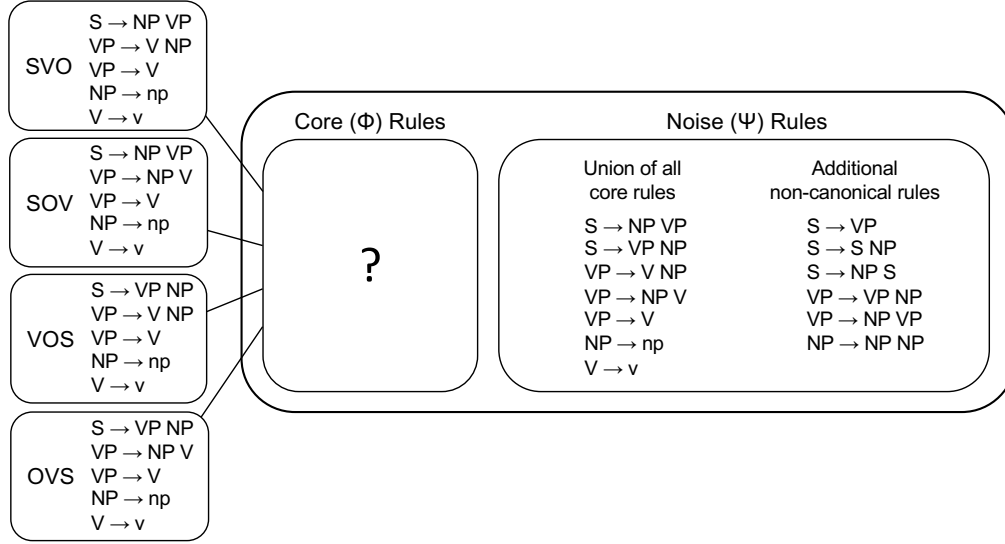


Figure 12. Hypothesis space of Noisy CFG learner for word order

4.1 Noisy CFG learner for word order

Our learner’s hypothesis space is an expanded version of the sample hypothesis space for word order in Figure 7b. It consists of four sets of Φ -rules and one shared set of Ψ -rules, giving rise to the four Noisy CFGs in Figure 12. Here, we explain more fully the structure of this hypothesis space as it pertains to the current case study.

Because the modeled learner is attempting to identify the canonical positions of subjects and objects in the language, here the learners’ Φ -rules generate the core predicate-argument structure of basic clauses with subjects and optionally objects. Following Section 3, we use the terms “subject” and “object” to express a structural asymmetry between core clause arguments that we assume is represented by a learner, and which is cashed out in the CFG rules here by defining “subject” as the NP daughter of S and sister of VP, and “object” as the NP daughter of VP and sister of V. This abstracts away from many finer-grained details of the ways in which these grammatical relations can be formally characterized, and we remain agnostic as to whether these grammatical relations and structural positions are derived or are primitives in a child’s or adult’s grammar; the important point for our purposes is that the learner’s hypothesis space encodes these

arguments in a particular structural relation, where objects form a constituent with the verb independent of the subject (Baker, 2001).

These Φ -rules have two important properties, encoding the learner’s assumptions about the grammatical phenomenon to be acquired. In the simulations in this section, we show that these properties play a key role in the learner’s ability to draw the correct word order generalizations. The first property is a set of substantive, domain-specific expectations about the nature of the rules governing basic clause syntax. These include the expectation that a sentence will obligatorily have an NP that is the sister of a VP, whereas a VP may or may not contain an NP that is the sister of its V; this encodes a widely-held assumption that subjects in canonical clauses are required in ways that objects are not (e.g. Keenan, 1976; Chomsky, 1982).¹⁰ Furthermore, noun phrases must occupy either canonical subject or canonical object position, reflecting the learner’s belief that adjunction is in some sense non-canonical. Since these assumptions are shared by all four sets of Φ rules, the learner has no way to consider departing from them. The second important property of the Φ rules is the ways in which the four candidate Φ components differ, which express the choice points available to the learner. Each of the four options has a degree of restrictiveness: each hypothesized set of core rules deterministically puts subjects before or after verb phrases and objects before or after verbs, yielding a four-way choice of canonical word order: SVO, SOV, VOS, OVS. We note that this two-by-two hypothesis space leaves aside questions about natural languages where the canonical word order separates the verb from the object (VSO and OSV), and we return to this in Section 6.

All four grammars share the same set of noise rules, encoding the learner’s hypotheses about the non-canonical processes that will introduce noise into the data. These Ψ -rules do

¹⁰ We abstract away from the various proposals that have been put forth for why subjects may be required, and note that this has also been debated. As McCloskey (1997) puts it, “It has sometimes been claimed that every clause must have a subject. This is not obviously correct, but it is clearly correct in some broad sense for some languages. There is no other argument-type or syntactic position for which this claim can be made with even remote plausibility. There are no languages, as far as I know, for which it has ever occurred to anyone to claim that every clause must have a direct object, or an indirect object or a prepositional complement or whatever” (p. 198).

not encode the full range of complex grammatical options for non-basic structures that a mature speaker of a language will eventually command (e.g., *wh*-movement, relativization, raising); instead, they encode “placeholder” rules that simply allow for all permutations and deletions of NP arguments, and for additions of NPs into non-argument positions. These are analogous to some of the Ψ -rules that we saw in Figure 7b in Section 3. This models a specific stage in the incremental process of grammar acquisition, motivated by empirical findings that basic word order acquisition appears to developmentally precede the acquisition of non-basic sentence types (Hirsh-Pasek & Golinkoff, 1996; Lidz et al., 2017; Gagliardi et al., 2016; Perkins & Lidz, 2020, 2021). The developmental stage that we model is one in which a learner is only attempting to acquire basic word order, leaving the full details of these other grammatical processes to be acquired at a later developmental stage; in Section 6, we return to the question of how these later developmental stages might proceed. The flexibility in the noise rules produces many more possibilities for expanding a given nonterminal than are provided by the core rules, mirroring the asymmetry between restrictive two-headed or two-tailed coins, and flexible head-tail coins.

Crucially, while the learner’s rules contain hypotheses about which canonical and non-canonical processes might be operative, the learner does not know ahead of time the ϕ , ψ and ϵ probabilities associated with these rules. It does not know how frequently it will encounter canonical transitive vs. intransitive clauses, and it does not know which kinds of non-canonical clauses it will encounter, or how frequently. We show that our learner is able to identify the correct Noisy CFG— the correct combination of core and noise rules— by marginalizing over all possible choices of these parameters, following the logic we saw above.

Our learner infers the posterior probabilities of the Noisy CFGs in its hypothesis space using distributions of imperfectly-identified noun phrases and verbs that a young infant might be able to represent. The learner performs this inference based on these string distributions alone, by using the inference mechanism described in Section 3. This does not require supplementary information about underlying clause structure. However, in restricting

	English	French	Japanese
Corpus	Brown	Lyon	Miipro
# Children	3	5	4
Ages	1;6 – 5;1	1;0 – 3;0	1;2 – 5;0
# Words	380,423	515,827	328,502
# Utterances	85,787	139,800	115,368

Table 1

Corpora of child-directed English, French, and Japanese

our attention to this type of distributional learning, we do not imply that children may not also make use of correlations between meaning and syntax, between prosody and syntax, or even between different syntactic phenomena (for instance, knowing that postpositions tend to correlate with pre-verbal objects) (Christophe, Millotte, Bernal, & Lidz, 2008; Morgan & Demuth, 1996; Pinker, 1984; Greenberg, 1970). We abstract away from these other sources of information in order to ask how much ground a learner might be able to gain by applying this learning mechanism to imperfectly-perceived string distributions, noting that a similar mechanism could be generalized to make use of correlated information from other domains.

4.2 Data

We used the CHILDES Brown, Lyon, and Miipro corpora (Brown, 1973; Demuth & Tremblay, 2008; Oshima-Takane, MacWhinney, Sirai, Miyata, & Naka, 1995), which contain speech directed to English, French, and Japanese learning children (see Table 1).

We followed a procedure for identifying verbs and noun phrases on the basis of distributional cues that an infant around the age of 15 months might be able to use; we provide an outline of the process here, but for full details see Appendix C. Based on empirical evidence, we assume that infants at this age can recognize a small number of functional elements of different sorts (e.g., Höhle, Weissenborn, Kiefer, Schulz, & Schmitz, 2004; Hicks, Maye, & Lidz, 2007; Shi & Melançon, 2010; Kim & Sundara, 2021; Mintz, 2013; Babineau, Shi, & Christophe, 2020; Marquis & Shi, 2012; Haryu & Kajikawa, 2016; He & Lidz, 2017); see Table C1 for a summary. For English, for example, we assume that an infant

can recognize *you* as a pronoun, *the* as a determiner, *-ed* as a verbal suffix and *will* as an auxiliary; these are four of the 75 elements that met our criteria for being recognizable, which included being among the 100 most frequent tokens in the corpus.

Once these recognizable elements have been tagged in the corpus, we use some simple heuristics to guess at the positions of verbs and noun phrases on the basis of only those tags. A noun phrase (**np**) is taken to appear wherever there is an element recognized as a pronoun or a name, and wherever there is an unrecognized element in a “**np**-cue position”; a verb (**v**) is taken to appear wherever there is an unrecognized element in a “**v**-cue position”. The definition of the cue positions differs by language. For English, for example, **np**-cue positions include those following a determiner and those preceding the suffix *-s*, and **v**-cue positions include those following an auxiliary and those preceding the suffix *-ed*. For Japanese, **np**-cue positions are those preceding a case-marker, and **v**-cue positions include those preceding the negation marker *-nai*. *Wh*-words and object clitics were not identified as **np**’s, because they may not be recognized as such by infants at the age when they are learning basic word order (Perkins & Lidz, 2021; Brusini et al., 2017). Object clitics in French that are homophonous with determiners were treated erroneously as determiners, to simulate the uncertainty that infants might have about their category. Case-markers in Japanese were used merely as cues for identifying **np**’s; their grammatical function (e.g., nominative or accusative) was not encoded, as we assume that a learner at this age does not yet know which case-markers mark subjects and which mark objects (a problem we consider in Section 5).

This allows us to map each corpus utterance to a string consisting of any number (possibly zero) of occurrences of **np** and **v**. From these we retain only those that contain exactly one **v**, since these are the ones relevant to the learner’s question of how the elements of a single clause are arranged. The proportions of the relevant string types that we arrive at for each language are shown in Table 2.

We created datasets ranging in size from 10–50 strings, sampled according to the distribution in Table 2 for each language. We chose our largest dataset to be 50 strings for

English	French	Japanese
0.34 np v	0.45 np v	0.63 v
0.27 np v np	0.20 np v np	0.23 np v
0.10 v	0.12 v	0.05 v np
0.08 v np	0.08 np np v	0.05 np np v
0.07 np v np np	0.05 v np	0.02 np v np
0.04 np np v	0.03 np np v np	
0.03 np np v np	0.03 np v np np	
0.02 v np np	0.01 np np np v	
0.01 np v np np np	0.01 v np np	
0.01 np np np v		
0.01 np np np v np		

Table 2

Proportions of most frequent string types in each language

reasons of computational tractability. We also tested a range of smaller datasets in order to explore how regularization behavior varies as a function of the size of the learner’s data: recall that the artificial-language simulations with the numerical regularization model in Section 2.2 showed that regularization was a property of learning from small datasets. Over 30% of the strings in each language are incompatible with the core rules of the target grammar (SVO for English and French, SOV for Japanese). As a whole, these data cannot be generated by the core rules of any single grammar in the learner’s hypothesis space, without considering the option of noise.

4.3 Results

4.3.1 Noisy CFG learner. Figure 13 displays our model’s inferred posterior probability distribution over the four Noisy CFGs in its hypothesis space, averaged over 10 runs of the model in each dataset. Visual inspection shows that the model’s posterior distributions take the same qualitative shapes across datasets of varying sizes within each language. Just as in the artificial language ‘ka’/‘bo’ simulations in Section 2.3, we see that asymmetries in these posterior distributions are weaker with smaller amounts of data, and become sharper as the amount of data increases. But while the amount of data presented to a learner can be controlled in an experimental setting, here it is an empirical question how

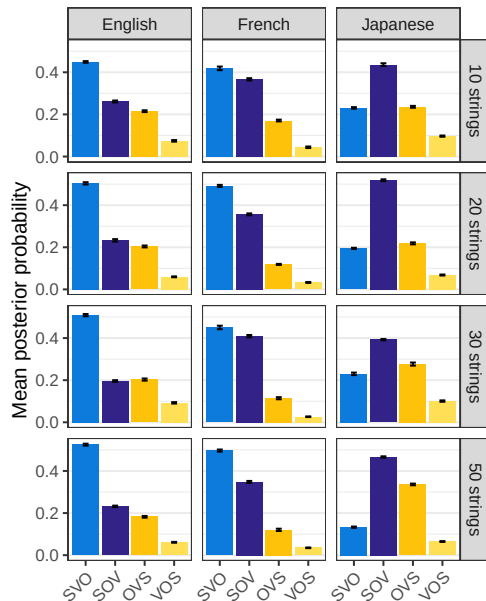


Figure 13. Posterior distribution over word-order grammars, Noisy CFG learner

many sentences a child is exposed to before identifying the target word order of the language; likely this number is much higher than 10 or even 50. For simplicity, we analyze the largest dataset presented to our model.

We find that in both English and French, the SVO grammar was assigned a higher posterior probability than any other grammar in the learner’s hypothesis space (all Welch’s $t > 17.24$, all $p < 0.001$, Bonferroni correction for multiple comparisons). In Japanese, the SOV grammar had highest posterior probability compared to all other grammars (all $t > 7.14$, all $p < 0.001$). Thus, the learner’s filtering mechanism allowed it to successfully overcome the large amount of noise in its data. The learner came to partition its data into noise and non-noise portions, in such a way that the non-noise portion provided a signal for the correct target word order.

Examining the behavior of the Japanese learner more closely, we find something interesting. With sufficient data, the learner in Japanese assigned the OVS grammar higher posterior probability than the SVO grammar, unlike in English and French. But this is puzzling: in Japanese, just like in English and French, the core rules of the OVS grammar

can generate *fewer* strings in the learner’s data than can the core rules of the SVO grammar. Only 7% of the strings in the Japanese dataset are compatible with the core OVS rules, compared to 25% compatible with the core SVO rules. Why, then, did the Japanese learner judge OVS more probable than SVO?

This difference reveals some subtleties in how the learner makes use of the distributions that it observes. Specifically, because Japanese allows argument-drop, the Japanese learner observed a relatively large number of bare *v* strings. This is the most common string type in the Japanese data, but comprises only about a tenth of the data in English and French. In order to analyze a string with only a *v* and no *np* satellites, the learner must make use of a noise rule that allows *S* to be rewritten as *VP* directly ($S \rightarrow VP$), without an *NP* subject. For Japanese, the learner came to identify that that this noise rule for omitting subjects was more useful than any of the core rules for introducing subjects, with a consequence for how it analyzed the *np v* strings in its data. Rather than taking these as evidence for a subject-initial grammar, the Japanese learner preferentially analyzed *np v* strings as subjectless clauses with an object-initial verb phrase, increasing the probability of OVS over SVO. In English and French, by contrast, the learner did not come to the same conclusion, because its data did not lead it to infer that the noise rule for omitting subjects had high probability. Thus, the different shapes of the posterior distributions in Fig. 13 arise in part because the learner correctly identified that subject-drop has high probability in Japanese, but not in English or French.

4.3.2 Comparison: A data-coverage heuristic. The results above illustrate that the learner’s conclusions can diverge from what one would expect from a simple “data coverage” heuristic, where the best-fitting grammar is simply the one whose core rules can account for the most data. A further demonstration comes from considering comparisons between hypotheses that differ in restrictiveness in their core rules. To examine this situation, we added a fifth “free-order” grammar to the learner’s hypothesis space, in which neither subjects nor objects have a fixed position. This grammar’s core ruleset is the union

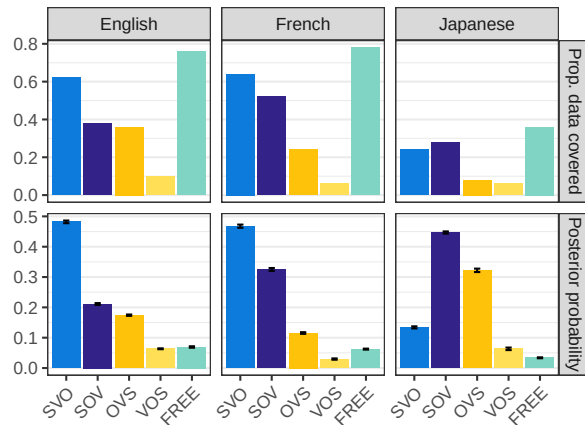


Figure 14. Five-way hypothesis space: Proportion data coverage vs. model’s posterior distribution (50-sentence datasets)

of the core rules in the learner’s four original restrictive grammars, and its noise rules are the same as those in the original four grammars.

Given a choice among the original restrictive grammars and this free-order grammar, the data-coverage heuristic will always favor the free-order grammar, since it generates the union of the stringsets generated by the original four. In Figure 14, we plot the predictions of the data-coverage heuristic and the model’s inferred posterior over this five-way hypothesis space, for the 50-sentence dataset in each language. In each of the top panels, where a comparison only among the leftmost four grammars would have SVO or SOV as the winner, we see that the more flexible grammar fares better by the data-coverage metric. But our learner still assigned SVO higher posterior probability than any other grammar in the hypothesis space in English and French, and SOV highest posterior probability in Japanese (Fig. 14, bottom, averaged across 10 runs of the learner; all $t > 4.80$, all $p < 0.001$).

Why does our learner still succeed at identifying the target word order in each language, even in the presence of another hypothesis that covers more of the data? Intuitively, our learner considers a tradeoff between fit to the data and restrictiveness of its hypotheses, an instance of what is often called “Bayesian Occam’s Razor” (Tenenbaum & Griffiths, 2001; see also Maitra & Perkins, 2023). Given the choice between a restrictive hypothesis that

provides a decent fit to the data, and a more flexible hypothesis that provides a slightly better fit, a preference emerges for the more restrictive option—just as we intuitively prefer to attribute as many coin flips as possible to a two-headed rather than a head-tail coin.

These findings demonstrate the flexibility and robustness of this learning mechanism. Our learner identifies the target strict word order as its preferred hypothesis not only in comparison with other equally-strict alternatives, but also when other less restrictive options are available. The fact that it settled on a restrictive word order in Figure 13 was not simply a by-product of the fact that we provided only restrictive options. An implicit tradeoff between a grammar’s restrictiveness and its fit to the data, and the expectation that this fit will be noisy, together enable the learner to identify the target restrictive word order among more flexible hypotheses.

4.3.3 Comparison: Numerical regularization. We now turn to the question of whether our model’s success depends on a choice of discrete canonical word-order grammars in the learner’s hypothesis space. To answer this question, we constructed a comparison learner whose hypothesis space collapses the distinction between canonical and non-canonical structures. This “fully-flexible” hypothesis space consists of a single standard PCFG comprising all of the word-order rules across our learner’s four grammars. For this fully-flexible model, learning canonical word order would mean identifying that some of its rules have probabilities near zero. Within this fully-flexible architecture, we impose a numerical regularization bias in the same manner as for the artificial language ‘ka’/‘bo’ simulation in Section 2.2, thereby assessing how our model compares to the general regularization bias approach taken in previous literature (Real & Griffiths, 2009; Culbertson et al., 2013; Perfors, 2012).

There are two important properties that distinguish our Noisy CFG model’s hypothesis space from that of the fully-flexible learner. First, our model’s hypothesis space encodes an expectation that the grammatical systems generating its data comprise a mixture of restrictive core rules and processes that introduce distortions of those rules. Second, it also

encodes substantive expectations about the content of those rules: specifically, it expects canonical clauses to have subjects and not to have adjuncts. Comparing our model to a learner that lacks these properties provides insight into the role that they played in our model’s success.

We tested two variants of the fully-flexible model. The first assumes that all rules in its hypothesis space are equally probable *a priori*, as in our original model. The second is numerically biased to regularize its rule weights. This regularization bias takes the form of a skewed prior over the rule weights $\vec{\theta}$ in the learner’s grammar (M. Johnson et al., 2007), analogous to the skewed prior used in Section 2.2. For each nonterminal A , we set all component parameters α_i^A of the model’s Dirichlet prior to a small value, 0.0001. Because the Dirichlet distribution is a generalization of the Beta distribution, when $\alpha < 1$, this again has the effect of symmetrically skewing the learner’s prior towards extreme values. Here, this biases the learner to put probability mass on only one expansion of a given nonterminal, and to push the probabilities of other expansions towards zero. Because the distribution is symmetric, the learner has no prior belief about which particular expansion is more likely. For simplicity, we test only one very extreme value of α in order to give the learner the best chance of regularizing successfully.

As this learner does not make a choice among discrete grammars, the way that it would arrive at the target canonical word order for the language is to put most probability mass on the appropriate rules for subject and object position in its posterior distribution over $\vec{\theta}$. As an indication of whether the posterior distribution over $\vec{\theta}$ displays these properties, we examine the learner’s posterior distribution over trees for its data. Our sampling process consists of just one of the steps in our original Gibbs sampler. We sample trees for the learner’s data from the posterior given its sole grammar, $P(\vec{t} \mid G, \vec{w})$, just as we sampled $P(\vec{t} \mid G, \vec{w})$ in our original model.

We assessed whether the fully-flexible learner had identified a canonical word order by analyzing the trees in which the learner had identified a subject NP (daughter of S, sister of

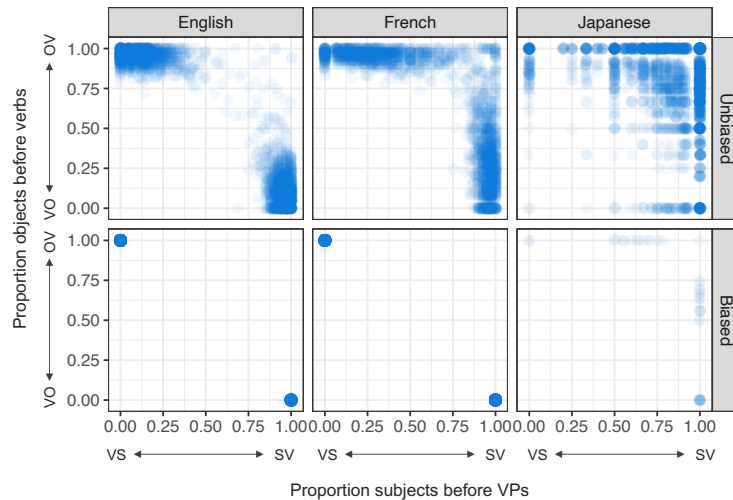


Figure 15. Posterior distribution over subject and object position in sampled treesets ($\vec{t}$), fully-flexible learner, 50-sentence dataset

VP) and/or an object NP (daughter of VP, sister of V). For each sampled treeset in which at least one tree had a subject and at least one tree had an object, we calculated the proportion of subjects that appeared before the verb phrase, and objects that appeared before the verb. These proportions are plotted in Figures 15-16, where each point corresponds to a sampled treeset, aggregated across ten runs of the model in each language. For the sake of space, we plot the unbiased and biased model results together only for the 50-sentence dataset (Fig. 15). Results from the biased model for the smaller datasets are plotted in Fig. 16.

These plotted distributions provide an indication of the learner’s inferred posterior probabilities of subject-initial and object-initial structures. The four possibilities for canonical word order correspond approximately to the four corners in each panel: clockwise from top left, these are OVS, SOV, SVO, and VOS. If the learner had successfully identified that English and French are canonically SVO, the majority of tree samples would lie close to the lower right corners of these graphs. For Japanese, we would expect to see the majority of tree samples close to the top right corner of the graph, corresponding to canonical SOV order.

To provide a direct comparison with the analysis of our Noisy CFG learner, we analyze the learner’s samples for the 50-sentence dataset (Fig. 15). For the unbiased learner, the number of sampled treesets that we are able to analyze is high across the three languages:

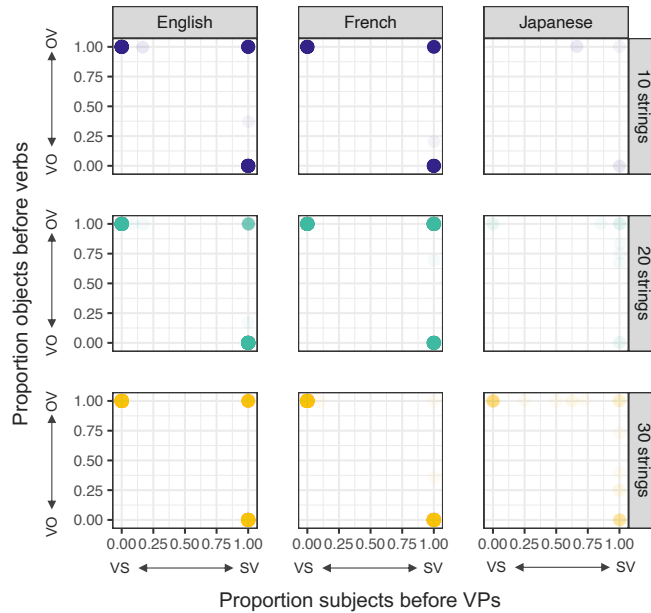


Figure 16. Posterior distribution over subject and object position in sampled treesets ($\vec{t}$), fully-flexible biased learner, smaller datasets

99% of the English samples, 90% of the French samples, and 94% of the Japanese samples contained at least one tree with a subject and at least one tree with an object. But across these analyzable samples, in each language the learner inferred distributions over tree structures that mirrored its noisy data. These ranged from the OVS (top-left) to the SVO (bottom-right) region in English, and across the OVS, SOV, and SVO regions in French and Japanese (Fig. 15, top). Thus, the unbiased learner failed to identify a restrictive word order for any of the three languages.

For the biased learner, the number of sampled treesets that we can analyze differs by language. Looking first at English and French, these numbers were high: 100% of the English samples and 99% of the French samples contained at least one tree with a subject and at least one tree with an object. In both languages, the biased learner inferred distributions over subject and object position that lie close to the corners corresponding to canonical word orders (Fig. 15, bottom). However, the learner assigned equal posterior probability to both OVS and SVO structures; the mean proportions of subject-initial and object-final trees were not significantly different from 0.5 in either language (English: both

mean subject-initial and mean object-final = 0.51; French: both mean subject-initial and mean object-final = 0.54; all $t < 0.71$, all $p > 0.49$). The learner’s numerical regularization bias led it towards the deterministic corners rather than the flexible middle, but it did not correctly identify SVO as a better corner than OVS. This same qualitative pattern is found with the smaller datasets as well (Fig. 16). With less data, the learner put additional probability mass on the SOV corner, but did not successfully discriminate SVO from OVS.

In Japanese, the number of sampled treesets that we can analyze for the biased learner is surprisingly low, resulting in very few data points plotted for that learner for any size of dataset in Figs. 15–16. For the 50-sentence dataset, only 1.6% of the learner’s samples contain at least one tree with a subject and at least one tree with an object. Instead, the Japanese biased learner overwhelmingly converged to analyses in which **np**’s occupied non-subject and non-object positions, introduced by recursive adjunction rules such as $S \rightarrow NP\ S$. For the 1.6% of the sampled treesets that do contain a subject and an object, we can perform the same analysis as for English and French, and find that more of these samples are in the SOV quadrant (mean subject-initial = 0.83, $t(9) = 13.00$, $p < 0.001$; mean object-initial = 0.68, $t(9) = 2.53$, $p < 0.05$). But this analysis of the relative positions of subjects and objects tells us less about the learner’s overall conclusions than it did for English and French, as the Japanese learner strongly preferred structures in which no subjects or objects were present. This difference arises again from the Japanese learner’s need to explain the prominence of bare **v** strings in its data. The Japanese learner concluded that the most probable way to introduce a VP is to use a rule with no NP subject, driving the probabilities of the subject-introducing rules towards zero. Similarly, it inferred that the most probable way to introduce a V is to use a rule with no NP object, driving the probabilities of the object-introducing rules towards zero. The flexibility in the learner’s hypothesis space, in combination with the argument-drop in its data, led the biased Japanese learner away from analyses in which **np**’s are clause arguments.

4.3.4 Role of substantive grammatical expectations. Unlike our Noisy CFG model, the fully-flexible learner did not identify the target canonical word order for its noisy data. Why does our approach perform better in this learning problem than the numerical regularization bias approach? Recall that there are two properties of our model’s hypothesis space that distinguish it from the fully-flexible learner, and may have allowed it to make better use of its observed data: (1) an expectation that its data are generated via a mixture of restrictive core rules and more flexible “noise” rules, and (2) an ability to encode substantive expectations about the nature of those core rules.

These two substantive expectations may have helped our learner in different ways across the languages tested. In English and French, we hypothesize that the crucial expectation is that canonical clauses require subjects. This may have allowed our learner to use one of the most common string types in its data— **np v**— as evidence for a subject-initial grammar. Given the choice between using its restrictive core rules to analyze the sole **np** as a canonical subject, versus using its noise rules to analyze the **np** in a different position (leaving the clause subjectless), a preference will emerge for the canonical-subject analysis, again paralleling our preference to analyze a sequence of heads as coming from a two-headed rather than a head-tail coin. In our comparison against the “data-coverage” heuristic above, we saw that this preference for restrictive hypotheses can inform the learner’s choice *across* grammars, when some are more restrictive than others. Here, we expect that this same mechanism applies *within* each grammar, governing the learner’s choice of attributing data to the restrictive core rules vs. the flexible noise rules. The fully-flexible learner does not distinguish between canonical structures in which subjects are required, and non-canonical structures in which they are not, so no preference emerges to analyze a sole **np** in a specific clausal position.

In Japanese, we hypothesize that the crucial assumption on the part of our learner was that adjuncts are non-canonical. As discussed above for the fully-flexible learner, the prevalence of bare **v** strings in Japanese provides evidence that unary-branching structures (S

dominating only VP, VP dominating only V) are common; and strings that do contain **np**'s can make use of those common substructures by analyzing the **np**'s as adjuncts. But for our learner, we expect that this temptation to analyze **np**'s as adjuncts will be balanced against a motivation to use restrictive core rules rather than more flexible noise rules. This is likely what prevented our learner from analyzing all **np**'s as adjuncts, and therefore allowed it to draw firmer conclusions about the positions of subjects and objects than the fully-flexible learner did.

Thus, the learner's two sorts of substantive expectations work in formally similar ways. Our learner's hypothesis space encodes an asymmetry between restrictive core rules and less restrictive noise rules, leading the learner to prefer core rule analyses. By allowing only particular sorts of core rules, we are able to express an expectation that canonical clauses have particular shapes. We suggest that this contributed in important ways to the learner's ability to draw sophisticated inferences from its noisy data.

We test this possibility with a further model comparison, focusing specifically on the learner's expectation that canonical clauses require subjects. We ask: does our learner's success in English and French depend on its substantive expectations about the core rules for subjects? Or would our learner succeed just as long as its hypothesis space encodes a distinction between restrictive core rules and flexible noise rules, regardless of the nature of those core rules? Because the learner's expectations about subjects and adjuncts have formally similar effects on its learning mechanism, answering this question for one of these cases will speak to the broader question of whether the nature of the core rules matters in our learner's inference.

We constructed a comparison Noisy CFG model whose hypothesis space lacks the requirement that canonical clauses have subjects. Each of the grammars in this hypothesis space now includes $S \rightarrow VP$ within its core ruleset, allowing the core rules to produce subjectless analyses. The hypothesis space is otherwise identical to that of our original model. By the reasoning above, we predict that this change should not affect our learner's

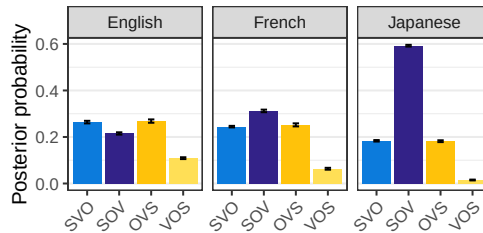


Figure 17. Posterior distribution over word-order grammars, Noisy CFG learner without subjects required canonically (50-sentence datasets)

success in Japanese, but it should lead to worse performance in English and French.

Figure 17 displays the resulting posterior probability distribution over grammars that this comparison learner inferred for the 50-sentence datasets. These results are consistent with our predictions. Changing the learner’s expectation about subjects in canonical clauses did not affect its success in Japanese: like our original model, this Japanese learner successfully assigned SOV highest posterior probability (all $t > 75.96$, all $p < 0.001$). But unlike our original model, this learner was unable to identify that English and French are SVO. In English, it inferred that SVO and OVS were tied as most probable (Welch’s $t(16.93) = 0.49, p = 0.63$). In French, it assigned highest posterior probability to SOV (all $t > 6.98$, all $p < 0.001$), and did not differentiate between SVO and OVS as the next-most-probable options (Welch’s $t(13.31) = 1.03, p = 0.32$). When the hypothesis space no longer encoded a requirement for subjects in canonical clauses, the English and French learners were not able to discriminate between subject-initial and object-initial grammars. Thus, it is not merely the presence of restrictive grammatical rules in our learner’s hypothesis space that matters for its success in this learning problem; the content of those rules also plays a large role.

4.4 Summary

Through this case study, we show that the approach of deciding among competing Noisy CFGs can be applied to a naturalistic language learning phenomenon which resembles the phenomenon of regularization observed in previous artificial language experiments. We

find that our model can successfully acquire basic word order from the immature sentence representations available to an infant at early stages of syntax acquisition. Using distributions of imperfectly-identified noun phrases and verbs, our model successfully infers that English and French are SVO and Japanese is SOV, without further cues to underlying sentence structure. It does so by separating signal for canonical word order from noise coming from non-canonical structures, thereby implementing a proposal that young learners “filter” non-canonical clauses from their data (Pinker, 1984; Perkins et al., 2022). Because the learner’s grammatical hypotheses allow only certain restrictive core rules, a preference emerges to use these core rules to explain the skews in its data when possible, rather than analyzing most of the data as noise. This provides the impetus for successful filtering, even though our learner does not know ahead of time the rate or properties of non-canonical clauses in the language.

Moreover, we find that our model out-performs a learner that instantiates the general regularization bias approach in prior literature (Real & Griffiths, 2009; Culbertson et al., 2013; Perfors, 2012). We show that two properties of our learner are crucial for its success. First, the learner’s hypothesis space encodes a distinction between restrictive core rules that produce canonical clause structures, and more flexible noise rules that introduce distortions into its data. Second, it encodes substantive expectations about the nature of those core rules: that canonical clauses have subjects, and that noun phrases canonically occupy argument positions within a clause. Without these properties, the learner is unable to identify signal for the target word order within its noisy data.

These findings suggest that, for this learning problem, it is important for learners to have specific expectations about the types of grammatical rules that may have generated their data, embedded within a broader system that introduces noise. By encoding a distinction between restrictive grammatical rules and noise processes, our learning architecture provides a natural way to express linguistically-motivated expectations about the nature of basic clause syntax. And by embedding these expectations within a

noise-tolerant system, the architecture makes it possible for a learners to recover the target basic clause structure for their language from data that appear inconsistent on the surface.

5 Simulation 2: Learning case-marking

In our second case study, we demonstrate how our approach generalizes to another learning problem within early morphosyntax acquisition. We consider the problem of acquiring a case-marking system in which subjects and objects are each marked with a particular affix, as in Japanese. Early in language development, children acquiring case-marking languages come to identify how specific affixes function as case-markers, despite sometimes variable and inconsistent evidence in their input (e.g., Fisher, Jin, & Scott, 2019; Suzuki & Kobayashi, 2017; Suzuki, 1999; Matsuo, Kita, Shinya, Wood, & Naigles, 2012; Göksun, Küntay, & Naigles, 2008). Here, we model a stage in development in which a child may have identified certain affixes as candidate case-markers, but does not know which marks subjects (nominative) and which marks objects (accusative). We show how the system that we have introduced above can be applied to this scenario. This case study demonstrates how learning problems that seem on the surface to be very different—acquiring morphological marking vs. clause structure—can both be expressed within the same kind of system, as a choice between Noisy CFGs.

We conducted simulations on a new dataset generated from child-directed Japanese, in which noun phrases sometimes co-occur with the *ga* (nominative) and *o* (accusative) suffixes. Because Japanese has scrambling and argument-drop, and the pronunciation of case-markers is optional, this dataset contains noisy and sparse evidence for the grammatical function of these affixes. We consider a learner that is attempting to identify which grammatical relation *ga* and *o* are each marking at the same time as identifying the canonical positions of subjects and objects. We show that our model simultaneously identifies *ga* as nominative and *o* as accusative, along with canonical SOV word order. Moreover, as in the previous section, our approach out-performs a learner that does not make a discrete choice among case-marking

grammars, but rather implements a numerical regularization bias over a fully-flexible hypothesis space.

5.1 Noisy CFG learner for case-marking

Our learner’s hypothesis space consists of augmented versions of the Noisy CFGs for the word order learner in the previous case study. We expanded the learner’s hypothesis space from Section 4 to include Φ rules producing all four possible orders of subjects and objects, crossed with both possible mappings of the case-markers *ga* and *o* to subjects vs. objects. This gives rise to the eight Noisy CFGs in Figure 18. The Φ -rules mirror the four grammars our word-order learner considered: they generate basic transitive and intransitive clause structures, with subjects obligatory and objects optional. But here subjects and objects are represented with distinct nonterminal symbols (NPS and NPO), and are re-written as terminal symbols with distinct case-marking (**np-ga** and **np-o**). Each case marker deterministically realizes either subjects or objects, yielding a 2-way choice for case-marking systems: either subjects are rewritten as **np-ga** and objects as **np-o**, or objects are rewritten as **np-ga** and subjects as **np-o**. Adjunct NPs are now represented by a distinct nonterminal symbol, NPA, and these are canonically rewritten as **np**, i.e., noun-phrases that are neither marked with *ga* nor with *o*.

All grammars again share the same set of Ψ rules. As before, these comprise a superset of the union of the core rules across the learner’s eight grammars, allowing for permutations and deletions of subjects and objects, and insertions of NPs into non-argument positions. In addition, the noise rules allow any type of NP to be rewritten by **np-ga**, **np-o**, or **np**. Thus, as before, we encode an asymmetry between highly restrictive core rules for case-marking and flexible noise rules that produce many more possibilities for expanding a given nonterminal. Here, the noise operates both internal to a tree, manipulating the position of NP nonterminals, and at a tree’s frontier, manipulating the position of **np** terminal symbols.

Similar to our previous case study, the learner observes strings of imperfectly-identified

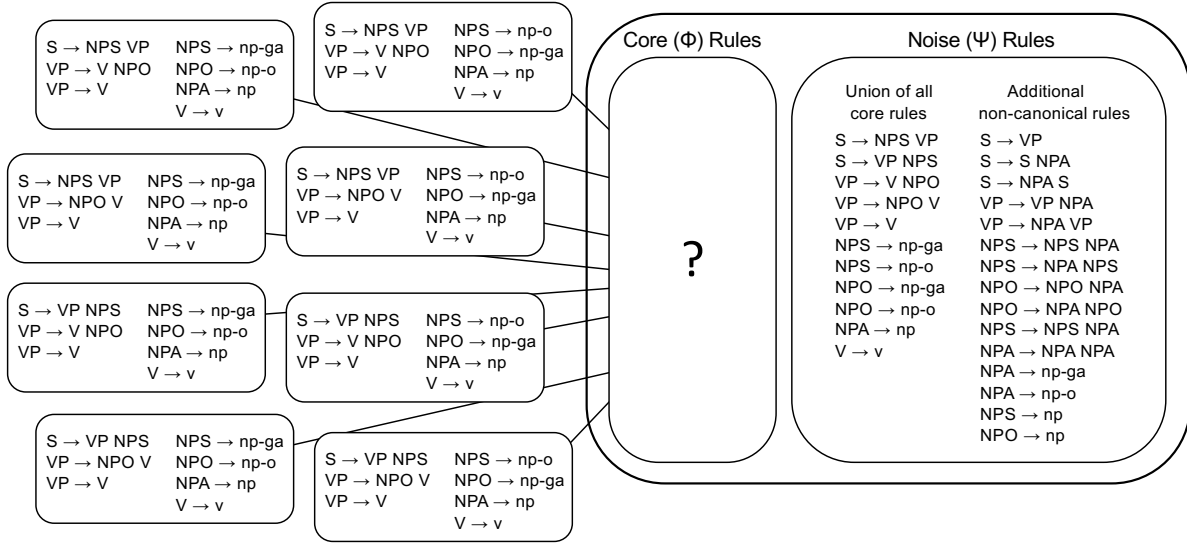


Figure 18. Hypothesis space of Noisy CFG learner for case-marking

verbs and noun phrases, some affixed with *ga* and *o*. Without knowing ahead of time the rate of noise or its properties, the learner evaluates how to partition its data into portions generated by the core case-marking and word-order component, and portions generated from noise. Comparing across the eight grammars, the learner selects the one that provides the best explanation for its data, thereby identifying both the target system of nominative/accusative case-marking and the basic word order of the language.

5.2 Data

We again used the CHILDES Miipro corpus of child-directed Japanese (Oshima-Takane et al., 1995), and followed a procedure similar to the one described in Section 4.2 to arrive at the distribution of strings shown in Table 3. The noun phrases that were all represented simply as **np** in the previous case study were subdivided into three types: those that were followed by ‘ga’ (**np-ga**), those that were followed by ‘o’ (**np-o**), and all others (**np**).¹¹ For

¹¹ This represents a simplification of the learning problem faced by a Japanese-learning child, who must identify the functions of *ga* and *o* among several other case-markers in the language, and come to distinguish case-markers from postpositions, which have very similar distributions. This also abstracts away from cases where certain Japanese predicates can mark objects with nominative *ga*, for instance when the subject is marked with dative case. How Japanese learners find their way around these complications is an important question that we leave for future work.

Japanese	
0.45 np-ga v	0.03 np np np-ga v
0.14 np np-ga v	0.02 np-ga np v
0.13 np-o v	0.02 np v np-ga
0.07 v np-ga	0.01 v np np-ga
0.05 np np-o v	0.01 v np-o
0.04 np-ga v np	

Table 3

Proportions of most frequent string types, case-marking learner

simplicity, we test our learner on one size of dataset, comprising 50 strings sampled in their relevant proportions. We restrict the dataset to strings in which there is exactly one **v**, as before, and furthermore retain only strings that contain at least one case-marked noun phrase (either **np-ga** or **np-o**), since the learner’s goal is to identify case-marking. Because case-markers are omitted very frequently in Japanese, this restriction is needed in order to ensure that our learner is able to observe case-marked noun-phrases in sufficient proportions in its 50-string dataset. This restriction means that this model does not learn from strings consisting of bare **v**, unlike the learner in Section 4. However, we do not attach any particular interpretation to the different makeups of these datasets: it is an empirical question how many sentences Japanese-learning children use for acquiring word order and case-marking, and therefore whether they might observe enough evidence for the distributions of case-marked noun-phrases without filtering their data in this manner. It is also an open empirical question whether word order and case-marking are acquired in tandem by Japanese learners, or sequentially. To the extent that our learner successfully infers that Japanese is SOV in this simulation, this shows us that this inference is possible in two settings: first, on the basis of the “unfiltered” data in Section 4, and second, on the basis of data that are filtered to only include sentences that are relevant for also learning case-marking.

From the distributions in Table 3, we observe that 55% of the strings in this learner’s dataset are incompatible with the core rules of the target grammar (SOV, with **np-ga** realizing subjects and **np-o** realizing objects), and no strings contain both case-markers simultaneously. Thus, the learner’s evidence for the target grammar is noisy and sparse.

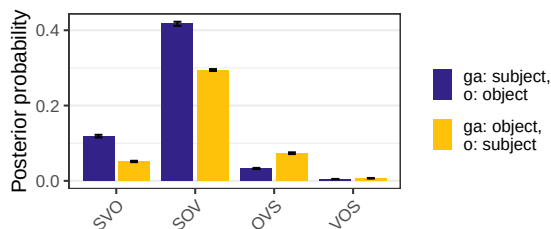


Figure 19. Posterior distribution over case-marking and word-order grammars, Noisy CFG learner

5.3 Results

5.3.1 Noisy CFG learner. Figure 19 displays the model’s inferred posterior probability distribution over the eight Noisy CFGs in its hypothesis space, averaged across 10 runs of the model. The SOV grammar in which **np-ga** realizes subjects and **np-o** realizes objects was assigned significantly higher posterior probability than any other grammar (all $t > 20.51$, all $p < 0.001$). Thus, the learner successfully chose SOV as the correct word order simultaneously with identifying the correct case-marking system in Japanese.

5.3.2 Comparison: Numerical regularization. To assess how our approach compares to the more common regularization approach in this learning problem, we again constructed a comparison learner with a “fully-flexible” hypothesis space. As before, this hypothesis space consists of a single grammar whose rules include all rules in the core and noise components of our learner’s eight Noisy CFGs. These rules allow all possible word order options and all possible mappings of case-marked **np**’s to clause arguments and adjuncts. This is in contrast to our learner’s hypothesis space, where the core rules for nominative/accusative case-marking encode the substantive expectation that *ga* and *o* each canonically mark a different argument, one marking NPS and the other marking NPO. This expectation is not present in the fully-flexible learner’s hypothesis space, which does not encode a distinction between canonical rules for case-marking and non-canonical noise processes. Identifying a case-marking system would mean identifying that **np-ga** is introduced with probability near 1 by only one clause argument (NPS or NPO), and that **np-o** is introduced with probability near 1 by the other.

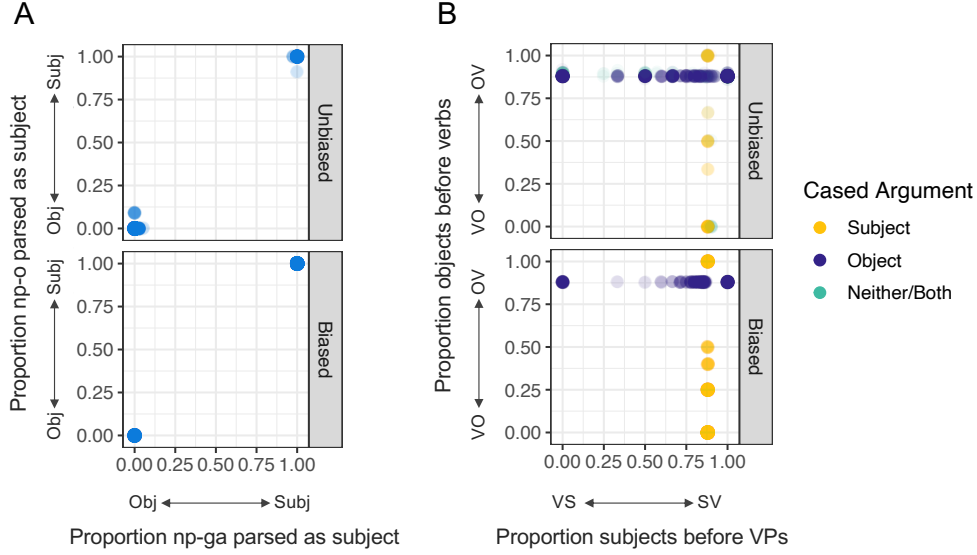


Figure 20. Posterior distribution in sampled treesets ($\vec{t}$) over (A) **np-ga** and **np-o** and (B) subject and object position, fully-flexible learner

We again tested two versions of this fully-flexible learner: one that assumes that all rules are equally probable *a priori*, and one with a strong numerical bias to put most probability mass on a single expansion for a given nonterminal (all α_i of the learner’s Dirichlet prior = 0.0001). For the learner to match our model’s success, we would expect to see it converge to a posterior in which it puts most probability mass on the correct rules for introducing **np-ga** and **np-o**, and also on the correct rules for subject and object position.

To assess the learner’s inference about case-marking, we examined its posterior distribution over trees for its data, $P(\vec{t} \mid G, \vec{w})$, following the same Gibbs sampling procedure described in the previous section. For each of these treesets, we calculated the proportion of **np-ga** that the model had parsed as NPS vs. NPO, and the proportion of **np-o** that the model had parsed as NPS vs. NPO. All of the treesets had at least one **np-ga** parsed as a subject or object, and at least one **np-o** parsed as a subject or object, so these proportions are well-defined. These proportions are plotted in Figure 20a.

If the learner had successfully mapped **np-ga** to subjects and **np-o** to objects, then we would expect to see the majority of tree samples in the lower right corners of the plots. Instead, we see that this fully-flexible learner failed to identify the target case-marking

system of Japanese. The learner’s posterior distribution over case-marked **np**’s lies in two different corners: analyses in which **np-ga** and **np-o** both realize subjects (upper right corner), or analyses in which they both realize objects (lower left corner). Both the biased and the unbiased learner showed an approximately similar pattern, with the biased learner converging more strongly to these deterministic corners. The unbiased learner placed significantly higher probability mass on analyses in which **np-ga** and **np-o** realize objects (mean proportion **np-ga** as subjects = 0.13; mean proportion **np-o** as subjects = 0.13; both significantly different from 0.5, $t > 9.04$, $p < 0.001$). The biased learner assigned approximately equal posterior probability to analyses in these two corners, with the mean proportion of **np-ga** and **np-o** parsed as subjects not significantly different from 0.5 (mean proportion **np-ga** as subjects = 0.73; mean proportion **np-o** as subjects = 0.73; both $t < 2.04$, both $p > 0.07$). Importantly, the learner did not treat **np-ga** and **np-o** differently in terms of the clause arguments that they realize; instead, it preferred to treat both as markers of the same argument.

Thus, while the learner’s numerical regularization bias led it to more strongly prefer certain deterministic corners of its hypothesis space, the pattern that emerges is not a case-marking system. Instead, the learner appears to have arrived at a solution in which **np-ga** and **np-o** were treated as members of the same category, and where the label of that category was either NPS or NPO. One possible explanation for the learner’s behavior might come from the fact that **np-ga** and **np-o** distribute similarly in the learner’s data, both tending to occur immediately to the left of **v**. These distributional similarities may have led the learner to categorize both of these case-marked **np**’s together. Using NPS or NPO as the label for this category would then capture another pattern in the data: case-marked noun-phrases only occur once per sentence, with no strings in the dataset containing both **np-ga** and **np-o** simultaneously. Because the learner’s rules allow only one NPS and one NPO to occur per derivation, assigning one of these to be the label of the “case-marked **np**” category provides a good account for why only one case-marked **np** would occur per string. This property of the data, which other learners may consider accidental, is one that the

fully-flexible learner treats as meaningful.¹²

We assessed the fully-flexible learner’s inference about word order by performing the same analysis described in Section 5.3.3 for our word order learner. For each sampled treeset in which at least one tree had a subject and at least one tree had an object, we calculated the proportion of subjects analyzed before the verb phrase, and the proportion of objects before the verb. These proportions are plotted in Figure 20b. Compared to the fully-flexible Japanese learner in Section 5, more of the sampled treesets had NP’s in argument positions and were therefore analyzable using this method: 56% of the sampled treesets for the unbiased learner, and 65% of the sampled treesets for the biased learner.

If the learner had identified that Japanese has SOV word order, we would expect to see the samples centered on the upper right corner of these plots. Instead, the learner inferred a different distribution, which is similar for both the biased and unbiased learner, with more spread for the unbiased learner. The learner converged to two different sorts of analyses: either subjects occur 88% of the time before the VP, with object position varying; or objects occur 88% of the time before the verb, with subject position varying. This does not resemble the fully-flexible learner’s inference about Japanese word order in Section 5. However, when we examine how subject and object position interacts with case-marking, a clearer picture emerges. Recall that the overwhelming majority of treesets either treated both *ga* and *o* as markers of subjects, or treated both as markers of objects. For treesets in which case-marked *np*’s were analyzed as subjects (in yellow), the learner preferred to analyze those arguments as 88% preverbal, and did not fix the position of its non-case-marked objects. For treesets in which case-marked *np*’s were analyzed as objects (in purple), the learner preferred to analyze those arguments as 88% preverbal, and did not fix the position of its non-cased-marked subjects. Thus, the learner converged to a solution in which case-marked *np*’s are 88%

¹² One might wonder if this is a consequence of the learner solving two problems in tandem: jointly acquiring case-marking and word order. We re-ran these simulations in a context where SOV word order is known, i.e., all rules for introducing clause arguments fix the subject before the VP and the object before the verb. We found the same qualitative pattern, both for the fully-flexible learner and for our Noisy CFG learner.

preverbal, and non-case-marked *np*'s vary in their position. This solution captures a pattern in the learner's dataset: approximately 88% of case-marked *np*'s appear before the verb.

Thus, the fully-flexible learner again displayed a form of regularization, coming to prefer a more deterministic rule system for its variable data. The learner preferred to analyze *ga* and *o* as markers of the same argument, and took "before the verb" to be the defining property of this marked argument. But these tendencies towards determinism did not lead it in the right direction: it did not regularize in such a way as to identify either a case-marking system or a canonical word order.

5.4 Summary

Here we show that our approach of choosing among Noisy CFGs generalizes to a second naturalistic phenomenon that looks on the surface qualitatively different from word-order acquisition: learning which of two candidate case-markers marks subjects and which marks objects in a language like Japanese, from data that provide noisy and sparse evidence. We find that our model successfully identifies the grammar of Japanese nominative/accusative case-marking, using only distributions of verbs and noun phrases affixed with *ga* and *o* in child-directed speech. Japanese allows clause arguments and case-markers to be dropped, introducing a large amount of noise into the learner's data. Our model nonetheless succeeds at separating out noise from signal for the canonical case-marking grammar, in tandem with learning the canonical word order of Japanese. By contrast, a learner with a numerical regularization bias and no discrete choice of grammars falls short. The biased fully-flexible learner identifies more deterministic analyses for its variable data, but it does not identify a system of case-marking, in which *ga* and *o* are identified as markers of different clause arguments.

Why does our model again out-perform the general regularization bias approach for this learning problem? We can again consider the contribution of the two properties that distinguish our approach. First, our learner's hypothesis space encodes a distinction between

restrictive core rules governing the canonical positions and morphology of clause arguments, and more flexible noise rules that distort those distributions. Second, it encodes substantive expectations about the nature of the core rules. Similar to our previous case study, the learner’s substantive expectations arise from knowledge of the grammatical system that it is trying to identify: here, knowledge that it is learning a nominative/accusative case-marking system, and therefore that *ga* and *o* each canonically mark a different clause argument. The fully-flexible learner’s hypothesis space does not distinguish between a canonical system of morphological marking and non-canonical noise processes, and therefore does not encode an assumption that the positions of *ga* and *o* are canonically exclusive. Our findings demonstrate that this assumption is needed. A learner with only a numerical regularization bias, and no substantive expectations about the nature of the grammatical system it is acquiring, does not spontaneously identify that these morphemes should be analyzed with different grammatical functions given the data that it observes.

6 General Discussion

We offer a general computational account for how children might in principle manage to draw systematic generalizations from messy data in the incremental process of acquiring their first language. We introduce a mechanism for noise-tolerant learning of restrictive grammatical hypotheses. The type of learner that we consider assumes that its data are generated by a complex system: the particular grammatical processes that the learner is currently trying to acquire at the current stage of development, and other independent processes that conspire to introduce divergences from those target grammatical processes into the data. We model the inference process as a special case of grammar learning, in which the learner evaluates a choice among different *Noisy Grammars*: composite grammars in which a restricted set of “core” rules operates alongside a less restricted set of “noise” rules. By partitioning the data into portions that provide signal for the core component and portions generated by noise, the learner identifies the grammar whose restrictive core rules

provide the best explanation for the skews in its data. It does so without knowing ahead of time the rate or properties of noise that it will encounter.

Our approach provides an alternative to a prominent proposal that learning in early development is driven by a domain-general bias to regularize variable data (Austin et al., 2022; Hudson Kam & Newport, 2009, 2005; Newport, 1999; Reali & Griffiths, 2009; Singleton & Newport, 2004; K. Smith & Wonnacott, 2010). We compare our learner to a common implementation of this general regularization bias approach, and show that both are able to account for results from a representative artificial language learning experiment (Austin et al., 2022; Reali & Griffiths, 2009; Perfors, 2012). However, we find that our learner out-performs the regularization bias approach in two naturalistic case studies in early syntax acquisition: learning the rules governing basic clause structure and those governing case morphology. We show that our learner succeeds because its architecture allows a natural way to express linguistically-motivated expectations about the character of those grammatical rules. A learner with a numerical regularization bias operating over a fully-flexible hypothesis space, and no expectation that it is acquiring a particular type of restrictive grammatical system, is not able to identify the correct canonical word order or case-marking system within its messy data.

These findings invite the possibility that other observed cases of regularization in grammar learning may be accounted for without adopting a fully-flexible hypothesis space—and that some cases may be better explained as noise-tolerant selection among discrete, restrictive grammatical hypotheses. We argue that this approach provides a straightforward way to encode knowledge about the specific types of regularities that a learner is expecting to encounter, which is important for learning to succeed in the case studies that we consider. In particular, our case studies showed that it was not sufficient to endow learners with a numerical bias towards probabilities close to zero or one, which is agnostic about the content of what is being learned and therefore does not favor any particular extreme within the gradient hypothesis space. However, another logical possibility is that a learner’s

domain-specific knowledge (for example, a preference for clauses to have subjects) could be expressed as a numerical bias that is asymmetrical, preferring certain extremes over others as a function of the particular learning problem being addressed.¹³ In other words, the current paper contrasts an approach to learning with a discrete hypothesis space and domain-specific knowledge against an approach with a gradient hypothesis space and few domain-specific expectations, but we note that these are not the only two possibilities in the space of theoretical options. We leave exploration of these issues to future work.

There are two sorts of questions that are raised by the particular domain-specific expectations that we posit here. One is whether these expectations about the structure of basic clauses can be empirically supported. This invites further empirical work investigating whether children possess this knowledge by the stage of development that we model here. The second question is where these particular domain-specific expectations might come from. One possibility is that they are part of the learner’s initial knowledge about the forms that grammars can take, i.e., what is often called Universal Grammar. If this is the case, it might explain how children arrive at the knowledge of word order that they seem to exhibit at this age, despite the problem of opacity that they face. It would also be in line with the tendencies of grammars to have these properties cross-linguistically (e.g., Baker, 2001; Keenan, 1976; Chomsky, 1982). However, it is also possible that these expectations could reflect linguistic knowledge that has been acquired at some earlier stage of development. If this is the case, it would require an account for how a child could arrive at this knowledge about the nature of clauses before identifying the position of clause arguments. The results

¹³ In particular, instead of adopting a symmetrical Dirichlet prior with a single α shape parameter, it is possible to encode asymmetrical preferences through a more general parameterization with distinct α values for each of the possible outcomes in the associated categorical distribution. The obligatoriness of subjects, for example, which we argued was an important part of the learner’s expectations in Section 4, could plausibly be encoded with an asymmetrical prior that prefers probabilities near 1 (over those near 0) for the rules $S \rightarrow NP VP$ and $S \rightarrow VP NP$, and prefers probabilities near 0 for $S \rightarrow VP$. It is less obvious how this kind of mechanism could express the assumptions about case-marking that we encoded in the learner’s hypothesis space in Section 5, however, because of the interdependence between different nonterminals’ rewrite options: the prior distribution would need to prefer points in the gradient hypothesis space that assign probabilities near one to both $NPS \rightarrow np\text{-}ga$ and $NPO \rightarrow np\text{-}o$, and similarly for $NPS \rightarrow np\text{-}o$ and $NPO \rightarrow np\text{-}ga$, while dispreferring points with high probabilities for both $NPS \rightarrow np\text{-}ga$ and $NPO \rightarrow np\text{-}ga$, for example.

that we have reported here do not distinguish these possibilities.

In contrasting how regularization behavior arises from these two approaches, we observe a difference in how learning interacts with the size of the learner’s data. On the numerical regularization account, regularization is a property of learning with small amounts of data: as a learner’s intake grows, its prior bias plays less of a role. Many previous accounts have therefore proposed that regularization arises in part from cognitive constraints that significantly limit the amount of data that children are able to take in for learning (Keogh et al., 2024; Perfors, 2012; Newport, 1999, 1990). On our restrictive hypotheses account, we see that limitations to the learner’s intake are not necessary in the same way. Our learner succeeds at identifying the correct canonical grammar with strikingly small amounts of data, but as more data are observed, its regularization tendencies do not weaken; instead, they become stronger as it becomes more confident in its guesses of how to partition data as coming from core vs. noise rules. The signal that our learner identifies for the core rules may indeed come from a small portion of its data, but observing more data allows it to determine with greater certainty which portion it should attend to. These differences invite questions about whether and how the strength of children’s grammatical generalizations vary relative to the amount of data that they encode.

The Noisy Grammar model we have introduced can straightforwardly be extended to deal with more complex grammatical phenomena than we have addressed here. Acquiring a particular piece of grammatical knowledge means selecting a hypothesis embodying that knowledge from some wider range of options; investigating the learning of a particular grammatical phenomenon therefore requires formulating a hypothesis space in which some but not all of the options embody that phenomenon. Recall that we first illustrated the basics of the system in Section 3.3 with a hypothesis space where the possible Φ components differed only in whether objects preceded the verb or followed it; in Section 4 we crossed this with a further two-way choice between whether subjects precede or follow the VP, presenting the learner with four possible basic word orders (SOV, SVO, OVS and VOS). This

hypothesis space does not make available the two other logically possible orders, VSO and OSV, and would therefore have to be extended to investigate the acquisition of word order in a language such as Irish, which has VSO word order. This could be done by instead crossing the OV/VO choice with a three-way choice that includes the option of assembling a sentence in such a way that the two components of the VP are discontinuous and surround the subject, rather than only to its left (as in VOS and OVS) or to its right (as in SVO and SOV).

A further direction to explore is the acquisition of word order in a language like German, where the core word order in general is SOV but this is only observed in embedded clauses because of the “verb second” (V2) property of main clauses (e.g. Thiersch, 1978). The learning difficulties posed by this combination of interacting mechanisms have been discussed by Gibson & Wexler, 1994, for example. To explore this scenario using our model, we could modify the hypothesis space in Figure 12 to include rules allowing embedded clauses (e.g. $VP \rightarrow V S$) in all four Φ components, and then cross the four-way choice there with an additional two-way choice for the V2 option (SVO-with-V2, SVO-without-V2, SOV-with-V2, etc.). The distinctive grammatical rules in the V2 Φ components would be rules that displace elements in ways that are formally analogous to those that would allow a verb and its object to be separated in VSO and OSV word orders. These sorts of choices of what to put in the hypothesis space reflect theoretical commitments about the content of the inductive bias that the learner brings to this particular stage of acquisition, and this raises the questions we noted above about the source of this inductive bias.

These are just two of many phenomena that might require moving from the simple context-free grammar rules that we have adopted in this paper to some more linguistically expressive formalism. The Noisy Grammar model that we have presented, and the probabilistic inference that our learner performs, is compatible with a wide range of grammar formalisms — specifically, it is compatible with any formalism that generates complex structures as a function of local choices about smaller subparts, where the likelihoods of the data are products of multinomials. Analyses of discontinuous verb phrases

(for VSO and OSV) and V2 word order from the contemporary syntax literature could both be straightforwardly formalized using “mildly context-sensitive” grammars that share the relevant mathematical characteristics with CFGs (e.g. Joshi, 1985; D. Weir, 1988; Stabler, 2011). With any such formalism, the important properties of a Noisy Grammar learner’s hypothesis space are (i) that flexible noise rules take the same form as the restrictive core rules (for example, for our Noisy CFG learner, they are also CFG productions), and (ii) that these noise rules are a superset of the union of all of the core rules in the learner’s hypothesis space. It is these relationships between the core rules and the noise rules, not the content or specific format of the rules themselves, that are crucial to the inferential logic that we have emphasized in this paper.

These future directions point towards ways in which our approach can be applied more widely to model the incremental growth of grammatical knowledge over the course of a child’s development. The developmental stage that we have modeled in the current case studies is one in which a child only attempts to acquire the basic positions of clause arguments, without yet attempting to explicitly acquire the grammatical processes through which those arguments may be displaced in “non-basic” constructions, such as *wh*-questions, relative clauses, and passives. This approach is motivated by empirical evidence that, at least in English, children do progress through two developmental stages with respect to these particular phenomena: they first identify the canonical positions of subjects and objects in their language, and only later identify the forms that argument movement can take (Gagliardi et al., 2016; Hirsh-Pasek & Golinkoff, 1996; Lidz et al., 2017; Perkins & Lidz, 2021, 2020). In the framework that we have put forward, a learner who has identified one of the current word-order hypotheses with some degree of certainty may then treat that knowledge as fixed, and then proceed to consider other hypotheses about Φ rules to account for the data that had been considered “noise” for the purposes of learning word-order. If such a learner settles on, say, SVO as its basic word order, it might discard the other three options in Figure 12 and move on to considering a new handful of candidate Φ components

that extend the chosen SVO basic word order with various different displacement possibilities; we pursue this in ongoing work.

This provides just one example of how our approach might account for two distinct stages of learning, out of the many that children eventually transit through on their way to acquiring a mature grammar. This gradual process of growing a grammar can be modeled as a process of introducing more grammatical rules into learners’ “core” rulesets, allowing them to understand the processes that were previously considered noise, with the result that less and less of their data is treated as noise over time. As discussed in Section 2, at no stage of development do these these core grammatical rules need to be fully deterministic; the learner at any particular developmental stage may consider that the core grammar includes variable processes that should not be “regularized away,” including those conditioned by contextual or sociolinguistic factors. Our framework provides a way for a learner to consider how the evidence for any grammatical phenomenon to be acquired, whether deterministic or nondeterministic, can be separated out from phenomena that are yet to be acquired at a particular stage of development.

In providing a way to explicitly account for the development of grammatical knowledge, our approach differs from other computational models of grammar acquisition, including those that apply forms of probabilistic learning. Our model is in some ways similar to other models of syntax learning that lean on the noise-tolerance of probabilistic Bayesian inference to address the issue of ambiguity and messiness in the learner’s data (Abend, Kwiatkowski, Smith, Goldwater, & Steedman, 2017; Kwiatkowski, Goldwater, Zettlemoyer, & Steedman, 2012; Perfors et al., 2011; Y. Yang & Piantadosi, 2022). However, on these previous approaches, grammars are learned wholesale: a learner considers a grammar that is complete even in its initial form, in the sense that it is intended to explicitly encode all of the interacting grammatical processes that could give rise to the observed data in any language. The process of learning is to gradually identify which of these rules have probability close to zero for the learner’s language, similar to the process taken by our “fully-flexible” learners.

Therefore, the the problem of being potentially misled by opaque interactions among grammatical phenomena is sidestepped by having the learner acquire all of these properties in tandem. Many other learning models take joint-learning approaches to various degrees (e.g., Elsner, Goldwater, Feldman, & Wood, 2013; Feldman, Griffiths, Goldwater, & Morgan, 2013; Gibson & Wexler, 1994; Howitt et al., 2021; Sakas & Fodor, 2001, 2012; Niyogi & Berwick, 1996; C. Yang, 2002), and it is possible that a certain amount of joint learning is quite useful for some phenomena. However, the empirical findings regarding the trajectory of basic word-order acquisition do not sit well with full joint learning in this particular domain, and point towards the need for a different sort of mechanism: one that can account for how at least certain pieces of a grammar are acquired one step at a time, from representations of data that are immature, incomplete, and even potentially misleading given the learner’s current grammatical knowledge. Our model provides a novel computational framework for characterizing this type of incremental learning.

Finally, returning to the two approaches to regularization that we compare in this paper, these relate more broadly to two general views of learning, in language and in other domains. On one view, learning involves summarizing the distributions in the learner’s data; this summary may be more or less veridical, if learners are generally biased towards certain distributions *a priori* (e.g., Elman et al., 1996; Aslin & Newport, 2012; Thelen & Smith, 2007). On another view, learning involves evaluating hypotheses about the generative systems that give rise to the distributions in the learner’s data in a specific domain. Learning is not an attempt to summarize those distributions, but rather to use them as indirect evidence to infer the underlying system that generated them (e.g., Chomsky, 1965, 1975; Lidz & Gagliardi, 2015; Lightfoot, 1991; C. Yang, 2002; Gallistel, 1990). We provide a formally explicit architecture for performing this inference in cases where the data are messy, because the generative system contains multiple components that interact in opaque ways. Solving this learning problem does not require assuming that learners bring with them a hypothesis space that is capable of encoding the full distribution of the data, in all of its

messiness. Instead, we argue that this problem can be solved if learners have specific assumptions about how their restrictive hypotheses will be noisily reflected in the data that they observe. We show that this approach can more readily account for the generalizations that infants draw in two areas of syntax acquisition. This provides support for theories in which learning, at least in certain domains, is underwritten by restrictive generative systems in a learner's hypothesis space, combined with a mechanism for filtering signal from noise.

7 References

- Abend, O., Kwiatkowski, T., Smith, N. J., Goldwater, S., & Steedman, M. (2017). Bootstrapping language acquisition. *Cognition*, 164, 116–143.
- Aslin, R. N., & Newport, E. L. (2012). Statistical learning: From acquiring specific items to forming general rules. *Current directions in psychological science*, 21(3), 170–176.
- Austin, A. C., Schuler, K. D., Furlong, S., & Newport, E. L. (2022). Learning a Language from Inconsistent Input: Regularization in Child and Adult Learners. *Language Learning and Development*, 18(3), 249–277.
- Babineau, M., Shi, R., & Christophe, A. (2020). 14-month-olds exploit verbs’ syntactic contexts to build expectations about novel words. *Infancy*, 25(5), 719–733.
- Baker, M. C. (2001). The Natures of Nonconfigurality. In R. K. Baltin & C. Collins (Eds.), *The Handbook of Contemporary Syntactic Theory* (pp. 407–438). Malden, Mass: Blackwell Publishers Ltd.
- Beech, C., & Swingley, D. (2023). Consequences of phonological variation for algorithmic word segmentation. *Cognition*, 235, 105401.
- Behrend, E. R., & Bitterman, M. E. (1961). Probability-matching in the fish. *The American Journal of Psychology*, 74(4), 542–551.
- Bever, T. G. (1982). Regression in the service of development. In *Regressions in Mental Development* (pp. 153–188). Routledge.
- Bickerton, D. (1981). *Roots of language*. Ann Arbor, MI: Karoma.
- Bickerton, D. (1984). The language bioprogram hypothesis. *Behavioral and brain sciences*, 7(2), 173–188.
- Bion, R. A., Miyazawa, K., Kikuchi, H., & Mazuka, R. (2013). Learning phonemic vowel length from naturalistic recordings of Japanese infant-directed speech. *PloS one*, 8(2), e51594.
- Booth, T. L., & Thompson, R. A. (1973). Applying probability measures to abstract languages. *IEEE Transactions on Computers*, C-22, 442–450.

- Brown, R. (1973). *A First Language: The Early Stages*. Cambridge, MA: Harvard University Press.
- Brusini, P., Dehaene-Lambertz, G., Van Heugten, M., De Carvalho, A., Goffinet, F., Fiévet, A.-C., & Christophe, A. (2017). Ambiguous function words do not prevent 18-month-olds from building accurate syntactic category expectations: An ERP study. *Neuropsychologia*, 98, 4–12.
- Bullock, D. H., & Bitterman, M. E. (1962). Probability-matching in the pigeon. *The American Journal of Psychology*, 75(4), 634–639.
- Cameron-Faulkner, T., Lieven, E., & Tomasello, M. (2003). A construction based analysis of child directed speech. *Cognitive science*, 27(6), 843–873.
- Chemla, E., Mintz, T. H., Bernal, S., & Christophe, A. (2009). Categorizing words using ‘frequent frames’: what cross-linguistic analyses reveal about distributional acquisition strategies. *Developmental science*, 12(3), 396–406. (Publisher: Wiley Online Library)
- Chomsky, N. (1957). *Syntactic Structures*. The Hague: Mouton.
- Chomsky, N. (1965). *Aspects of the Theory of Syntax*. Cambridge, MA: MIT Press.
- Chomsky, N. (1975). *Reflections on language*. New York, NY: Pantheon.
- Chomsky, N. (1980). *Rules and representations*. New York: Columbia University Press.
- Chomsky, N. (1982). *Some concepts and consequences of the theory of government and binding*. MIT Press.
- Chomsky, N. (1986). *Knowledge of language: Its nature, origins and use*. New York: Praeger.
- Christophe, A., Millotte, S., Bernal, S., & Lidz, J. (2008). Bootstrapping Lexical and Syntactic Acquisition. *Language and Speech*, 51(1-2), 61–75.
- Craig, G. J., & Myers, J. L. (1963). A developmental study of sequential two-choice decision making. *Child Development*, 483–493.
- Cristia, A. (2018). Can infants learn phonology in the lab? A meta-analytic answer. *Cognition*, 170, 312–327.

- Cristia, A., Dupoux, E., Ratner, N. B., & Soderstrom, M. (2019). Segmentability differences between child-directed and adult-directed speech: A systematic test with an ecologically valid corpus. *Open Mind*, 3, 13–22.
- Culbertson, J., & Kirby, S. (2016). Simplicity and Specificity in Language: Domain-General Biases Have Domain-Specific Effects. *Frontiers in Psychology*, 6.
- Culbertson, J., Smolensky, P., & Wilson, C. (2013). Cognitive biases, linguistic universals, and constraint-based grammar learning. *Topics in cognitive science*, 5(3), 392–424.
- Dautriche, I., Cristia, A., Brusini, P., Yuan, S., Fisher, C., & Christophe, A. (2014). Toddlers Default to Canonical Surface-to-Meaning Mapping When Learning Verbs. *Child Development*, 85(3), 1168–1180.
- Demuth, K., & Tremblay, A. (2008). Prosodically-conditioned variability in children’s production of French determiners. *Journal of child language*, 35(1), 99–127.
- Derks, P. L., & Paclisanu, M. I. (1967). Simple strategies in binary prediction by children and adults. *Journal of Experimental Psychology*, 73(2), 278.
- Elman, J. L., Bates, E. A., Johnson, M. H., Karmiloff-Smith, A., Plunkett, K., & Parisi, D. (1996). *Rethinking innateness: A connectionist perspective on development*. MIT press.
- Elsner, M., Goldwater, S., Feldman, N., & Wood, F. (2013). A joint learning model of word segmentation, lexical acquisition, and phonetic variability. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing* (pp. 42–54). Association for Computational Linguistics.
- Estes, W. K. (1964). Probability learning. In *Categories of human learning* (pp. 89–128). Elsevier.
- Estes, W. K. (1976). The cognitive side of probability learning. *Psychological Review*, 83(1), 37.
- Feldman, N. H., Griffiths, T. L., Goldwater, S., & Morgan, J. L. (2013). A role for the developing lexicon in phonetic category acquisition. *Psychological review*, 120(4), 751. (Publisher: American Psychological Association)

- Ferdinand, V., Kirby, S., & Smith, K. (2019). The cognitive roots of regularization in language. *Cognition*, 184, 53–68.
- Fisher, C., Jin, K.-S., & Scott, R. M. (2019). The developmental origins of syntactic bootstrapping. *Topics in Cognitive Science*, 12(1), 48–77.
- Fodor, J. D. (1998). Parsing to learn. *Journal of Psycholinguistic research*, 27(3), 339–374.
- Franck, J., Millotte, S., Posada, A., & Rizzi, L. (2013). Abstract knowledge of word order by 19 months: An eye-tracking study. *Applied Psycholinguistics*, 34(2), 323–336.
- Frank, R., & Kapur, S. (1996). On the use of triggers in parameter setting. *Linguistic Inquiry*, 623–660.
- Gagliardi, A., Mease, T. M., & Lidz, J. (2016). Discontinuous development in the acquisition of filler-gap dependencies: Evidence from 15-and 20-month-olds. *Language Acquisition*, 23(3), 1–27.
- Gallistel, C. R. (1990). *The organization of learning*. The MIT Press.
- Gardner, R. A. (1957). Probability-learning with two and three choices. *The American Journal of Psychology*, 70(2), 174–185.
- Gavarró, A., Leela, M., Rizzi, L., & Franck, J. (2015). Knowledge of the OV parameter setting at 19 months: Evidence from Hindi–Urdu. *Lingua*, 154, 27–34.
- Geman, S., & Geman, D. (1984). Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images. *IEEE Transactions on pattern analysis and machine intelligence*(6), 721–741.
- Gertner, Y., Fisher, C., & Eisengart, J. (2006). Learning words and rules: abstract knowledge of word order in early sentence comprehension. *Psychological Science*, 17(8), 684–691.
- Geyer, C. J. (1991). Markov chain Monte Carlo maximum likelihood. *Computing Science and Statistics: Proceedings of the 23rd Symposium on the Interface*, 156–163. (New York American Statistical Association)
- Gibson, E., & Wexler, K. (1994). Triggers. *Linguistic Inquiry*, 25, 407–454.

- Gleitman, L. R. (1990). The structural sources of verb meanings. *Language acquisition*, 1(1), 3–55.
- Greenberg, J. H. (1970). *Language Universals*. De Gruyter.
- Gutman, A., Dautriche, I., Crabbé, B., & Christophe, A. (2015). Bootstrapping the syntactic bootstrapper: probabilistic labeling of prosodic phrases. *Language Acquisition*, 22(3), 285–309.
- Göksun, T., Küntay, A. C., & Naigles, L. R. (2008). Turkish children use morphosyntactic bootstrapping in interpreting verb meaning. *Journal of child language*, 35(2), 291–323.
- Haryu, E., & Kajikawa, S. (2016). Use of bound morphemes (noun particles) in word segmentation by Japanese-learning infants. *Journal of Memory and Language*, 88, 18–27. (Publisher: Elsevier)
- Hastings, W. K. (1970). Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57(1), 97–109.
- He, A. X., & Lidz, J. (2017). Verb learning in 14-and 18-month-old English-learning infants. *Language Learning and Development*, 1–22.
- Hicks, J., Maye, J., & Lidz, J. (2007). The role of function words in infants’ syntactic categorization of novel words. *Linguistic Society of America Annual Meeting*.
- Hirsh-Pasek, K., & Golinkoff, R. M. (1996). The intermodal preferential looking paradigm: A window onto emerging language comprehension. In D. McDaniel, C. McKee, & H. S. Cairns (Eds.), *Methods for assessing children’s syntax* (pp. 105–124). Cambridge, MA: The MIT Press.
- Hitczenko, K., & Feldman, N. H. (2022). Naturalistic speech supports distributional learning across contexts. *Proceedings of the National Academy of Sciences*, 119(38), e2123230119.
- Howitt, K., Dey, S., & Sakas, W. G. (2021, January). Gradual syntactic triggering: The gradient parameter hypothesis. *Language Acquisition*, 28(1), 65–96.
- Hudson Kam, C. L., & Newport, E. L. (2005). Regularizing Unpredictable Variation: The

- Roles of Adult and Child Learners in Language Formation and Change. *Language Learning and Development*, 1(2), 151–195.
- Hudson Kam, C. L., & Newport, E. L. (2009). Getting it right by getting it wrong: When learners change languages. *Cognitive psychology*, 59(1), 30–66.
- Hyams, N. (1986). *Language Acquisition and the Theory of Parameters*. Dordrecht: D. Reidel Publishing Company.
- Höhle, B., Weissenborn, J., Kiefer, D., Schulz, A., & Schmitz, M. (2004). Functional elements in infants' speech processing: The role of determiners in the syntactic categorization of lexical elements. *Infancy*, 5(3), 341–353.
- Jin, K.-S., & Fisher, C. (2014). Early evidence for syntactic bootstrapping: 15-month-olds use sentence structure in verb learning. In *Proceedings of the 38th Boston University Conference on Language Development*. Boston, MA: Cascadia Press.
- Johnson, J. S., Shenkman, K. D., Newport, E. L., & Medin, D. L. (1996). Indeterminacy in the grammar of adult language learners. *Journal of memory and language*, 35(3), 335–352.
- Johnson, M., Griffiths, T., & Goldwater, S. (2007). Bayesian inference for PCFGs via Markov Chain Monte Carlo. In *Human Language Technologies 2007: The Conference of the North American Chapter of the Association for Computational Linguistics; Proceedings of the Main Conference* (pp. 139–146). Association for Computational Linguistics.
- Joshi, A. (1985). How much context-sensitivity is necessary for characterizing structural descriptions? In D. Dowty, L. Karttunen, & A. Zwicky (Eds.), *Natural language processing: Theoretical, computational and psychological perspectives* (pp. 206–250). New York: Cambridge University Press.
- Jusczyk, P. W., & Aslin, R. N. (1995). Infants' detection of the sound patterns of words in fluent speech. *Cognitive psychology*, 29(1), 1–23.
- Keenan, E. L. (1976). Towards a Universal Definition of 'Subject'. In C. Li (Ed.), *Syntax and Semantics: Subject and Topic*. New York: Academic Press.

- Keogh, A., Kirby, S., & Culbertson, J. (2024). Predictability and Variation in Language Are Differentially Affected by Learning and Production. *Cognitive Science*, 48(4), e13435.
- Kim, Y. J., & Sundara, M. (2021, July). 6-month-olds are sensitive to English morphology. *Developmental Science*, 24(4), e13089.
- Kwiatkowski, T., Goldwater, S., Zettlemoyer, L., & Steedman, M. (2012). A probabilistic model of syntactic and semantic acquisition from child-directed utterances and their meanings. In *Proceedings of the 13th Conference of the European Chapter of the Association for Computational Linguistics* (pp. 234–244).
- Labov, W. (1989). The child as linguistic historian. *Language variation and change*, 1(1), 85–97. (Publisher: Cambridge University Press)
- Lewis, J. D., & Elman, J. L. (2001). Learnability and the statistical structure of language: Poverty of stimulus arguments revisited. In *Proceedings of the 26th annual Boston University conference on language development* (Vol. 1, pp. 359–370). Citeseer.
- Lidz, J., & Gagliardi, A. (2015). How nature meets nurture: Universal grammar and statistical learning. *Annu. Rev. Linguist.*, 1(1), 333–353.
- Lidz, J., White, A. S., & Baier, R. (2017). The role of incremental parsing in syntactically conditioned word learning. *Cognitive Psychology*, 97, 62–78.
- Lightfoot, D. (1991). *How to set parameters: Arguments from language change*. Cambridge, MA: MIT Press.
- Maitra, S., & Perkins, L. (2023). Filtering Input for Learning Constrained Grammatical Variability: The Case of Spanish Word Order. *Proceedings of the Society for Computation in Linguistics*, 6(1), 108–120.
- Manzini, M. R., & Wexler, K. (1987). Parameters, Binding Theory, and Learnability. *Linguistic Inquiry*, 18(3), 413–444.
- Marquis, A., & Shi, R. (2012). Initial morphological learning in preverbal infants. *Cognition*, 122(1), 61–66. (Publisher: Elsevier)
- Matsuo, A., Kita, S., Shinya, Y., Wood, G. C., & Naigles, L. (2012). Japanese two-year-olds

- use morphosyntax to learn novel verb meanings. *Journal of child language*, 39(3), 637–663.
- Mattys, S. L., & Jusczyk, P. W. (2001). Phonotactic cues for segmentation of fluent speech by infants. *Cognition*, 78(2), 91–121.
- Mattys, S. L., Jusczyk, P. W., Luce, P. A., & Morgan, J. L. (1999). Phonotactic and prosodic effects on word segmentation in infants. *Cognitive psychology*, 38(4), 465–494.
- Maye, J., Werker, J. F., & Gerken, L. (2002). Infant sensitivity to distributional information can affect phonetic discrimination. *Cognition*, 82(3), 101–111.
- McCloskey, J. (1997). Subjecthood and subject positions. In L. Haegeman (Ed.), *Elements of grammar: Handbook of generative syntax* (pp. 197–235). Springer.
- Miller, K. (2013). Acquisition of variable rules:/s/-lenition in the speech of Chilean Spanish-speaking children and their caregivers. *Language variation and change*, 25(3), 311–340. (Publisher: Cambridge University Press)
- Miller, K., & Schmitt, C. (2012). Variable input and the acquisition of plural morphology. *Language Acquisition*, 19(3), 223–261. (Publisher: Taylor & Francis)
- Mintz, T. H. (2003). Frequent frames as a cue for grammatical categories in child directed speech. *Cognition*, 90(1), 91–117.
- Mintz, T. H. (2013). The segmentation of sub-lexical morphemes in English-learning 15-month-olds. *Frontiers in Psychology*, 4(24).
- Morgan, J. L., & Demuth, K. (Eds.). (1996). *Signal to syntax: Bootstrapping from speech to grammar in early acquisition*. Lawrence Erlbaum.
- Myers, J. L. (1976). Probability learning and sequence learning. *Handbook of Learning and Cognitive Processes*, ed. WK Estes, 171–205.
- Newport, E. L. (1990). Maturational constraints on language learning. *Cognitive science*, 14(1), 11–28.
- Newport, E. L. (1999). Reduced input in the acquisition of signed languages: Contributions to the study of creolization. In M. Degraff (Ed.), *Creolization, diachrony, and language*

- acquisition*. Cambridge, MA: MIT Press.
- Niyogi, P., & Berwick, R. C. (1996). A language learning model for finite parameter spaces. *Cognition*, 61(1-2), 161–193. (Publisher: Elsevier)
- Oshima-Takane, Y., MacWhinney, B., Sirai, H., Miyata, S., & Naka, N. (1995). *CHILDES manual for Japanese* (Tech. Rep.). Montreal: McGill University.
- Perfors, A. (2012). When do memory limitations lead to regularization? An experimental and computational investigation. *Journal of Memory and Language*, 67(4), 486–506.
- Perfors, A., Tenenbaum, J. B., & Regier, T. (2011). The learnability of abstract syntactic principles. *Cognition*, 118(3), 306–338. (Publisher: Elsevier)
- Perfors, A., Tenenbaum, J. B., & Wonnacott, E. (2010). Variability, negative evidence, and the acquisition of verb argument constructions. *Journal of child language*, 37(3), 607–642.
- Perkins, L., Feldman, N. H., & Lidz, J. (2022). The power of ignoring: filtering input for argument structure acquisition. *Cognitive Science*, 46(1).
- Perkins, L., & Lidz, J. (2020). Filler-gap dependency comprehension at 15 months: The role of vocabulary. *Language Acquisition*, 27(1), 98–115.
- Perkins, L., & Lidz, J. (2021). 18-month-old infants represent non-local syntactic dependencies. *Proceedings of the National Academy of Sciences*, 118(41), e2026469118.
- Pinker, S. (1984). *Language Learnability and Language Development*. Cambridge, MA: Harvard University Press.
- Real, F., & Christiansen, M. H. (2005). Uncovering the richness of the stimulus: Structure dependence and indirect statistical evidence. *Cognitive Science*, 29(6), 1007–1028. (Publisher: Wiley Online Library)
- Real, F., & Griffiths, T. L. (2009). The evolution of frequency distributions: Relating regularization to inductive biases through iterated learning. *Cognition*, 111(3), 317–328.
- Roberts, J. (1997). Acquisition of variable rules: a study of (-t, d) deletion in preschool

- children. *Journal of child language*, 24(2), 351–372. (Publisher: Cambridge University Press)
- Roberts, J., & Labov, W. (1995). Learning to talk Philadelphian: Acquisition of short a by preschool children. *Language Variation & Change*, 7(1).
- Sagae, K. (2021). Tracking child language development with neural network language models. *Frontiers in Psychology*, 12, 674402. (Publisher: Frontiers Media SA)
- Sakas, W. G., & Fodor, J. D. (2001). The structural triggers learner. In S. Bertolo (Ed.), *Language acquisition and learnability* (pp. 172–233). Cambridge: Cambridge University Press. (Publisher:)
- Sakas, W. G., & Fodor, J. D. (2012). Disambiguating syntactic triggers. *Language Acquisition*, 19(2), 83–143. (Publisher: Taylor & Francis)
- Sambridge, M. (2014). A parallel tempering algorithm for probabilistic sampling and multimodal optimization. *Geophysical Journal International*, 196(1), 357–374.
- Schulz, L. E., & Sommerville, J. (2006). God Does Not Play Dice: Causal Determinism and Preschoolers’ Causal Inferences. *Child Development*, 77(2), 427–442.
- Seki, H., Matsumara, T., Fujii, M., & Kasami, T. (1991). On multiple context-free grammars. *Theoretical Computer Science*, 88, 191–229.
- Senghas, A., & Coppola, M. (2001). Children creating language: How Nicaraguan Sign Language acquired a spatial grammar. *Psychological science*, 12(4), 323–328.
- Shi, R., & Melançon, A. (2010). Syntactic Categorization in French-Learning Infants. *Infancy*, 15(5), 517–533.
- Shin, N., & Miller, K. (2022). Children’s Acquisition of Morphosyntactic Variation. *Language Learning and Development*, 18(2), 125–150.
- Singleton, J. L., & Newport, E. L. (2004). When learners surpass their models: The acquisition of American Sign Language from inconsistent input. *Cognitive Psychology*, 49(4), 370–407.
- Smith, J., & Durham, M. (2019). *Sociolinguistic variation in children’s language: Acquiring*

- community norms*. Cambridge University Press.
- Smith, J., Durham, M., & Richards, H. (2013, January). The social and linguistic in the acquisition of sociolinguistic norms: Caregivers, children, and variation. *Linguistics*, 51(2).
- Smith, K., Perfors, A., Fehér, O., Samara, A., Swoboda, K., & Wonnacott, E. (2017). Language learning, language use and the evolution of linguistic variation. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 372(1711), 20160051.
- Smith, K., & Wonnacott, E. (2010). Eliminating unpredictable variation through iterated learning. *Cognition*, 116(3), 444–449.
- Song, J. Y., Shattuck-Hufnagel, S., & Demuth, K. (2015). Development of phonetic variants (allophones) in 2-year-olds learning American English: A study of alveolar stop/t, d/codas. *Journal of Phonetics*, 52, 152–169. (Publisher: Elsevier)
- Stabler, E. P. (2004). Varieties of crossing dependencies: structure dependence and mild context sensitivity. *Cognitive Science*, 28, 699–720.
- Stabler, E. P. (2011). Computational perspectives on minimalism. In C. Boeckx (Ed.), *The oxford handbook of linguistic minimalism*. Oxford: Oxford University Press.
- Stevenson, H. W., & Weir, M. W. (1959). Variables affecting children’s performance in a probability learning task. *Journal of Experimental Psychology*, 57(6), 403.
- Stromswold, K. (1995). The acquisition of subject and object wh-questions. *Language Acquisition*, 4(1-2), 5–48.
- Suzuki, T. (1999). *Two aspects of Japanese case in acquisition* (Doctoral dissertation). University of Hawai’i at Manoa.
- Suzuki, T., & Kobayashi, T. (2017). Syntactic Cues for Inferences about Causality in Language Acquisition: Evidence from an Argument-Drop Language. *Language Learning and Development*, 13(1), 24–37.
- Swingley, D. (2019). Learning Phonology from Surface Distributions, Considering Dutch and English Vowel Duration. *Language Learning and Development*, 15(3), 199–216.

- Tenenbaum, J. B., & Griffiths, T. L. (2001). Generalization, similarity, and Bayesian inference. *Behavioral and brain sciences*, 24(04), 629–640.
- Thelen, E., & Smith, L. B. (2007). Dynamic Systems Theories. In W. Damon & R. M. Lerner (Eds.), *Handbook of Child Psychology* (1st ed.). Wiley.
- Thiersch, C. L. (1978). *Topics in german syntax* (Unpublished doctoral dissertation). MIT.
- Torrego, E. (1989). Unergative-unaccusative alternations in Spanish. *MIT Working Papers in Linguistics*, 10, 253–272.
- Valian, V. (1990). Logical and psychological constraints on the acquisition of syntax. In L. Frazier & J. G. De Villiers (Eds.), *Language Processing and Language Acquisition*. Dordrecht: Kluwer.
- Weir, D. (1988). *Characterizing mildly context-sensitive grammar formalisms* (Unpublished doctoral dissertation). University of Pennsylvania.
- Weir, M. W. (1964). Developmental changes in problem-solving strategies. *Psychological review*, 71(6), 473.
- Wetherell, C. S. (1980). Probabilistic Languages: A Review and Some Open Questions. *Computing Surveys*, 12, 361–379.
- Wolfram, W. (1985). Variability in tense marking: A case for the obvious. *Language Learning*, 35(2), 229–253.
- Yang, C. (2002). *Knowledge and learning in natural language*. Oxford: Oxford University Press.
- Yang, Y., & Piantadosi, S. T. (2022, February). One model for the learning of language. *Proceedings of the National Academy of Sciences*, 119(5), e2021865119.
- Zhu, J., Franck, J., Rizzi, L., & Gavarró, A. (2022). Do infants have abstract grammatical knowledge of word order at 17 months? Evidence from Mandarin Chinese. *Journal of Child Language*, 49(1), 60–79.

Appendix A

Details of the coin-flipping example

Recall the scenario with just Bag H: this bag contains an unknown number of Φ coins, which always come up heads, and an unknown number of Ψ coins, which all have some single unknown probability ψ of coming up heads. Ten times, a coin is chosen from the bag and flipped; this produces eight heads and two tails. How many of these ten flips might we guess came from Φ coins, and how many from Ψ coins?

In the main text we contrasted two hypotheses: one where there are six Φ -flips and four Ψ -flips ($N_\Phi = 6$), and one where there are zero Φ -flips and ten Ψ -flips ($N_\Phi = 0$). Conditioned upon the unknown probability ψ , the likelihood of the $N_\Phi = 6$ hypothesis is $\binom{4}{2}\psi^2(1-\psi)^2$, and the likelihood of the $N_\Phi = 0$ hypothesis is $\binom{10}{8}\psi^8(1-\psi)^2$. To compare the likelihoods not conditioned on ψ , we can marginalize over this parameter, as in (14) and (15).

$$(14) \quad \begin{aligned} P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N_\Phi = 0, \text{Bag H}\right) &= \int_0^1 P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N_\Phi = 0, \text{Bag H}, \psi\right) p(\psi) d\psi \\ &= \int_0^1 \binom{10}{8} \psi^8 (1-\psi)^2 p(\psi) d\psi \end{aligned}$$

$$(15) \quad \begin{aligned} P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N_\Phi = 6, \text{Bag H}\right) &= \int_0^1 P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N_\Phi = 6, \text{Bag H}, \psi\right) p(\psi) d\psi \\ &= \int_0^1 \binom{4}{2} \psi^2 (1-\psi)^2 p(\psi) d\psi \end{aligned}$$

These expressions make reference to $p(\psi)$, which encodes our prior beliefs about how likely the various possible values of ψ are. If we had reason to believe that some values were *a priori* more likely than others, then we might want to assign more “weight” to those values. On a numerical regularization approach, such a belief could in principle be expressed by assuming that $p(\psi)$ takes the form of a skewed Beta distribution. But on our approach, any skewed learning outcomes will arise from a choice among discrete restrictive systems, rather than a numerically skewed prior. So, here we assume that $p(\psi)$ follows a uniform Beta(1,1) distribution, meaning that all possible values of ψ are equally likely *a priori*.

This assumption of a uniform Beta prior makes available particularly simple

closed-form solutions for the integrals in (14) and (15). Specifically, for any m and k :

$$(16) \quad \int_0^1 \binom{m}{k} \theta^k (1 - \theta)^{m-k} p(\theta) d\theta = \frac{1}{m+1} \quad (\text{when } p(\cdot) \text{ is the flat/uniform prior over } [0, 1])$$

This tells us that if a θ -weighted coin is tossed m times, then the probability of heads appearing k times is $\frac{1}{m+1}$. It may be surprising at first that this does not depend on k , but since we make no assumptions at all about θ , any value of k from the set of options $\{0, 1, \dots, m\}$ is equally likely; and since there are $m+1$ options, each has probability $\frac{1}{m+1}$.

The integrals in (14) and (15) therefore evaluate to $\frac{1}{11}$ and $\frac{1}{5}$; these are the likelihoods of the data given $N_\Phi = 0$ and $N_\Phi = 6$, respectively, after marginalizing over ψ . As we saw in the main text, if we hypothesize n_Φ flips of Φ coins, then we must invoke $10 - n_\Phi$ flips of Ψ coins, and so the likelihood is $\frac{1}{(10-n_\Phi)+1} = \frac{1}{11-n_\Phi}$. Even better than positing six Φ -flips, then, is positing eight Φ -flips, and leaving only two Ψ flips, for a likelihood of $\frac{1}{3}$. But this is the highest we can go: if there are only eight heads in the data, then the two-headed coins could not have been flipped more than eight times. The cap on the number of Φ -flips is even lower under Bag T, where the Φ -coins produce only tails; here, the largest number of Φ -flips is two.

$$(17) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N_\Phi = n_\Phi, \text{Bag H}\right) = \begin{cases} \frac{1}{11 - n_\Phi} & \text{if } n_\Phi \leq 8 \\ 0 & \text{otherwise} \end{cases}$$

$$(18) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N_\Phi = n_\Phi, \text{Bag T}\right) = \begin{cases} \frac{1}{11 - n_\Phi} & \text{if } n_\Phi \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

To compare the likelihoods of Bag H and Bag T not conditioned on a particular choice of n_Φ , we can marginalize over all choices of $n_\Phi \in \{0, 1, \dots, 10\}$. Stepping more slowly through what we saw in the main text, (17) and (18) tell us that only certain subsets of that range will contribute non-zero values to the sum:

$$\begin{aligned}
(19) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag H}\right) &= \sum_{n_\Phi=0}^{10} \left[P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid N_\Phi = n_\Phi, \text{Bag H}\right) \times P(N_\Phi = n_\Phi \mid \text{Bag H}) \right] \\
&= \sum_{n_\Phi=0}^8 \left[\frac{1}{11 - n_\Phi} \times P(N_\Phi = n_\Phi \mid \text{Bag H}) \right]
\end{aligned}$$

$$(20) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag T}\right) = \sum_{n_\Phi=0}^2 \left[\frac{1}{11 - n_\Phi} \times P(N_\Phi = n_\Phi \mid \text{Bag T}) \right]$$

The remaining factors $P(N_\Phi = n_\Phi \mid \dots)$ in (19) and (20) are the probability that n_Φ of the ten draws yield Φ coins. We have assumed that we know nothing about the ratio of Φ coins to Ψ coins, which means that the probability of each value of n_Φ is uniform across the eleven values on the x -axis of Figure 4 in the main text, following the logic from (16).¹⁴

$$(21) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag H}\right) = \sum_{n_\Phi=0}^8 \left[\frac{1}{11 - n_\Phi} \times \frac{1}{11} \right] = 0.138$$

$$(22) \quad P\left(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag T}\right) = \sum_{n_\Phi=0}^2 \left[\frac{1}{11 - n_\Phi} \times \frac{1}{11} \right] = 0.027$$

Given the likelihoods in (21) and (22) and a prior probability for each bag, we can apply Bayes' Rule to calculate each bag's posterior probability. Assuming an equal prior of 0.5 for each bag, we find that Bag H is about five times more likely to have generated the data than Bag T.

$$\begin{aligned}
(23) \quad P(\text{Bag H} \mid \begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix}) &= \frac{P(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag H})P(\text{Bag H})}{P(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag H})P(\text{Bag H}) + P(\begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix} \mid \text{Bag T})P(\text{Bag T})} \\
&= \frac{0.138 \times 0.5}{0.138 \times 0.5 + 0.027 \times 0.5} \\
&= 0.834
\end{aligned}$$

$$P(\text{Bag T} \mid \begin{smallmatrix} 8 \text{ heads} \\ 2 \text{ tails} \end{smallmatrix}) = 0.166$$

¹⁴ Given different assumptions about the ratio of Φ coins to Ψ coins in the bags, we might want some portions of the x -axis of Figure 4 to be weighted more heavily than others. But the subset-superset relationship shown on the graph makes it clear that as long as our assumptions about this ratio are *the same* for the two bags, there is no way to distribute this weight that will make Bag T's total larger than Bag H's.

Appendix B

Details of Gibbs sampling

In the first step of sampling, we use Bayes' Rule to calculate the posterior probability of each grammar given the observed strings $\vec{w}$ and a collection of hypothesized trees $\vec{t}$ for those strings:

$$(24) \quad P(G|\vec{t}, \vec{w}) = \frac{P(\vec{t}, \vec{w}|G)P(G)}{\sum_{G'} P(\vec{t}, \vec{w}|G')P(G')}$$

Bayes' Rule tells us that the posterior probability of any grammar is proportional to the product of the likelihood (the probability of $\vec{t}$ and $\vec{w}$ under that grammar) and the prior probability of that grammar. We assume that all grammars have equal prior probability.

Because we are only considering trees that could have yielded the strings in the data, the joint likelihood of the trees and strings, $P(\vec{t}, \vec{w}|G)$, is equivalent to the likelihood of the trees alone, $P(\vec{t}|G)$. Calculating this likelihood requires summing over the unknown ways that each portion of these trees might be analyzed as stemming from either a core (Φ) or noise (Ψ) rewrite, i.e., the choices of “articulated trees” for a given tree. The specific core vs. noise choices are interchangeable for each particular nonterminal given a grammar, so we make this calculation tractable by considering how *many* core vs. noise rewrites might have occurred for each nonterminal. This follows similar logic to the illustration in the simple example in Section 3; here we show how this applies to the more general case.

We divide the n^A total observations of a particular nonterminal A into $n_1^A \dots n_m^A$ observations of the 1st through the m^{th} possible rewrites (collapsing across Φ -rewrites and Ψ -rewrites of A). The full likelihood of the set of trees, $P(\vec{t}|G)$, is the product over all nonterminals A of $P(n_1^A \dots n_m^A | G)$. We divide each of the observed rewrites of a nonterminal into some number of core rewrites (Φ) and some number of noise rewrites (Ψ).¹⁵

¹⁵ In Section 3, these choices occurred, perhaps more intuitively, in the reverse order: first we divided the total observations of a given nonterminal into Φ vs. Ψ observations, and then we partitioned the observations coming from each component (Φ vs. Ψ) among the different ways that the nonterminal could be rewritten. These two orders produce the same final result as long as the ranges for the summations are calculated appropriately, but the order that we demonstrate here scales better to larger grammars.

The n_1^A occurrences of the first type of rewrite for A are divided into $n_1^{A^\phi}$ core occurrences and $n_1^{A^\psi}$ noise occurrences. More generally, the n_m^A occurrences of the m^{th} rewrite type are divided into $n_m^{A^\phi}$ core occurrences and $n_m^{A^\psi}$ noise occurrences. We can calculate the likelihood by marginalizing over $n_1^{A^\phi} \dots n_m^{A^\psi}$:

$$(25) \quad P(\vec{t}|G) = \prod_A P(n_1^A \dots n_m^A | G) = \prod_A \left[\sum_{n_1^{A^\phi}=0}^{n_1^A} \dots \sum_{n_m^{A^\phi}=0}^{n_m^A} \left[P(n_1^{A^\phi} \dots n_m^{A^\phi} | n^{A^\phi}, G) \right. \right. \\ \left. \left. \times P(n_1^{A^\psi} \dots n_m^{A^\psi} | n^{A^\psi}, G) \right. \right. \\ \left. \left. \times P(n^{A^\phi} | n^A, G) \right] \right]$$

The first term in the summation is the probability of observing $n_1^{A^\phi} \dots n_m^{A^\phi}$ core occurrences of each rewrite type, out of n^{A^ϕ} total core occurrences of A . This follows a multinomial distribution with parameter $\vec{\phi}^{A_G}$. Because $\vec{\phi}^{A_G}$ is unknown, we integrate over all possible values of $\vec{\phi}^{A_G}$ to obtain

$$(26) \quad \frac{B(\vec{\alpha}_\phi^{A_G} + (n_1^{A^\phi} \dots n_m^{A^\phi}))}{B(\vec{\alpha}_\phi^{A_G})}$$

for this first term, where $\vec{\alpha}_\phi^{A_G}$ represents the parameters of the Dirichlet prior over $\vec{\phi}^{A_G}$, and $B(\cdot)$ is the multivariate Beta function. As we noted in Section 3, when the prior over rule weights is uniform (all components of $\vec{\alpha}_\phi^{A_G}$ are equal to 1), this is equivalent to 1 divided by the number of ways to partition n^{A^ϕ} core occurrences of A into m^{A^ϕ} possible rewrite types:

$$(27) \quad \frac{1}{\binom{n^{A^\phi} + m^{A^\phi} - 1}{m^{A^\phi} - 1}} = \frac{n^{A^\phi}!(m^{A^\phi} - 1)!}{(n^{A^\phi} + m^{A^\phi} - 1)!}$$

The second term in the sum in (25) is analogous: this is the probability, given n^{A^ψ} total noisy occurrences of A , of observing $n_1^{A^\psi} \dots n_m^{A^\psi}$ noisy occurrences of each rewrite type, which follows a multinomial distribution with parameter $\vec{\psi}^{A_G}$. The third term is the probability of observing n^{A^ϕ} total core occurrences out of n^A overall occurrences of A . This follows a binomial distribution with parameter $(1 - \epsilon^{A_G})$. We again integrate over all

possible values of $\vec{\psi}^{A_G}$ and ϵ^{A_G} , obtaining results analogous to (26).

This allows us to calculate the likelihood $P(\vec{t} | G)$ for each G in our hypothesis space, and (since we assume a flat prior over grammars) sample a new G with probability proportional to this likelihood.

After re-sampling a new grammar G , we then use a component-wise Hastings proposal to sample a new set of trees $\vec{t}$ for the observed strings, given G . Following M. Johnson et al. (2007), we consider the probability of a tree structure t_i for corresponding string w_i , given G and the current hypotheses about trees $\vec{t}_{-i}$ for all the other strings. We can define a function f that is proportional to the posterior distribution over t_i , $f(t_i) \propto P(t_i | w_i, \vec{t}_{-i}, G)$, as

$$(28) \quad f(t_i) = P(w_i | t_i) P(t_i | \vec{t}_{-i}, G)$$

The probability of a string being the yield of a given tree, $P(w_i | t_i)$, is always 1 or 0. The probability of a tree given all other trees and G , $P(t_i | \vec{t}_{-i}, G)$, is

$$(29) \quad P(t_i | \vec{t}_{-i}, G) = \frac{P(\vec{t} | G)}{P(\vec{t}_{-i} | G)}$$

Both $P(\vec{t} | G)$ and $P(\vec{t}_{-i} | G)$ can be calculated according to (25). In practice, $P(\vec{t}_{-i} | G)$ does not need to be calculated because it will cancel out in the acceptance function (30), below.

We can use this function f to sample $\vec{t}$ given G and $\vec{w}$ as follows. Within each iteration of the Gibbs sampler, we first initialize a new set of trees $\vec{t}$ for the Hastings sampler that we will then use to re-sample $\vec{t}$ given G and $\vec{w}$. We do so by randomly initializing the parameters θ for the compiled-out PCFG given by the current choice of grammar G , then generating full “articulated” trees for each string based on this choice of θ , and finally collapsing the articulated trees to remove the noise/non-noise distinction. Given these initialized trees, we then re-sample $\vec{t}$ using a procedure modified from M. Johnson et al. (2007). First, we choose a string w_i and its current corresponding t_i at random. Second, we take the other trees $\vec{t}_{-i}$, to be the output of a simple PCFG which generates these structures directly, rather than generating them via articulated trees that distinguish between noise vs. non-noise rewrites.

We estimate the probabilities of each rewrite for a given nonterminal in this simple PCFG, $\vec{\theta}^s$, by sampling from a multivariate Gaussian whose mean is set to the relative frequencies of each observed rewrite, using add-one smoothing to account for accidental gaps. Third, we generate a new proposed tree t_i' for w_i by sampling from this simple grammar’s distribution using $\vec{\theta}^s$. Finally, we decide to accept this proposal with probability

$$(30) \quad A(t_i') = \min \left(1, \frac{f(t_i')P(t_i|w_i, \vec{\theta}^s)}{f(t_i)P(t_i'|w_i, \vec{\theta}^s)} \right)$$

We ran multiple chains from different starting places to test convergence. For the simulations reported in Sections 4 and 5, we ran chains of 50,000 iterations of Gibbs sampling each, and analyzed every 10th iteration from the last quarter of each chain. We report averages across 10 chains as estimates of the posterior over G and $\vec{t}$.

To simulate the “fully-flexible” learners in these sections, we estimate the posterior distribution over $\vec{t}$ by using a component-wise Hastings sampler analogous to that for estimating $P(\vec{t}|G, \vec{w})$ in our original model. To improve convergence within the learner’s multimodal hypothesis space, we use parallel tempering (Geyer, 1991). We run 10 chains in parallel, of which 9 sample from a “tempered” version of the target posterior: the posterior is raised to a power between 0 and 1, flattening the distribution and allowing the chain to mix quickly. At each iteration, a state swap between the sampled trees $\vec{t}$ of two random chains is proposed, and this proposal is accepted using a Metropolis-Hastings update, which preserves the joint target posterior distributions for each chain; see Sambridge (2014) for detail. Only the chain that samples from the true target posterior $P(\vec{t}|G, \vec{w})$ is analyzed at the end of the run. We ran 10 such target chains of 50,000 Hastings iterations each, and analyzed every 10th iteration from the last quarter of each chain.

Appendix C

Appendix on data pre-processing

The procedure described here is a simple implementation of the idea that an infant has learned to recognize certain classes of functional elements, and can use those functional elements as cues to the positions of nouns and verbs. This idea receives substantial empirical support from findings that functional elements are recognized early in development and are used to classify content words by young infants, at or before the ages that we model here (approximately 15 months); see Table C1. These results primarily come from English, French, and German. We extrapolate from these results in modeling Japanese, noting that much less is known about infants’ early word class and word order development in that language, and also noting that the possibility that this development may not proceed within the same timeframe across all of the languages that we model. There are many ways in which the idea of using functional elements to identify noun phrases and verbs could be fleshed out (see e.g., Mintz, 2003; Chemla, Mintz, Bernal, & Christophe, 2009; Gutman, Dautriche, Crabbé, & Christophe, 2015), and different proposals will in general make distinct fine-grained predictions about exactly what syntactic information an infant can extract from a given utterance. But at the level of granularity that is relevant for this study, where our goal is to explore what can be learned from messy distributions of strings like those in Table 2 in the main text, we assume that many of these different options would yield roughly similar results.

We begin with CHILDES’s tokenization of corpus utterances, treating the “clitics” in the CHILDES annotation as separate tokens (e.g. CHILDES analyzes *you wanna* as *you want~to*, which we treat as three tokens: *you want to*). We also split off as separate tokens apparent occurrences of a small number of hand-identified affixes for each language. In English these are *-s*, *-ed* and *-ing*; in French, *-é(e)(s)*, *-er* and *-ons*; and in Japanese, *-nai*, *-te*, *-de*, and *-ba*. These were split off entirely mechanically, so that *corpus*, for example, would be separated into two tokens, *corpu* and *s*.

We then assume that an infant (at around the age of 15 months) can recognize certain

Determiners used to classify nouns	Shi and Melançon (2010); Hicks et al. (2007); Höhle et al. (2004); Babineau et al. (2020); He and Lidz (2017)
Pronouns treated as clause arguments (NPs)	Babineau et al. (2020); Jin and Fisher (2014)
Auxiliaries used to classify verbs	Hicks et al. (2007)
English <i>-s</i> , <i>-ing</i> , and <i>-ed</i> recognized as suffixes	Kim and Sundara (2021); Mintz (2013)
French <i>-é</i> recognized as suffix	Marquis and Shi (2012)
Japanese <i>ga</i> recognized as bound morpheme	Haryu and Kajikawa (2016)

Table C1

Summary of selected empirical findings suggesting infants’ abilities at or prior to 15 months to use functional elements to classify nouns and verbs.

functional elements from among these tokens. The full list of such recognizable elements, for each language, is given in Table C2. All of these recognizable elements were among the 100 most frequent tokens in the corresponding language’s corpus. We assume that some of these are recognized by the infant as belonging to some syntactically-meaningful category; in English, for example, we assume that an infant can recognize **you** as PRONOUN, **the** as a DET (determiner), **-ed** as a V-SUFFIX (verbal suffix) and **will** as an AUX. Others are simply given the “meaningless” category OTHER. Elements with messy or ambiguous distributions (such as **is**, which might be an auxiliary or a copula main verb, and **to**, which might be an auxiliary or a preposition), we tended to classify as OTHER, in order to be conservative in our assumptions about the infant’s syntactic knowledge. The ambiguous English suffix **-s** (plural or third-person present) was not disambiguated: each time it occurs, it was counted as the plural suffix, because this is more frequent according to CHILDES’s tagging.

Once these recognizable elements have been tagged in the corpus, we use some simple heuristics to guess at the positions of verbs and nouns on the basis of only those tags. For each language, we define a small set of “NOUN-cue” positions, such that any unrecognized element in one of these positions is tagged as a NOUN. For example, the position immediately following a DETERMINER is a NOUN-cue position in English and French. There are also “VERB-cue” positions for each language. The full list is given in Table C3. With the

initial tagging (based on Table C2) now supplemented with NOUN and VERB tags based on functional cues, the final step is to extract a sequence of **np** and **v** tokens. This final sequence includes an **np** wherever something has been tagged as a PRONOUN, NAME or NOUN, and includes a **v** wherever something has been tagged as a VERB. The full process is illustrated with two English example sentences in Figure C1.

English		French		Japanese	
you	PRONOUN	tu	PRONOUN	-te	V-SUFFIX
is	OTHER	est	OTHER	ne=TAG	SFP
what	OTHER	ce	OTHER	no=QUD	OTHER
the	DET	que	OTHER	yo=ASSERT	SFP
it	PRONOUN	ça	PRONOUN	da&PRES=be	OTHER
that	OTHER	le	DET	wa=TOP	CASE-MARKER
-s	N-SUFFIX	il	PRONOUN	no=GEN	CASE-MARKER
a	DET	là	OTHER	kore=this	PRONOUN
not	NEG	pas	NEG	-nai	NEG
do	AUX	je	PRONOUN	ga=NOM	CASE-MARKER1
I	PRONOUN	un	DET	nani=what	OTHER
to	OTHER	on	PRONOUN	tte=QUOT	OTHER
are	OTHER	cm	OTHER	ni=DAT	CASE-MARKER
-ing	V-SUFFIX	et	OTHER	ka=Q	SFP
no	OTHER	la	OTHER	Asato-kun_MFAM=Asato	NAME
and	OTHER	de	OTHER	mo=ADD	CASE-MARKER
oh	OTHER	ah	OTHER	ni&ADV=be	OTHER
your	DET	oui	OTHER	kanaa=DOUBT	SFP
in	OTHER	a	OTHER	kara=because	OTHER
on	OTHER	non	OTHER	koko=here	PRONOUN
did	AUX	le	DET	sore=this	PRONOUN
he	PRONOUN	hein	OTHER	da&PRES:na=be	OTHER
yeah	OTHER	y	OTHER	Natchan=Natchan	NAME
there	OTHER	les	DET	doko=where	OTHER
will	AUX	est-ce	OTHER	naa=TAG	SFP
this	OTHER	oh	OTHER	Nanami=Nanami	NAME
can	AUX	à	OTHER	de=LOC	CASE-MARKER
one	OTHER	va	AUX	kotchi=here	PRONOUN
she	PRONOUN	te	OTHER	o=ACC	CASE-MARKER2
here	OTHER	as	OTHER	Kakka=Mum	NAME
yes	OTHER	une	DET	de&wa-NEG-PRES=be	OTHER
-ed	V-SUFFIX	alors	OTHER	Okaasan=HON_Mother_HON	NAME
where	OTHER	voilà	OTHER	da&POL-PRES=be	OTHER
of	OTHER	en	OTHER	soo=like_that	OTHER
me	PRONOUN	elle	PRONOUN	koo=like_this	OTHER
for	OTHER	qui	OTHER	de&CONN=be	OTHER
we	PRONOUN	comme	OTHER	to=and	OTHER
with	OTHER	mais	OTHER	ne=APPEAL	OTHER
they	PRONOUN	quoi	OTHER	to=COMT	CASE-MARKER
right	OTHER	des	DET	daroo&POL=INFR	OTHER
up	OTHER	dans	OTHER	Juri-chan_FAM=Juri	NAME
how	OTHER	ben	OTHER	sa=FINA	SFP
well	OTHER	toi	PRONOUN	-de	V-SUFFIX
does	AUX	moi	PRONOUN	dore=which	OTHER
Sarah	NAME	où	OTHER	Natchi	NAME
now	OTHER	pour	OTHER	-ba	V-SUFFIX
Adam	NAME	bon	OTHER	da&POL-HORT=be	OTHER
why	OTHER	me	OTHER	dare=who	OTHER
huh	OTHER	allez	OTHER	jan=TAG	SFP
who	OTHER	la	OTHER	nande=why	OTHER
was	OTHER	ouais	OTHER	kono=this	DET
all	OTHER	faut	AUX	kedo=although	OTHER
out	OTHER	du	DET	doo=how	OTHER
her	OTHER	se	OTHER	yan=TAG	SFP
alright	OTHER	hop	OTHER	mon=ASSERT	SFP
be	OTHER	vas	AUX	wa=FINA	CASE-MARKER
down	OTHER	sur	OTHER	ya&PRES=be	OTHER
them	PRONOUN	si	OTHER		
just	OTHER	mh	OTHER		
at	OTHER	es	OTHER		
some	DET	bah	OTHER		
so	OTHER	nhm	OTHER		
okay	OTHER	ne	OTHER		
am	OTHER	ai	AUX		
hm	OTHER	d'accord	OTHER		
if	OTHER	ils	PRONOUN		
when	OTHER	maman	NAME		
would	AUX	ton	DET		
my	DET	sont	OTHER		
nhm	OTHER	Nanou	NAME		
too	OTHER	vais	AUX		
two	OTHER	aux	OTHER		
his	DET	lui	PRONOUN		
him	PRONOUN	euh	OTHER		
Eve	NAME	oui	OTHER		

Table C2

The elements we assume are recognizable by the learner for each language. Of the 75 recognizable English elements, 29 play a role in the definition of cue positions; the remaining 46 are recognized only as OTHER. In French, there are also 75 recognizable elements, of which 27 are used as cues (with 49 OTHER); in Japanese, there are 57 recognizable elements, of which 24 are used as cues (with 23 OTHER).

English and French	
NOUN-cue positions	VERB-cue positions
following a DET	following an AUX
preceding a N-SUFFIX	following a NEG that follows an AUX
	preceding a V-SUFFIX
Japanese	
NOUN-cue positions	VERB-cue positions
preceding a CASE-MARKER	preceding a NEG
	preceding a SFP (sentence-final particle)
	preceding a V-SUFFIX

Table C3

The “cue positions” defined for each language

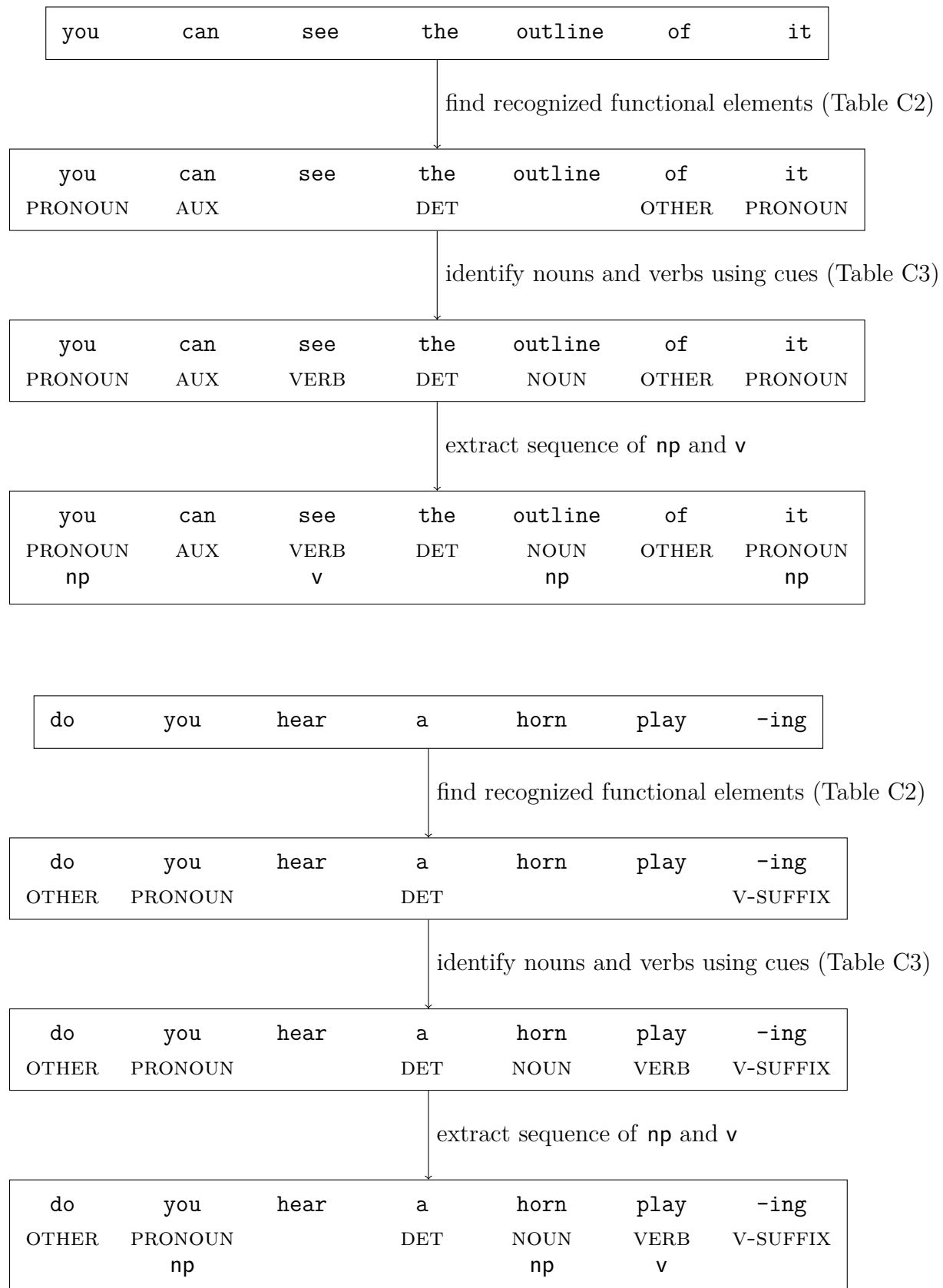


Figure C1